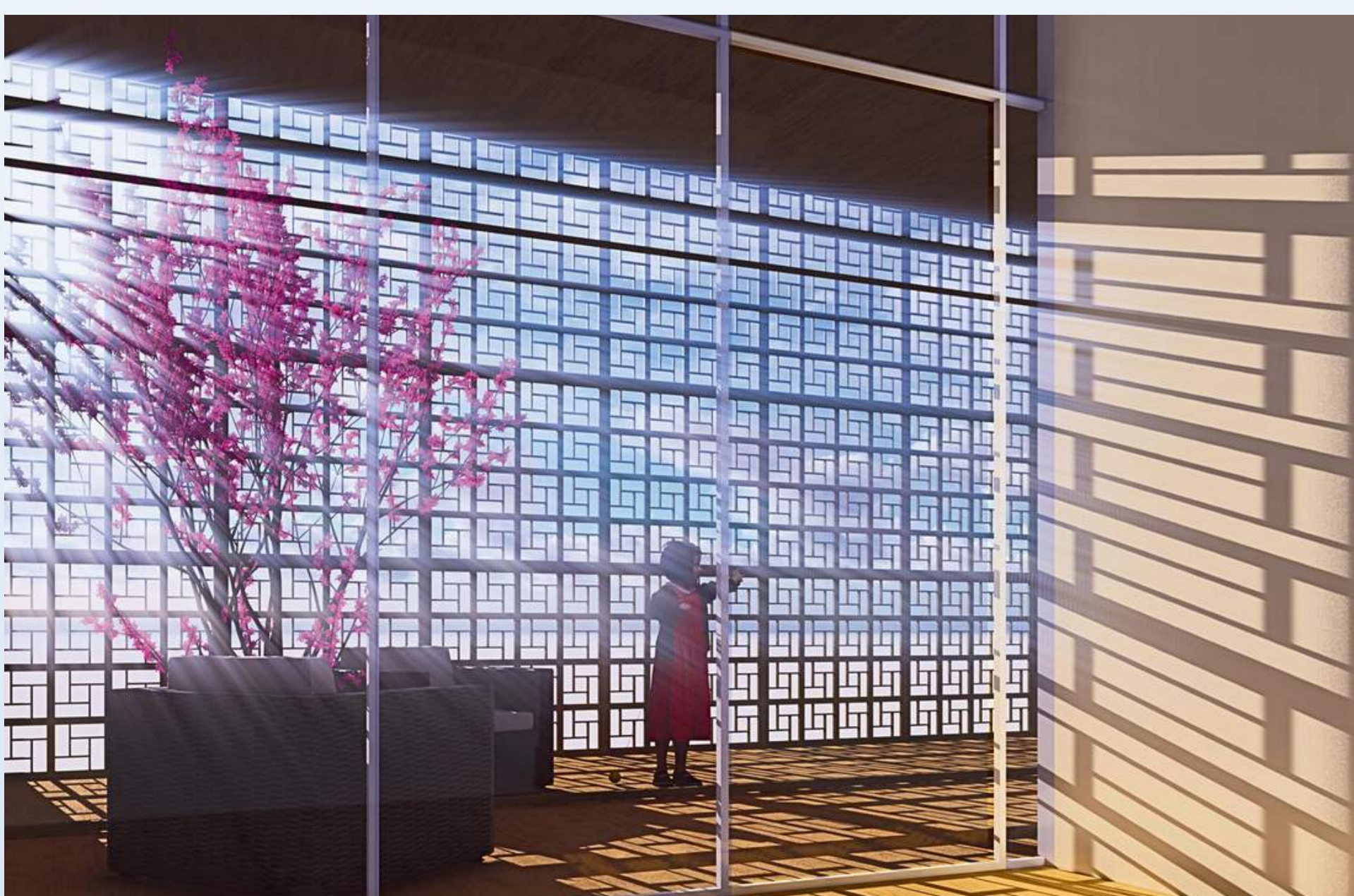




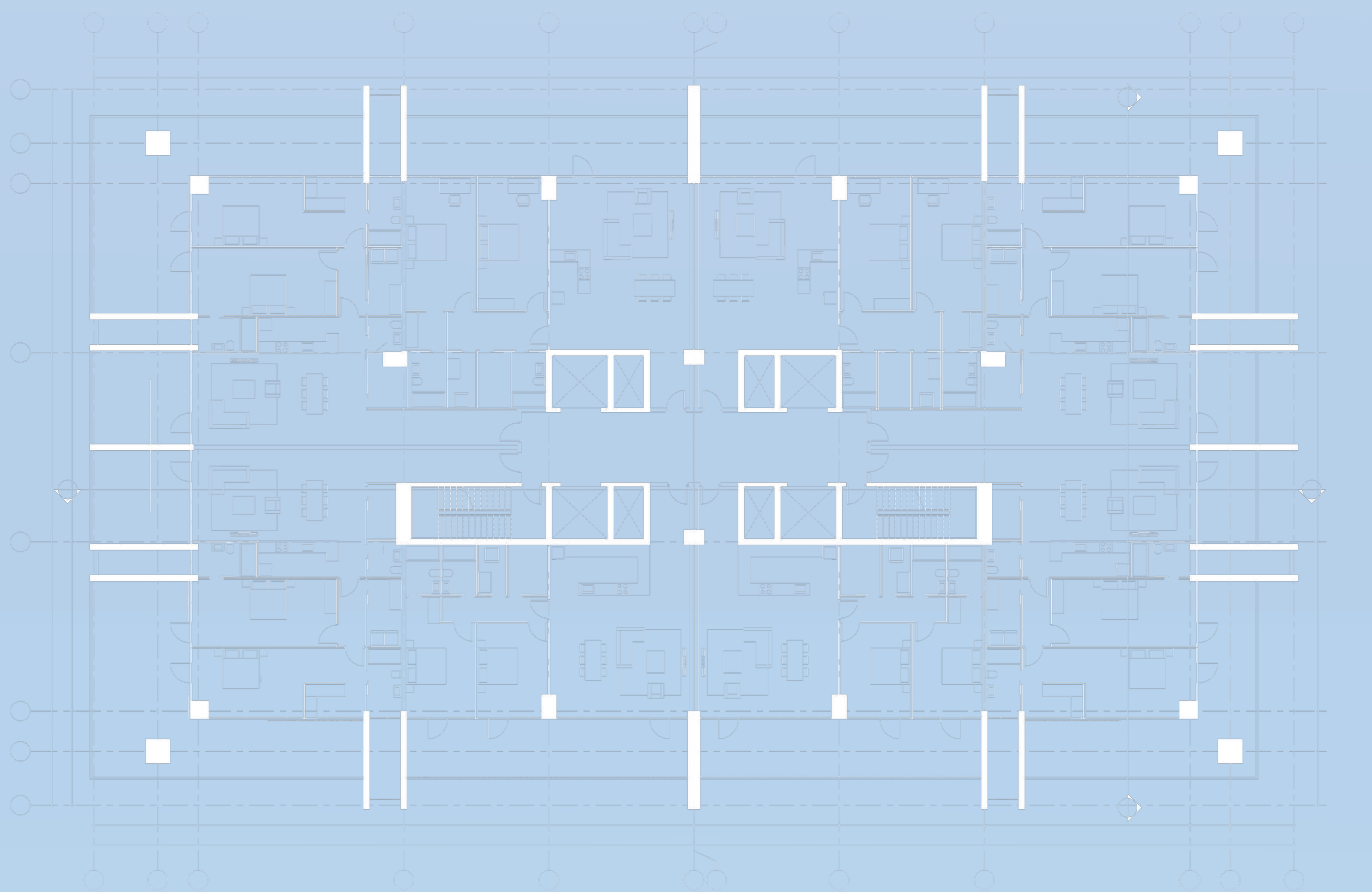
MASAR RESIDENCES

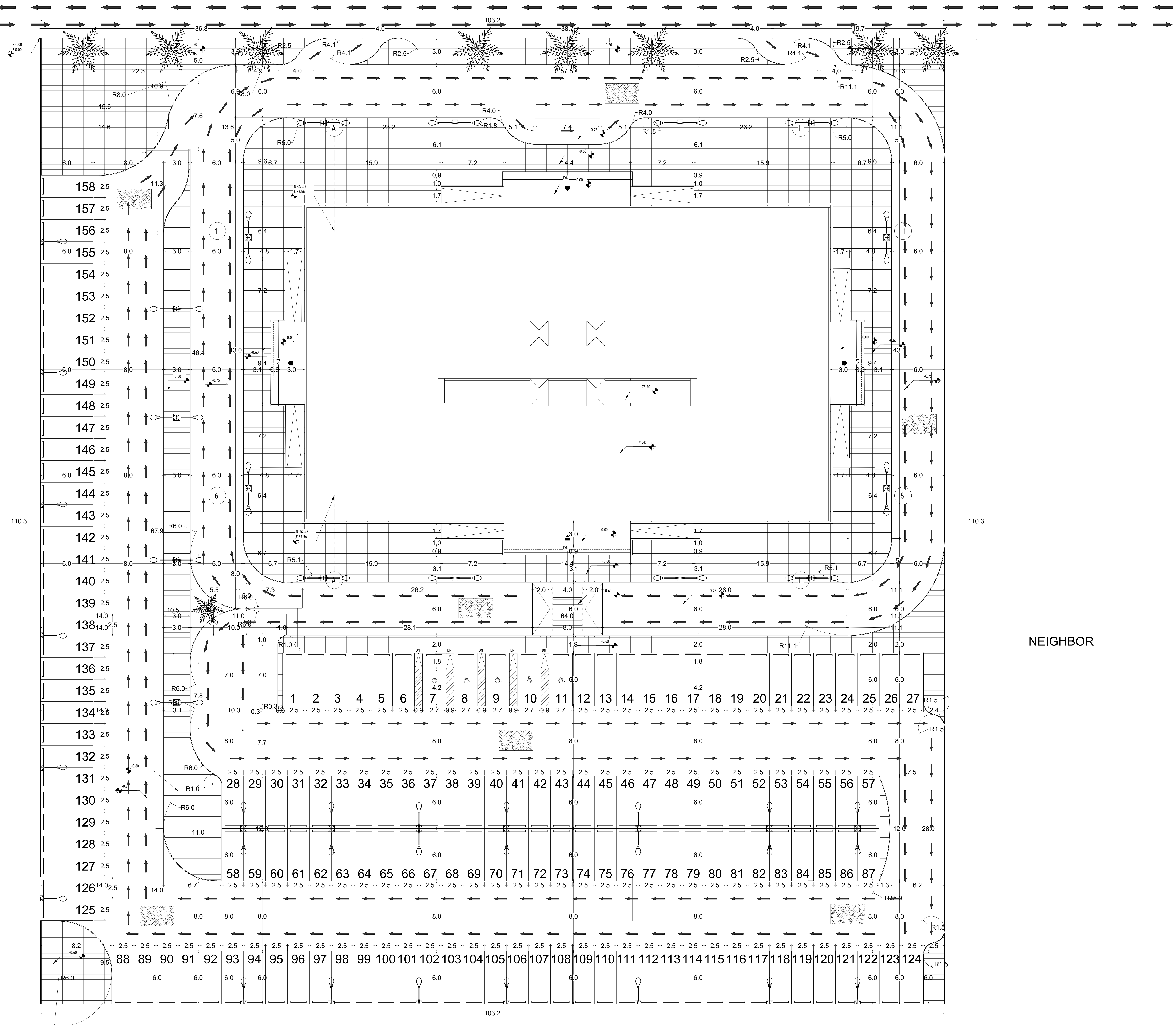
THIS 20-STORY RESIDENTIAL TOWER IS DESIGNED TO BLEND MODERN AESTHETICS WITH FUNCTIONAL LIVING SPACES. THE EXTERIOR FEATURES A DYNAMIC INTERPLAY OF GLASS, CONCRETE, AND PERFORATED METAL PANELS, CREATING A STRIKING FAÇADE THAT BALANCES TRANSPARENCY AND PRIVACY. THE BOLD STRUCTURAL ELEMENTS EMPHASIZE VERTICALITY WHILE ENSURING A VISUALLY ENGAGING COMPOSITION.

- **FAÇADE DESIGN:** THE TOWER INTEGRATES CONTEMPORARY GEOMETRIC PATTERNS WITH SHADING ELEMENTS TO ENHANCE ENERGY EFFICIENCY. THE USE OF PERFORATED SCREENS ALLOWS FOR CONTROLLED NATURAL LIGHT PENETRATION WHILE MAINTAINING PRIVACY.
- **BALCONIES & OPEN SPACES:** EACH UNIT FEATURES SPACIOUS BALCONIES, PROMOTING INDOOR-OUTDOOR LIVING AND OFFERING PANORAMIC VIEWS OF THE SURROUNDINGS.
- **NATURAL VENTILATION & LIGHT:** LARGE FLOOR-TO-CEILING WINDOWS OPTIMIZE DAYLIGHT EXPOSURE, REDUCING RELIANCE ON ARTIFICIAL LIGHTING AND IMPROVING OCCUPANT WELL-BEING.

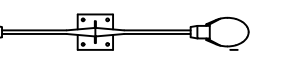
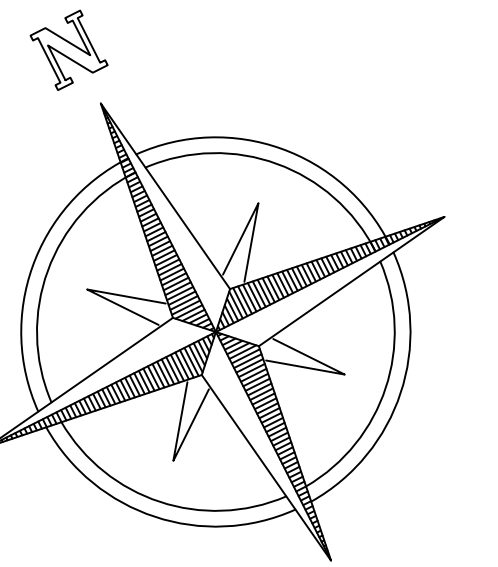


FORM FOLLOWS ELEVATION.





COLLEGE OF ARCH. & PLAN
DEPT. OF BUILDING ENGINEERING



Design Studio (8), Technical
Project, 2024/2025

INSTRUCTORS:

Prof. M. Essam Shaawat Co.
Dr. Faris Almaziad
Dr. M. Farooqi
Dr. M. Al Mulhim
Dr. AL Tayeb
Dr. M. Al hater
Eng. Meqdad
Eng. Rehan Gamil
Eng. Noman Ashraf

DONE BY:

ABDULELAH ALQABBANI

PROJECT:

Apartment Building

DRAWING NAME:

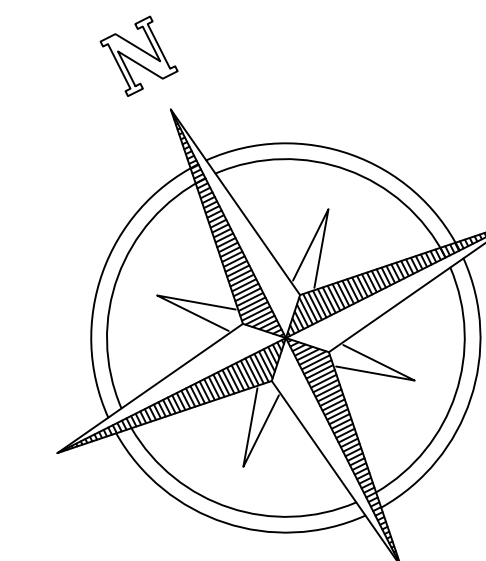
Site Plan

SCALE:

1:150

SHEET NUMBER:

A1



ALL UNITS IN METER

Design Studio (8), Technical
Project, 2024/2025

INSTRUCTORS:

Prof. M. Essam Shaawat Co.
Dr. Faris Almaziad
Dr. M. Farooqi
Dr. M. Al Mulhim
Dr. AL Tayeb
Dr. M. Al hater
Eng. Meqdad
Eng. Rehan Gamil
Eng. Noman Ashraf

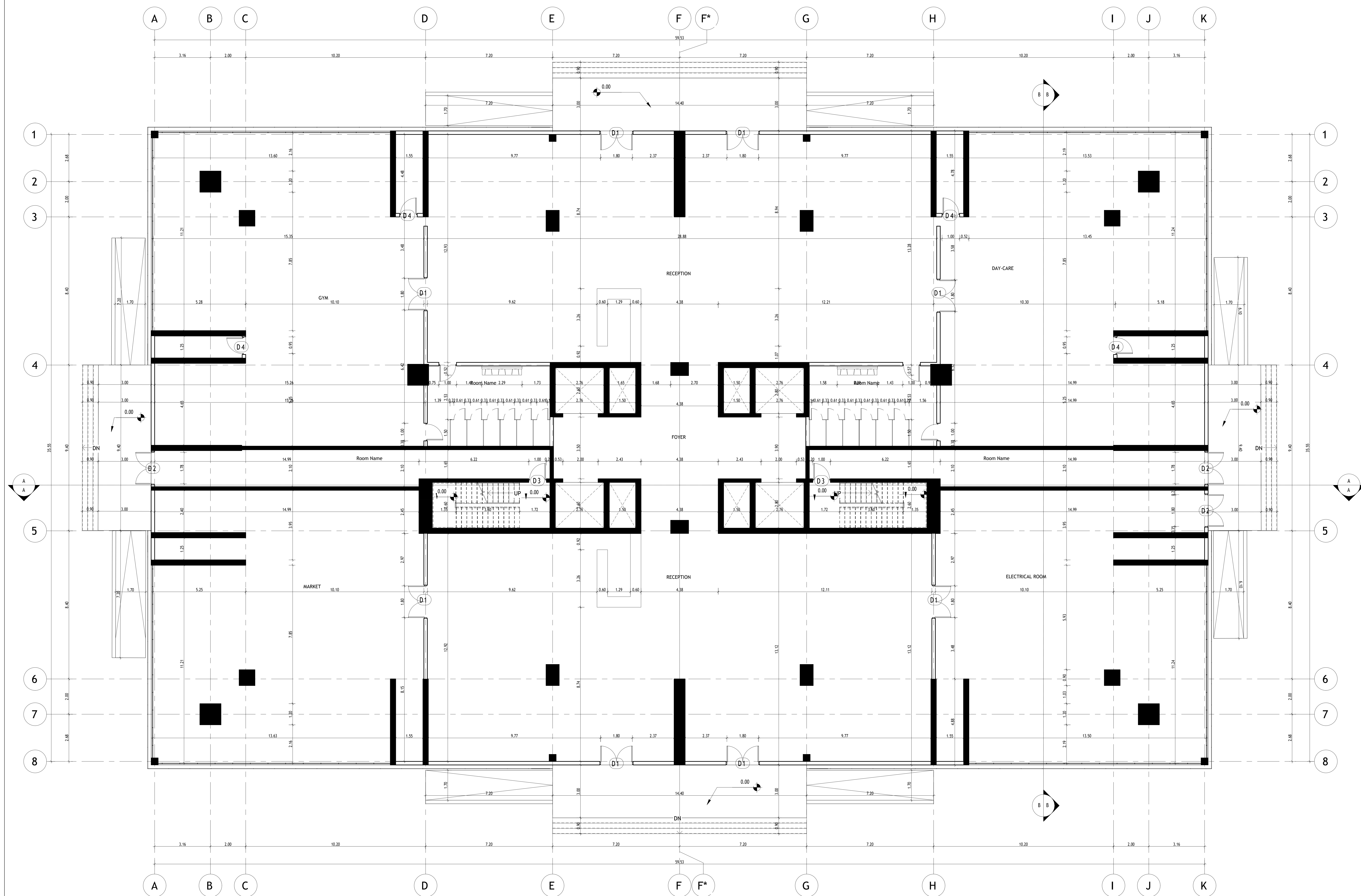
DONE BY:
ABDULELAH ALQABBANI

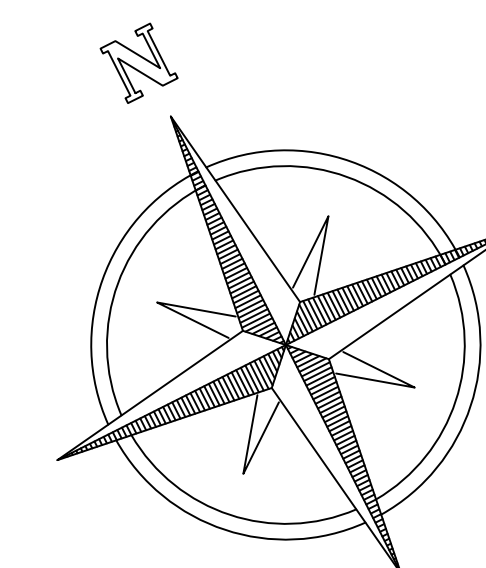
PROJECT:
Apartments Building

DRAWING NAME:
Ground Floor Plan

SCALE:
1:75

SHEET NUMBER:
A2





ALL UNITS IN METER

Design Studio (8), Technical
Project, 2024/2025

INSTRUCTORS:

Prof. M. Essam Shaawat Co.
Dr. Faris Almaziad
Dr. M. Farooqi
Dr. M. Al Mulhim
Dr. AL Tayeb
Dr. M. Al hater
Eng. Meqdad
Eng. Rehan Gamil
Eng. Noman Ashraf

DONE BY:

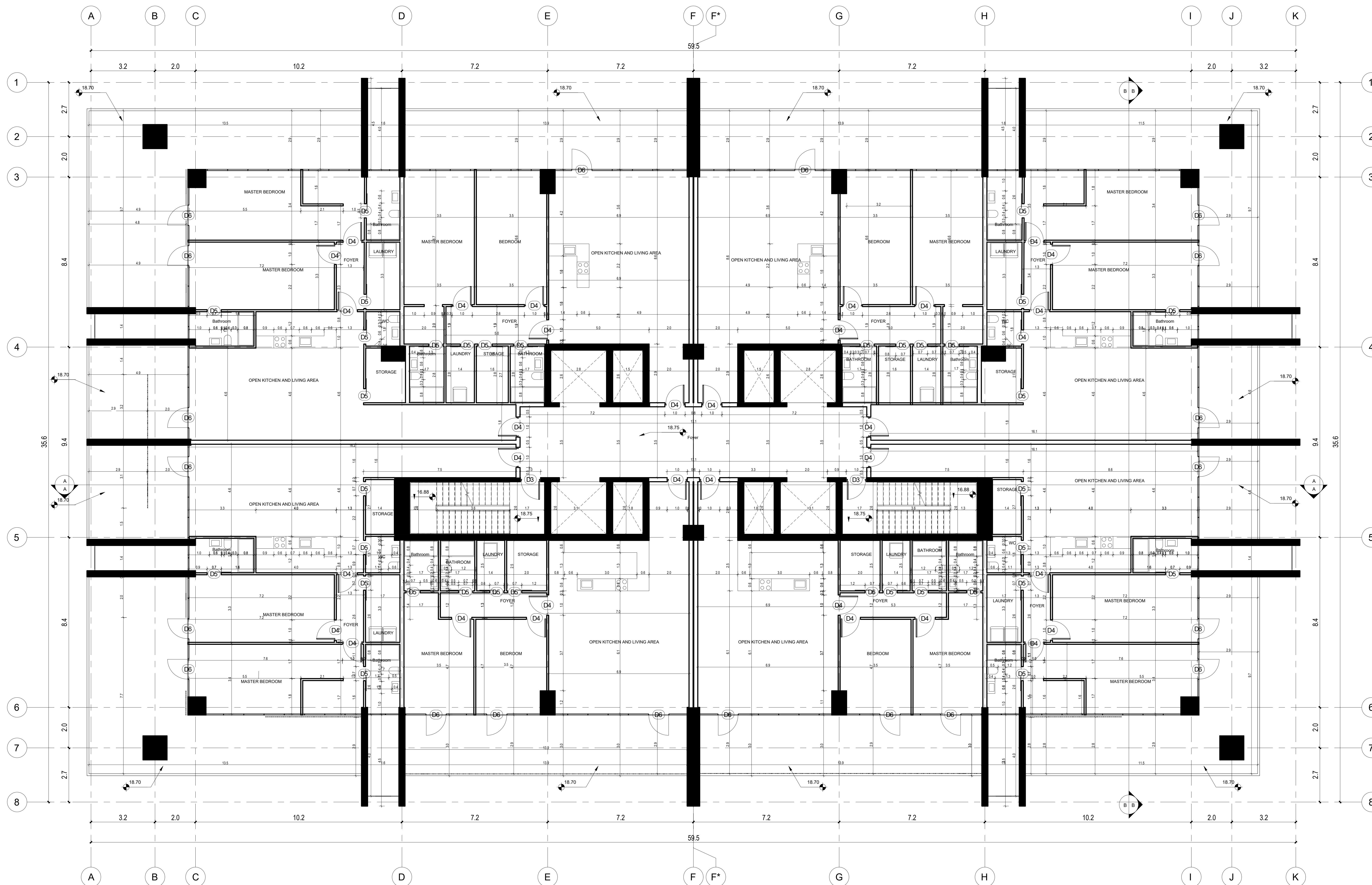
ABDULELAH ALQABBANI

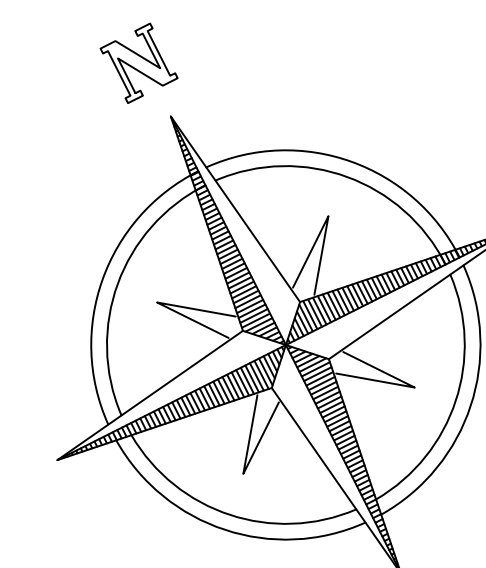
PROJECT:
Apartments Building

DRAWING NAME:
Typical Floor Plan (1st floor-19th floor)

SCALE:
1:75

SHEET NUMBER:
A3





ALL UNITS IN METER

Design Studio (8), Technical
Project, 2024/2025

INSTRUCTORS:

Prof. M. Essam Shaawat Co.
Dr. Faris Almaziad
Dr. M. Farooqi
Dr. M. Al Mulhim
Dr. AL Tayeb
Dr. M. Al hater
Eng. Meqdad
Eng. Rehan Gamil
Eng. Noman Ashraf

DONE BY:

ABDULELAH ALQABBANI

PROJECT:

Apartments Building

DRAWING NAME:

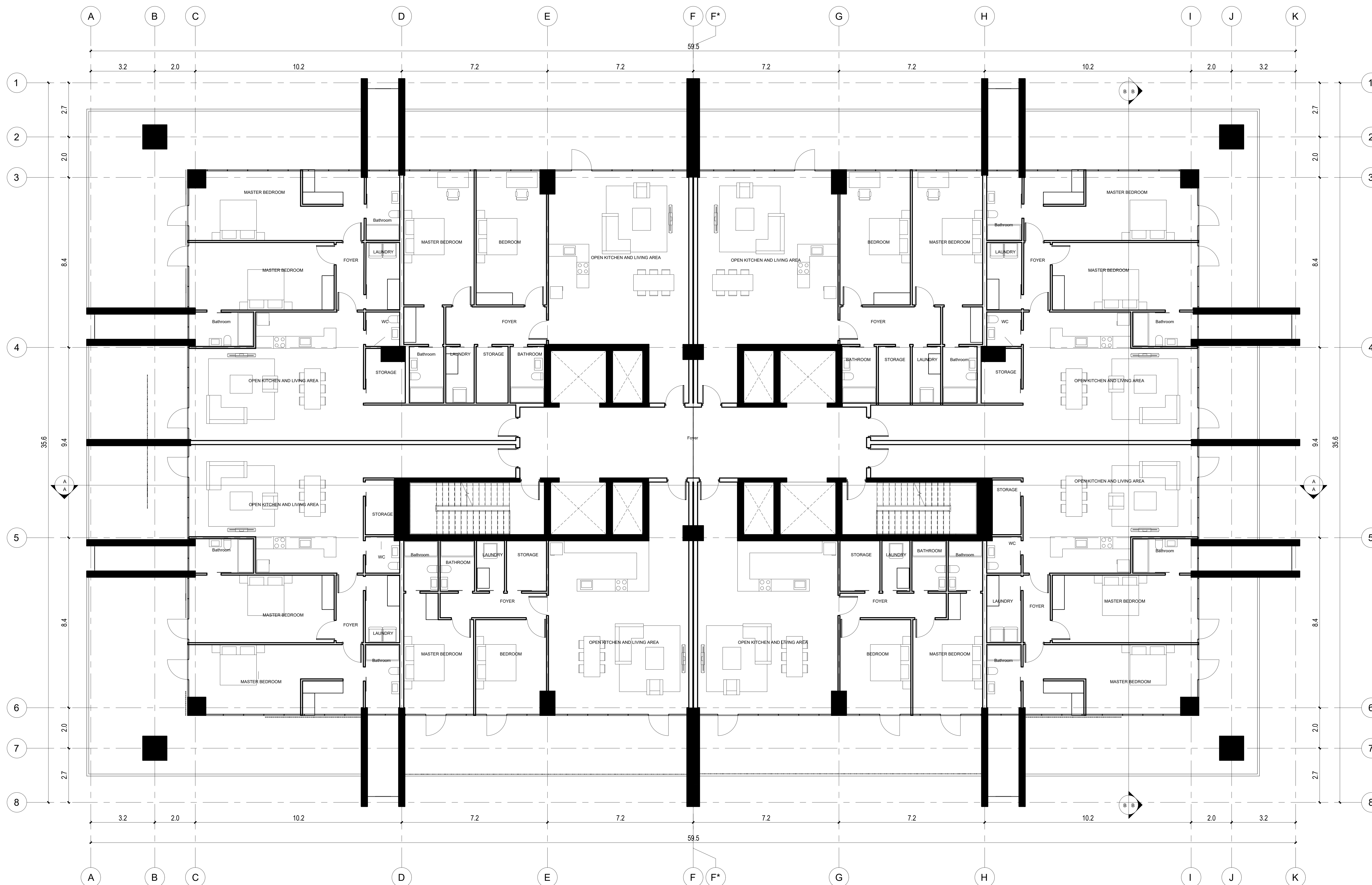
Typical Floor Plan (1st floor-19th floor)
(furniture)

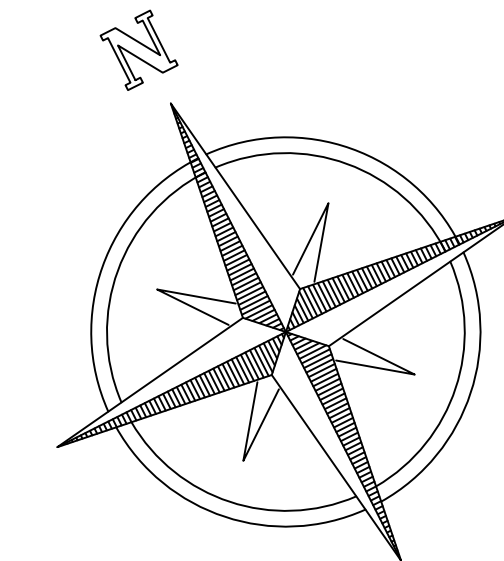
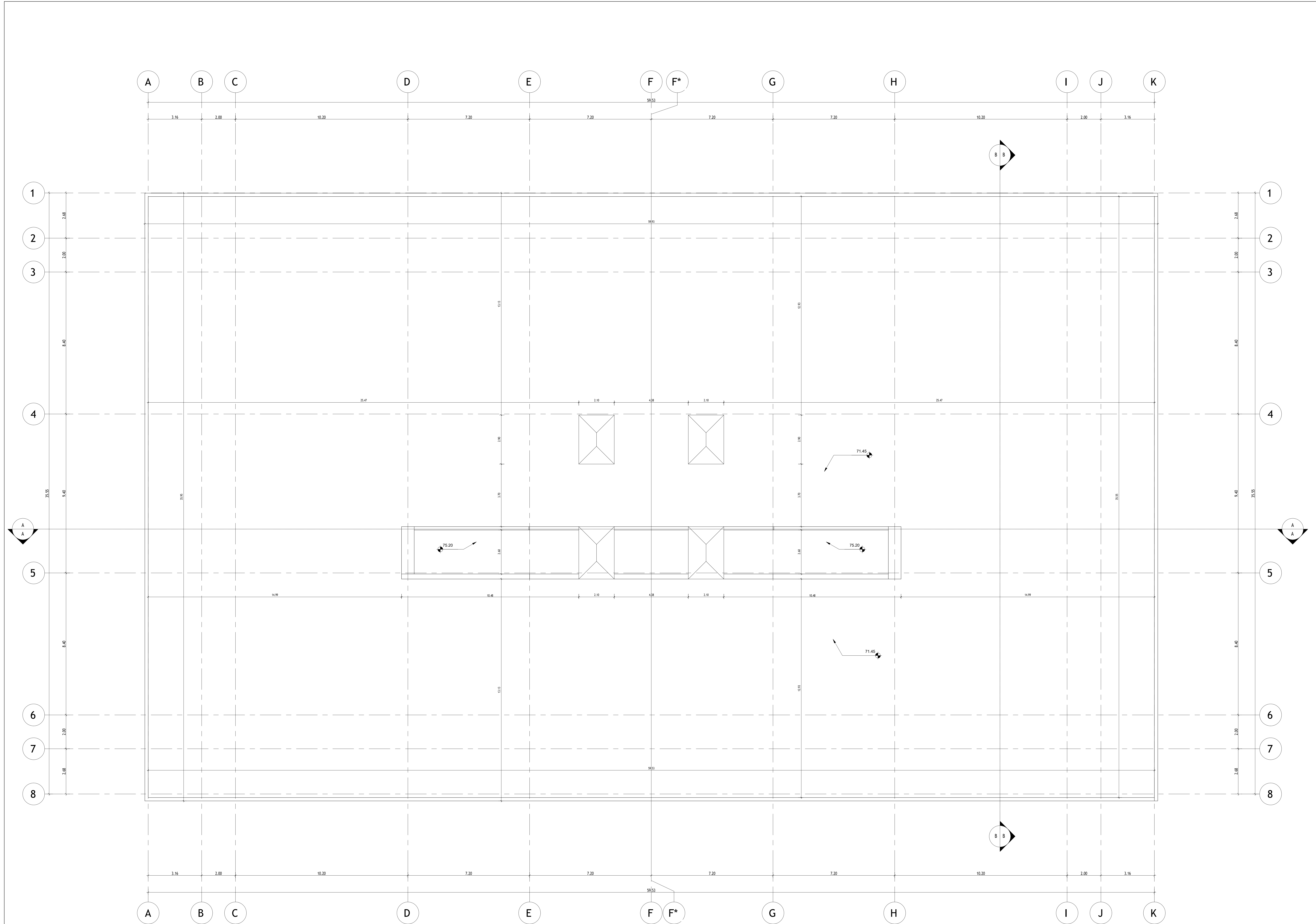
SCALE:

1:75

SHEET NUMBER:

A4





ALL UNITS IN METER

Design Studio (8), Technical
Project, 2024/2025

INSTRUCTORS:

Prof. M. Essam Shaawat Co.
Dr. Faris Almaziad
Dr. M. Farooqi
Dr. M. Al Mulhim
Dr. AL Tayeb
Dr. M. Al hater
Eng. Meqdad
Eng. Rehan Gamil
Eng. Noman Ashraf

DONE BY:
ABDULELAH ALQABBANI

PROJECT:
Apartments Building

DRAWING NAME:
Roof Plan

SCALE:
1:75

SHEET NUMBER:
A5



ALL UNITS IN METER

Design Studio (8), Technical
Project, 2024/2025

INSTRUCTORS:

Prof. M. Essam Shaawat Co.
Dr. Faris Almaziad
Dr. M. Farooqi
Dr. M. Al Mulhim
Dr. AL Tayeb
Dr. M. Al hater
Eng. Meqdad
Eng. Rehan Gamil
Eng. Noman Ashraf

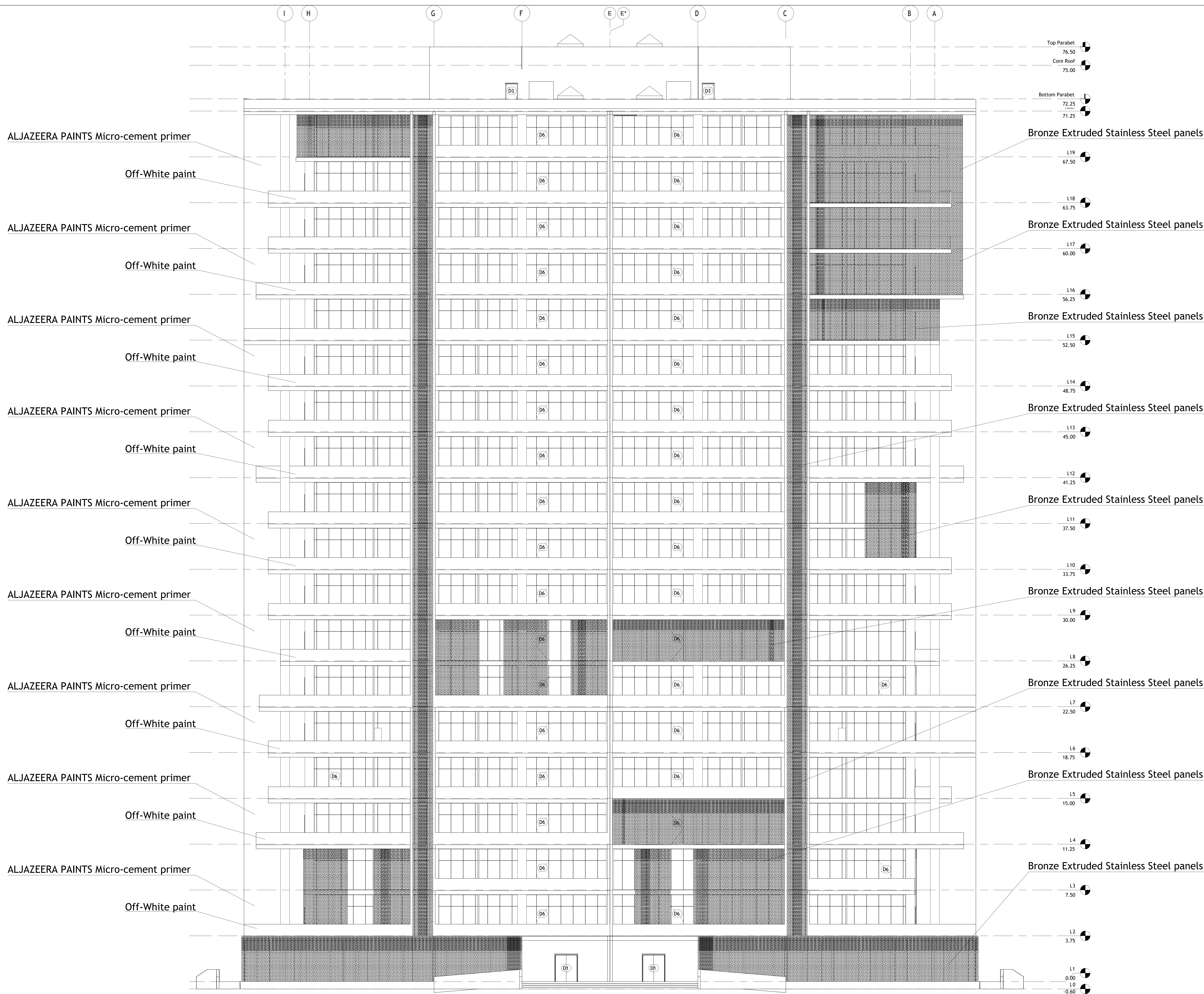
DONE BY:
ABDULELAH ALQABBANI

PROJECT:
Apartments Building

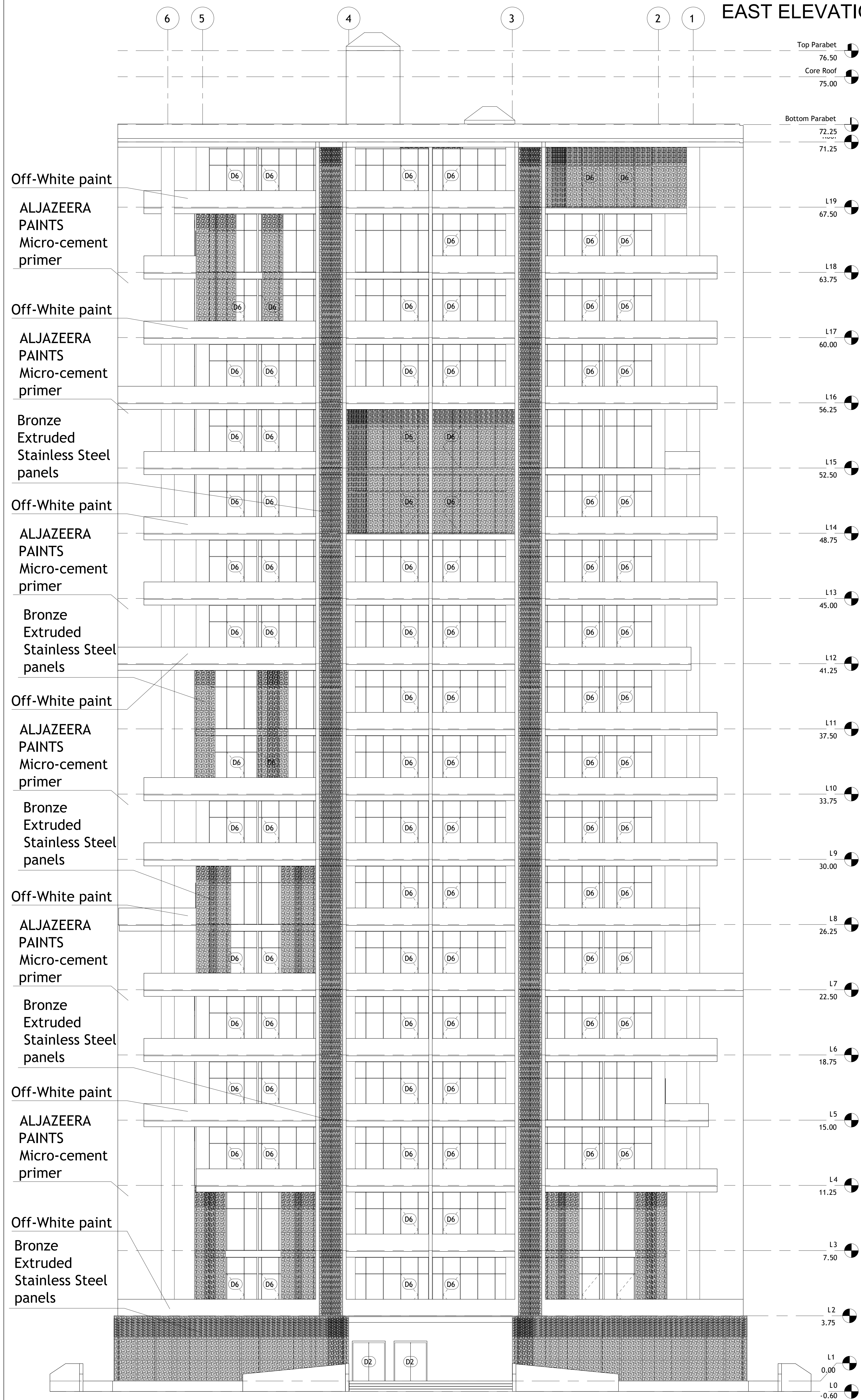
DRAWING NAME:
North Elevation

SCALE: 1:100

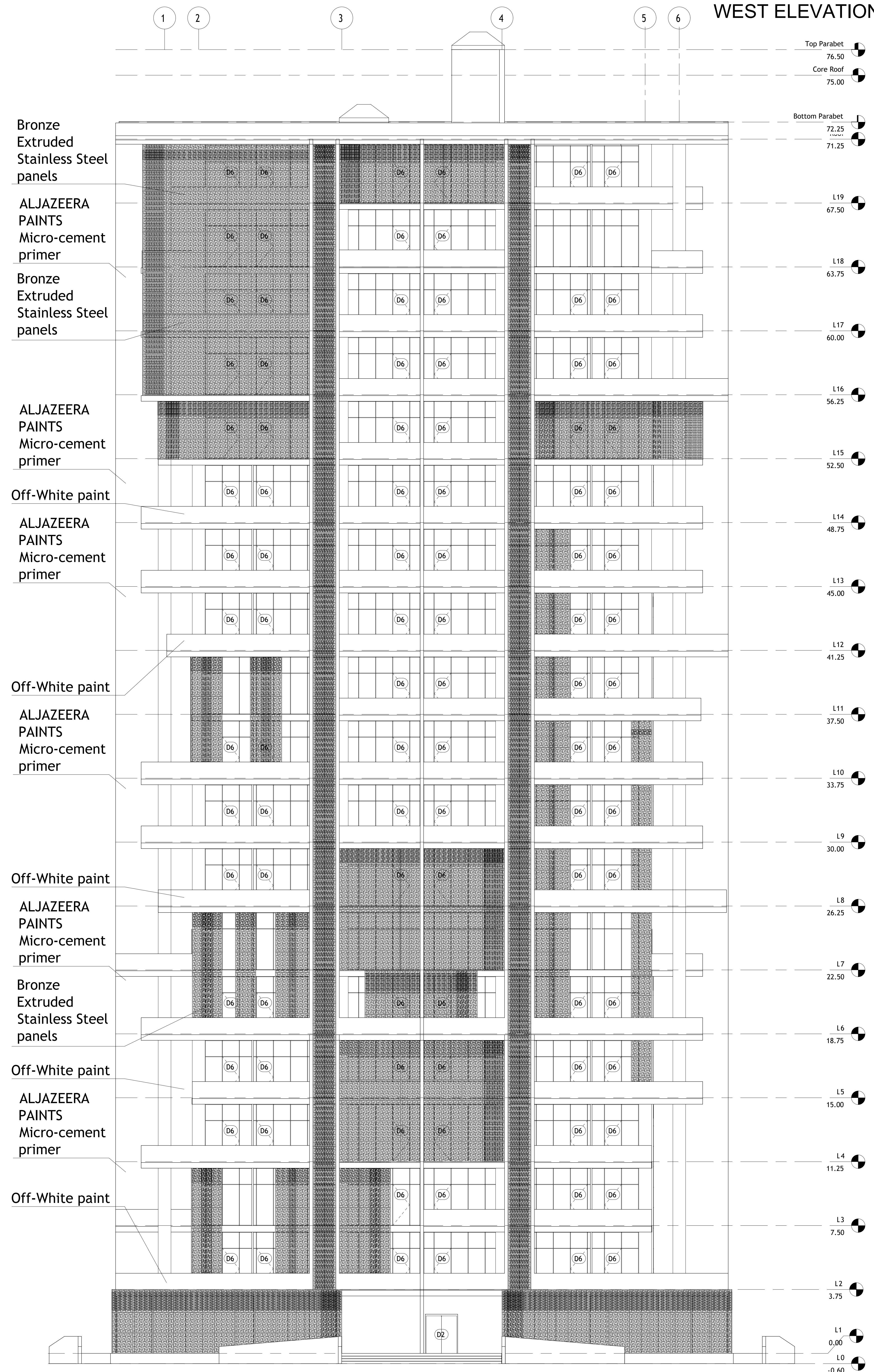
SHEET NUMBER:
A6



EAST ELEVATION



WEST ELEVATION



COLLEGE OF ARCH. & PLAN
DEPT. OF BUILDING ENGINEERING

ALL UNITS IN METER

Design Studio (8), Technical
Project, 2024/2025

INSTRUCTORS:

Prof. M. Essam Shaawat Co.
Dr. Faris Almaziad
Dr. M. Farooqi
Dr. M. Al Mulhim
Dr. AL Tayeb
Dr. M. Al hater
Eng. Meqdad
Eng. Rehan Gamil
Eng. Noman Ashraf

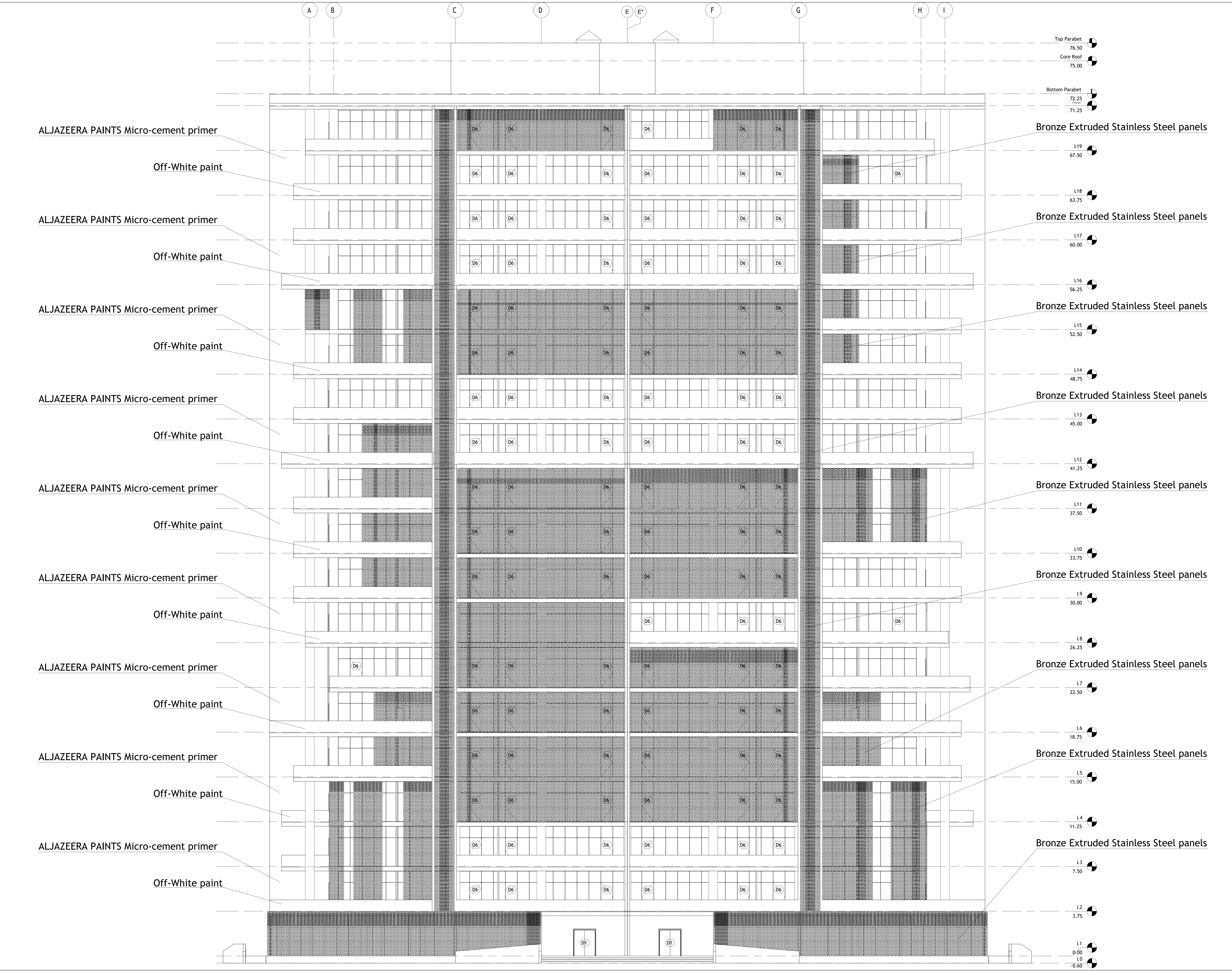
DONE BY:
ABDULELAH ALQABBANI

PROJECT:
Apartments Building

DRAWING NAME:
East & West Elevations

SCALE: 1:100

SHEET NUMBER:
A7



ALL UNITS IN METER

Design Studio (8), Technical
Project, 2024/2025

INSTRUCTORS:

Prof. M. Essam Shaawat Co.
Dr. Faris Almaziad
Dr. M. Farooqi
Dr. M. Al Mulhim
Dr. AL Tayeb
Dr. M. Al hater
Eng. Meqdad
Eng. Rehan Gamil
Eng. Noman Ashraf

DONE BY:
ABDULELAH ALQABBANI

PROJECT:
Apartments Building

DRAWING NAME:
South Elevation

SCALE:
1:100

SHEET NUMBER:
A8

ALL UNITS IN METER

Design Studio (8), Technical
Project, 2024/2025

INSTRUCTORS:

Prof. M. Essam Shaawat Co.
Dr. Faris Almaziad
Dr. M. Farooqi
Dr. M. Al Mulhim
Dr. AL Tayeb
Dr. M. Al hater
Eng. Meqdad
Eng. Rehan Gamil
Eng. Noman Ashraf

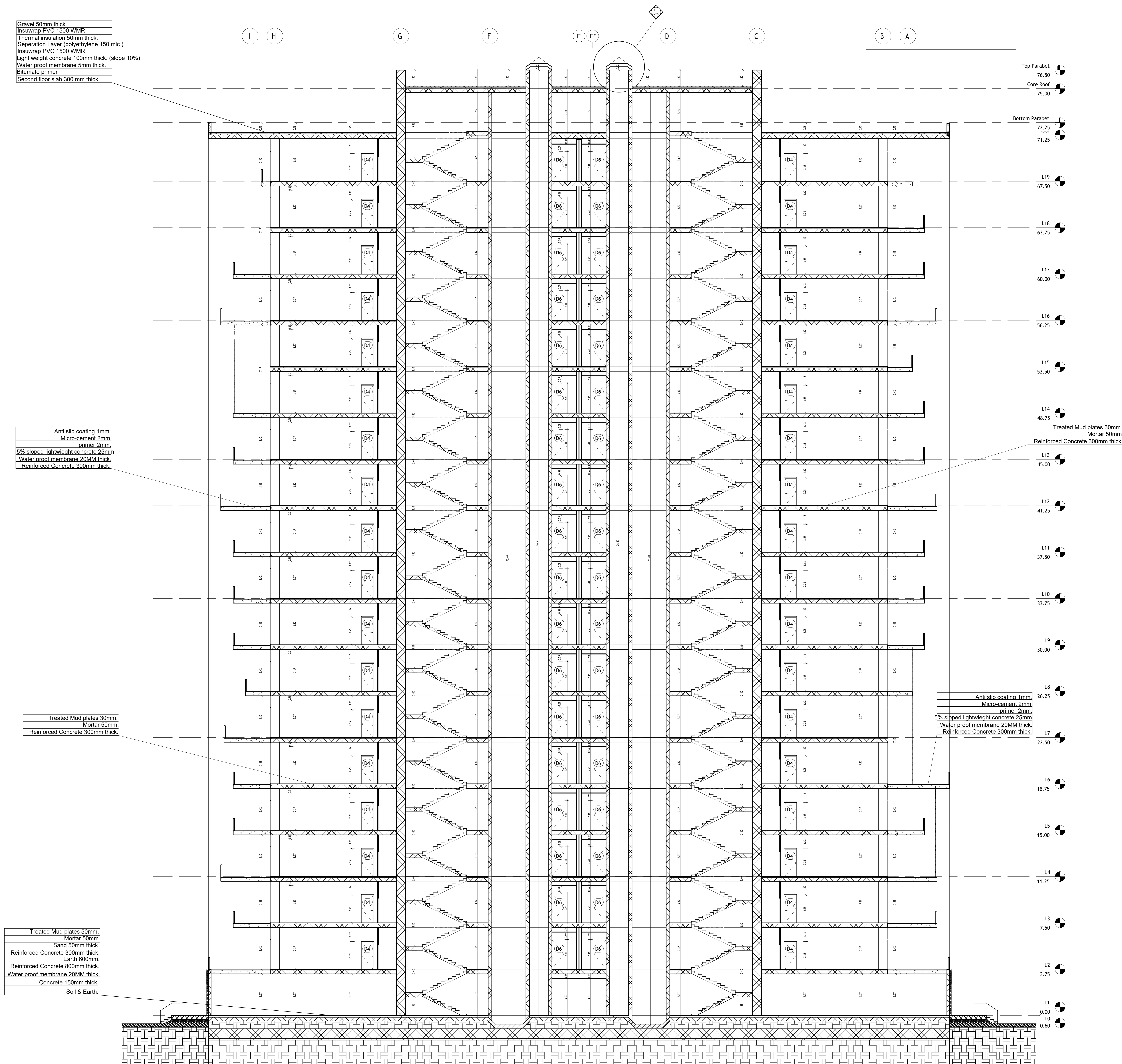
DONE BY:
ABDULELAH ALQABBANI

PROJECT:
Apartments Building

DRAWING NAME:
Section A-A

SCALE:
1:90

SHEET NUMBER:
A9



ALL UNITS IN METER

Design Studio (8), Technical
Project, 2024/2025

INSTRUCTORS:

Prof. M. Essam Shaawat Co.
Dr. Faris Almaziad
Dr. M. Farooqi
Dr. M. Al Mulhim
Dr. AL Tayeb
Dr. M. Al hater
Eng. Meqdad
Eng. Rehan Gamil
Eng. Noman Ashraf

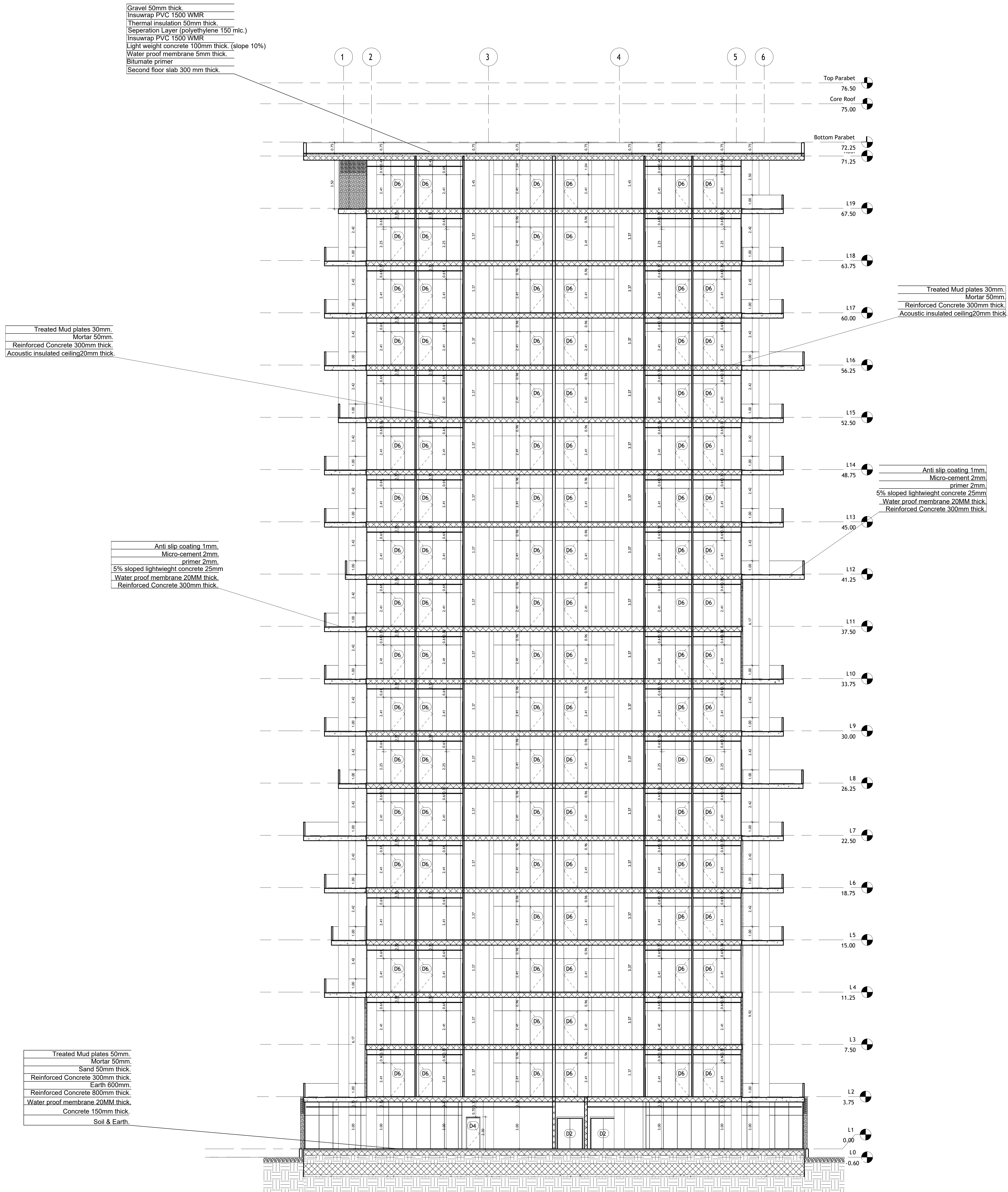
DONE BY:
ABDULELAH ALQABBANI

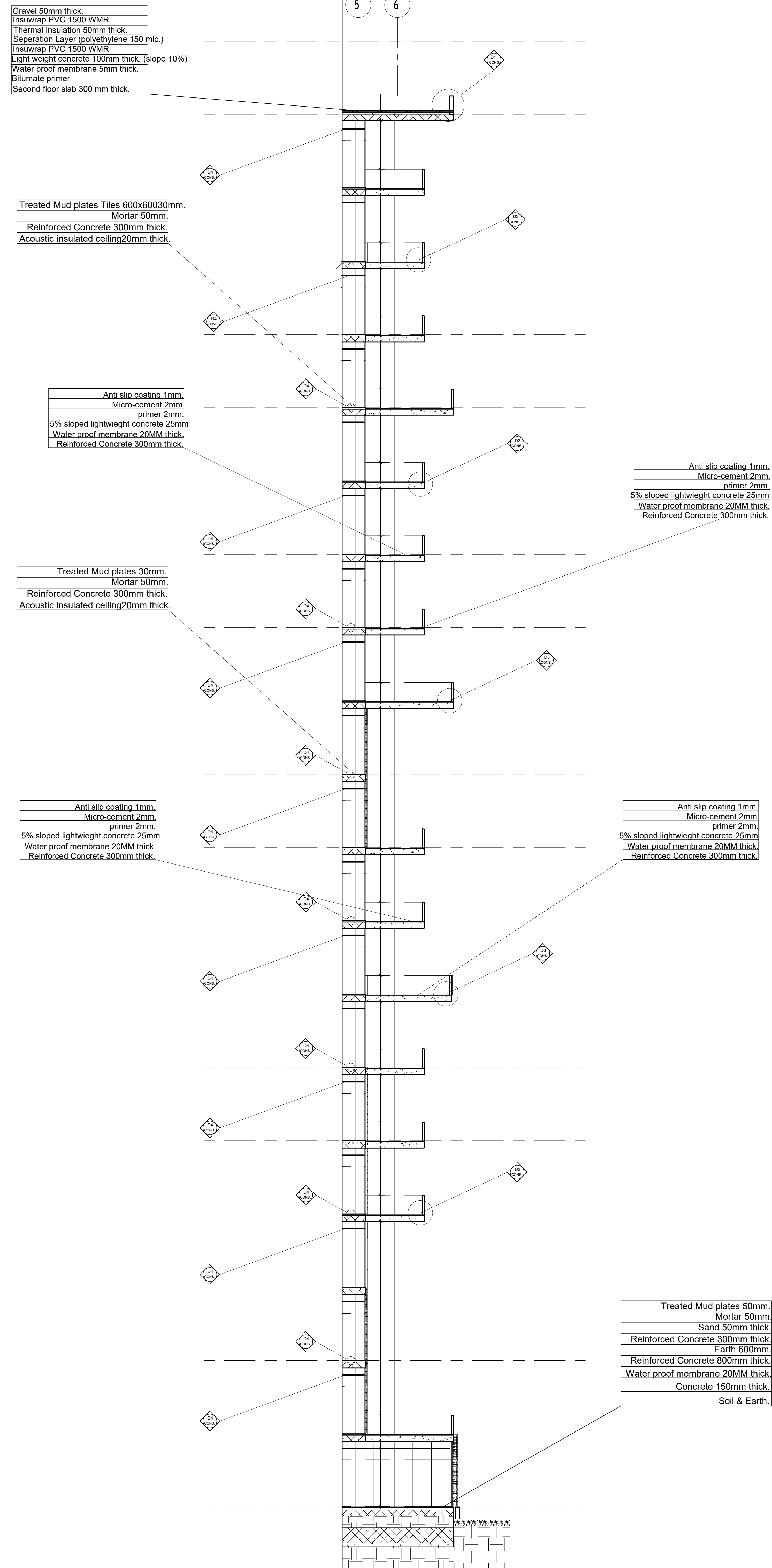
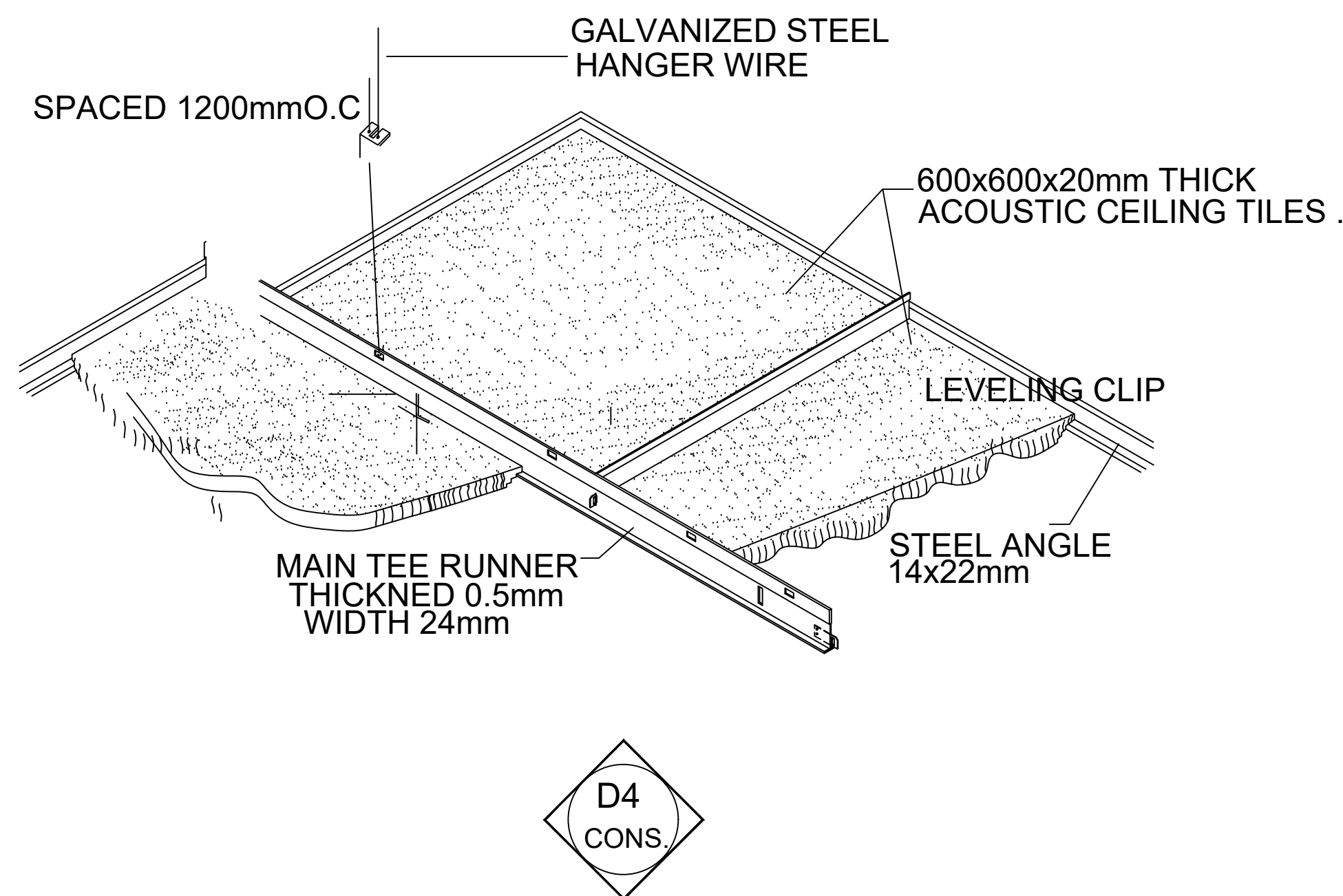
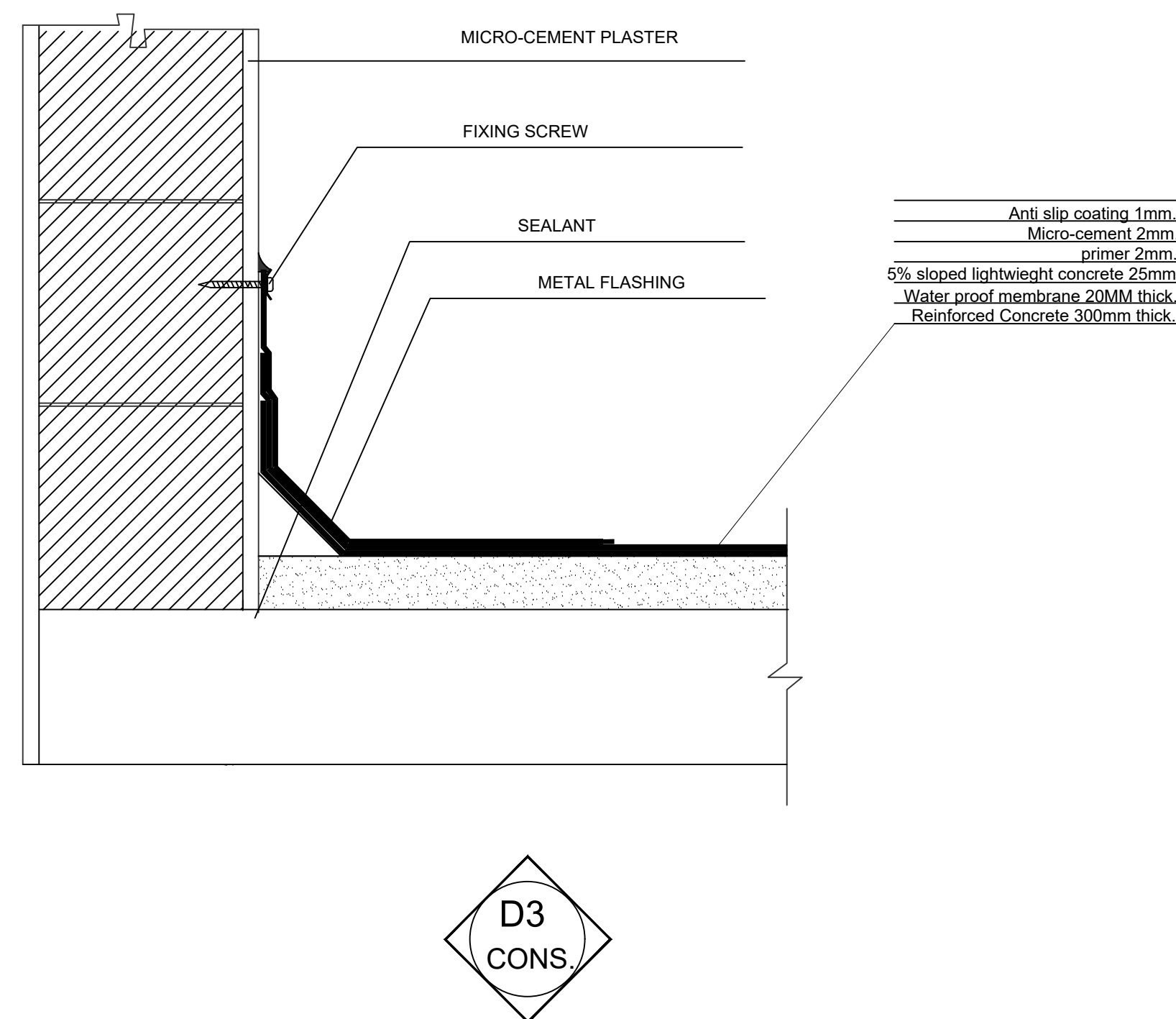
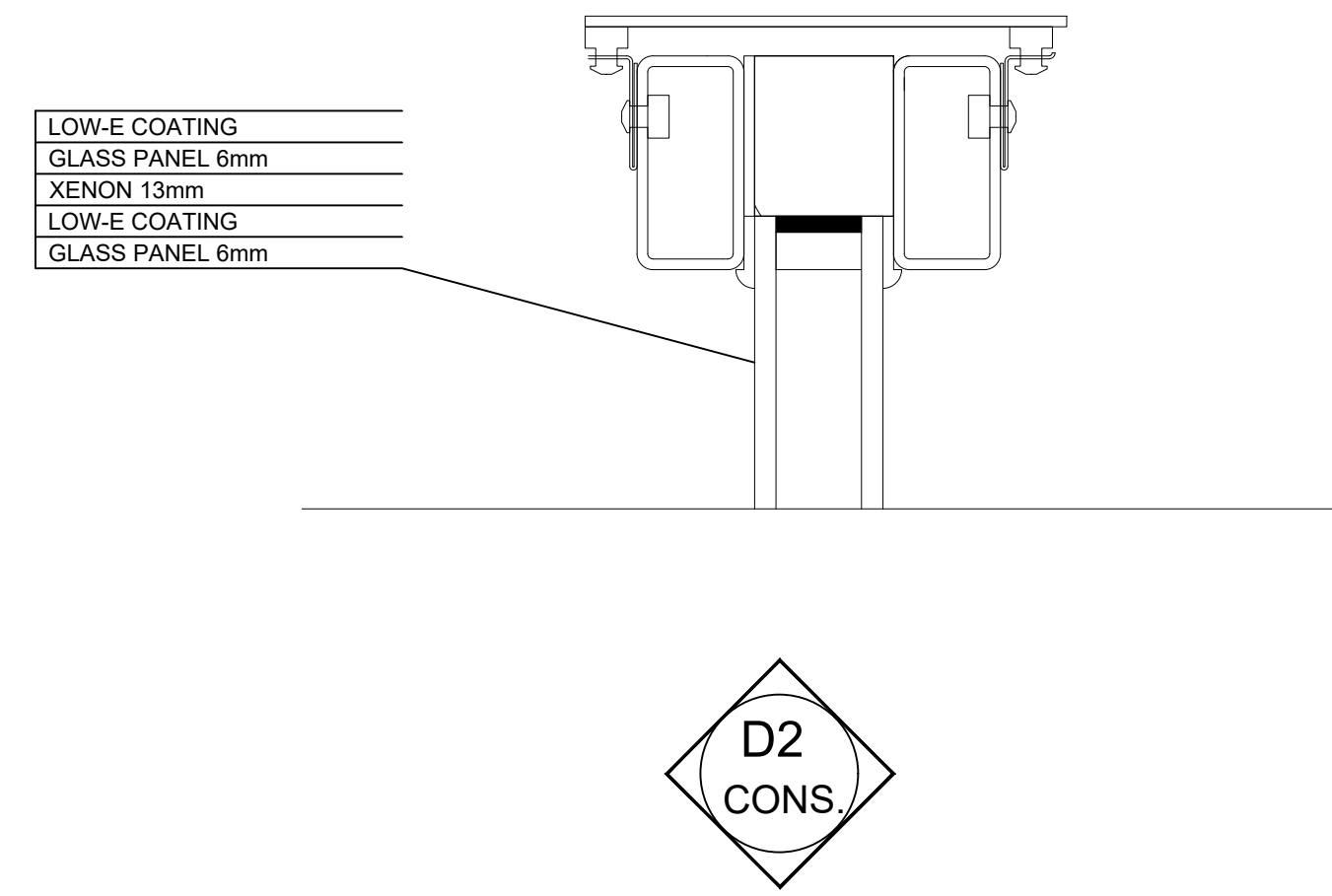
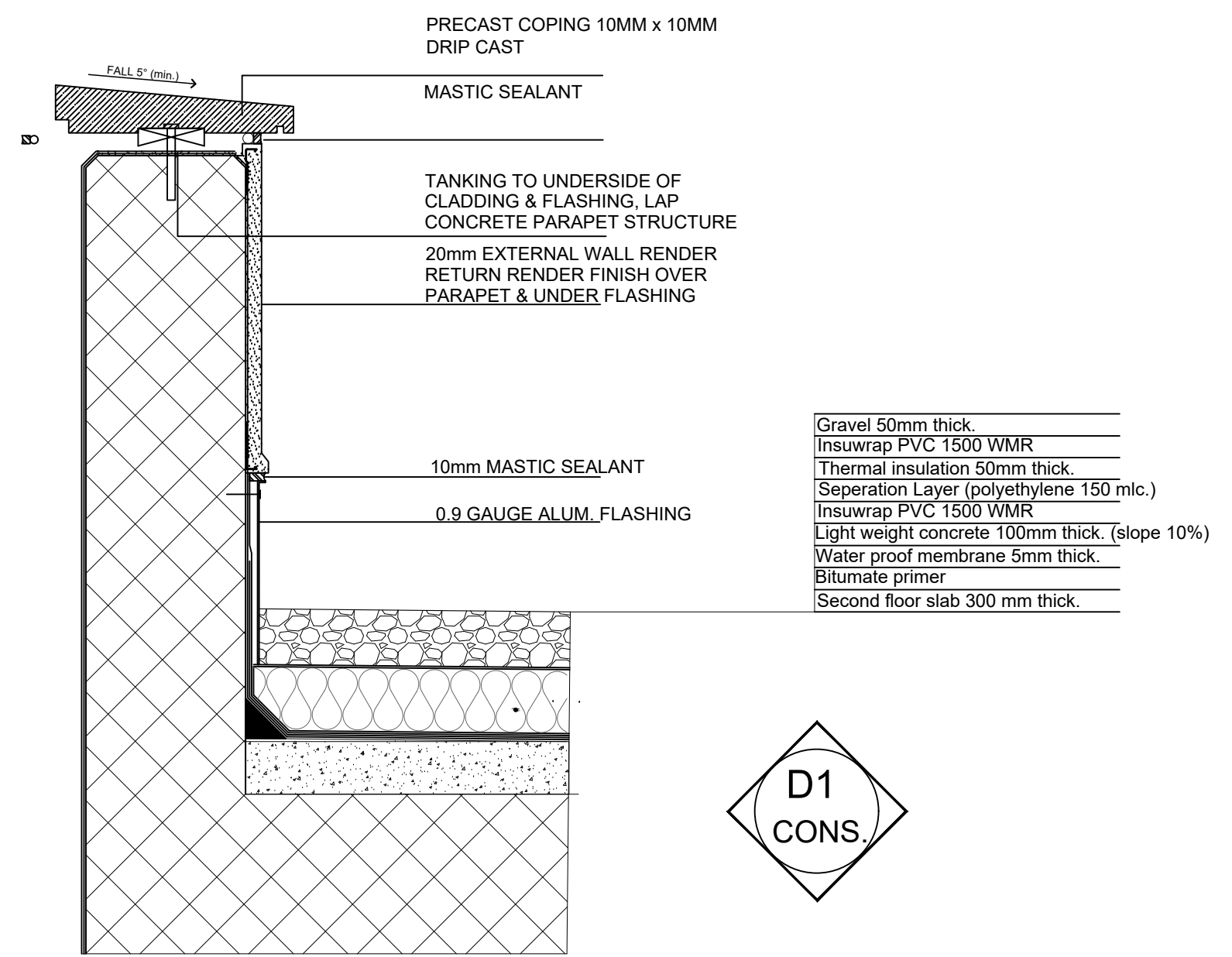
PROJECT:
Apartments Building

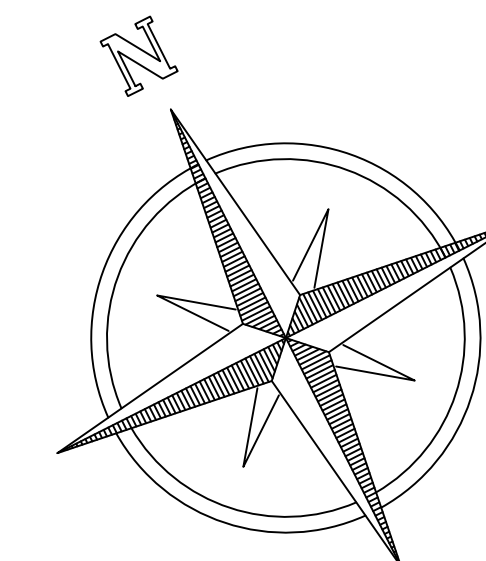
DRAWING NAME:
Section B-B

SCALE: 1:90

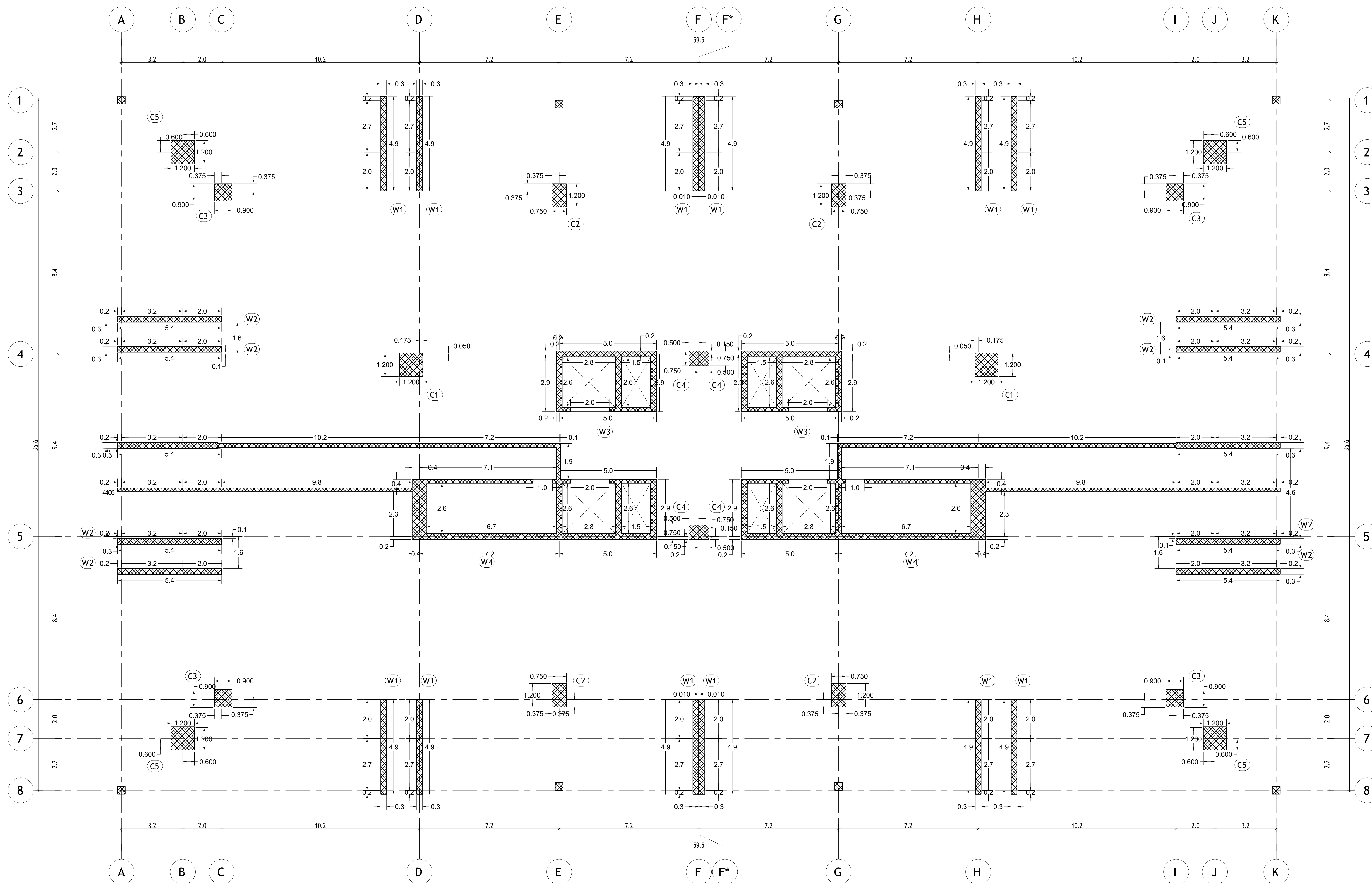
SHEET NUMBER:
A10







ALL UNITS IN METER



Design Studio (8), Technical
Project, 2024/2025

INSTRUCTORS:

Prof. M. Essam Shaawat Co.
Dr. Faris Almaziad
Dr. M. Farooqi
Dr. M. Al Mulhim
Dr. AL Tayeb
Dr. M. Al hater
Eng. Meqdad
Eng. Rehan Gamil
Eng. Noman Ashraf

DONE BY:

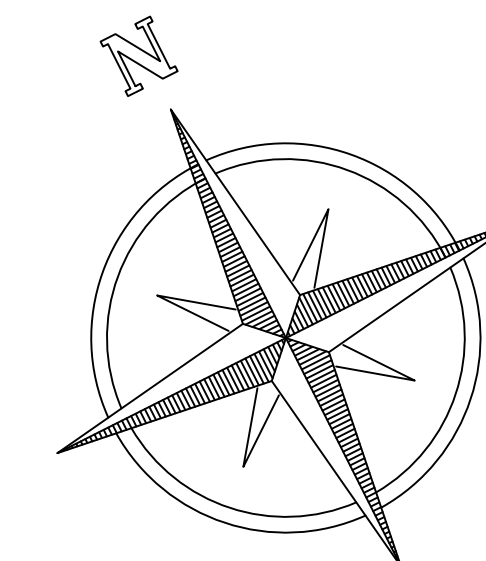
ABDULELAH ALQABBANI

PROJECT:
Apartments Building

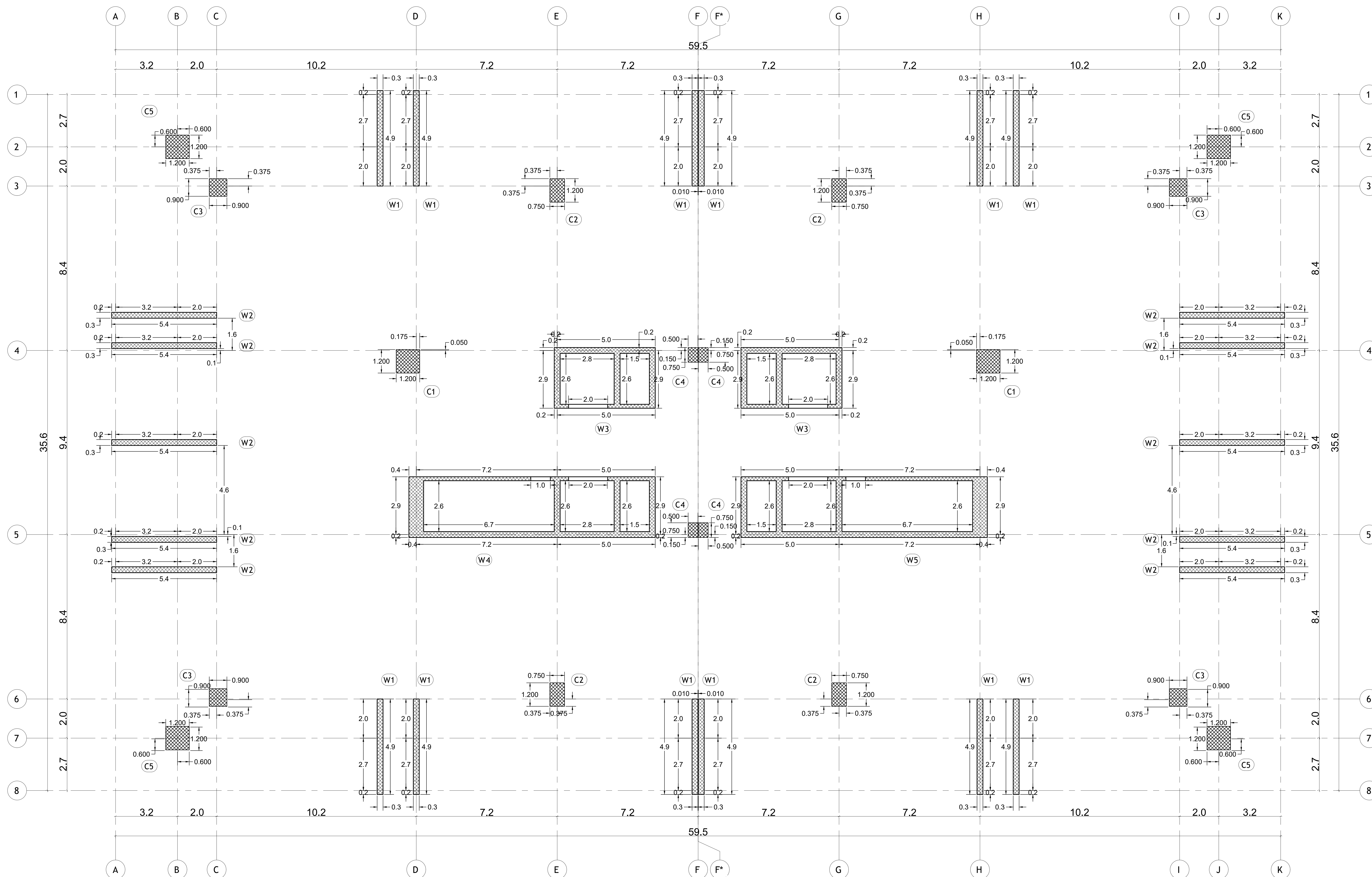
DRAWING NAME:
GF Coloumn & Axis

SCALE:
1:75

SHEET NUMBER:
A12



ALL UNITS IN METER



Design Studio (8), Technical
Project, 2024/2025

INSTRUCTORS:

Prof. M. Essam Shaawat Co.
Dr. Faris Almaziad
Dr. M. Farooqi
Dr. M. Al Mulhim
Dr. AL Tayeb
Dr. M. Al hater
Eng. Meqdad
Eng. Rehan Gamil
Eng. Noman Ashraf

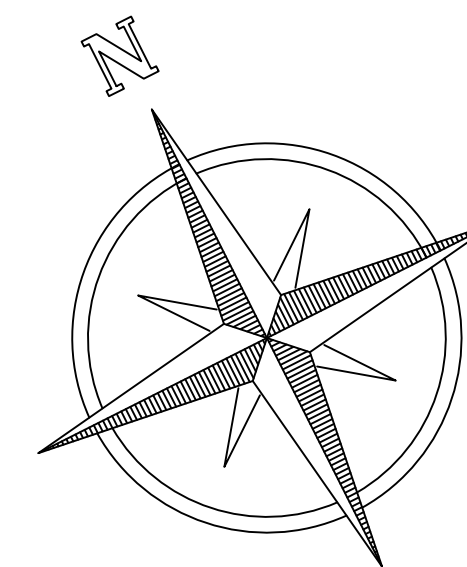
DONE BY:
ABDULELAH ALQABBANI

PROJECT:
Apartments Building

DRAWING NAME:
Typical Coloumn & Axis

SCALE:
1:75

SHEET NUMBER:
A13



ALL UNITS IN METER

Design Studio (8), Technical
Project, 2024/2025

INSTRUCTORS:

Prof. M. Essam Shaawat Co.
Dr. Faris Almaziad
Dr. M. Farooqi
Dr. M. Al Mulhim
Dr. AL Tayeb
Dr. M. Al hater
Eng. Meqdad
Eng. Rehan Gamil
Eng. Noman Ashraf

DONE BY:

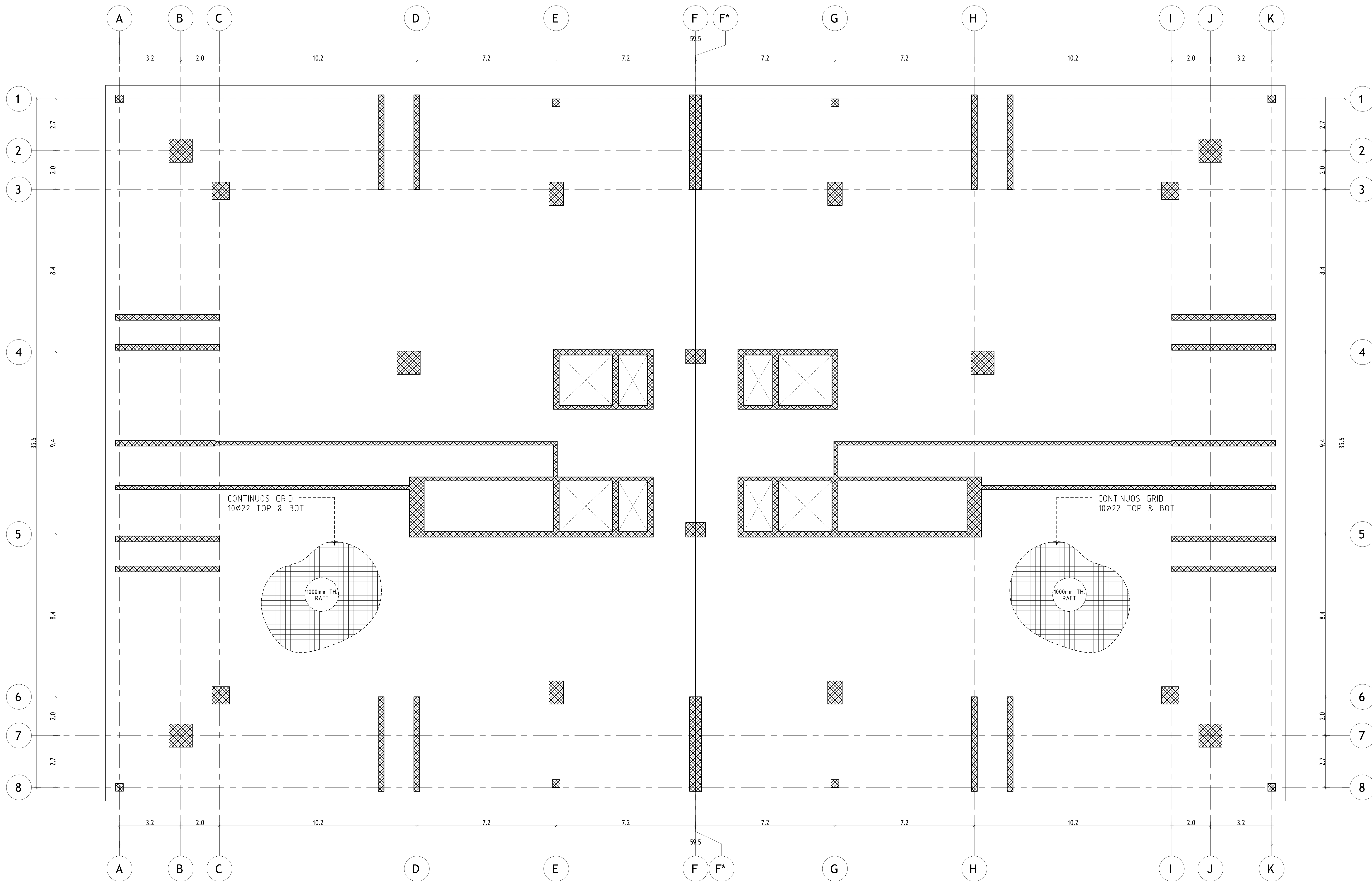
ABDULELAH ALQABBANI

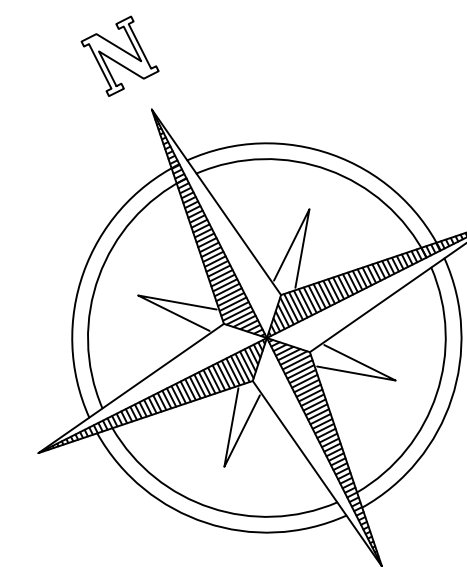
PROJECT:
Apartments Building

DRAWING NAME:
Foundation Framing

SCALE:
1:75

SHEET NUMBER:
A14





ALL UNITS IN METER

Design Studio (8), Technical
Project, 2024/2025

INSTRUCTORS:

Prof. M. Essam Shaawat Co.
Dr. Faris Almaziad
Dr. M. Farooqi
Dr. M. Al Mulhim
Dr. AL Tayeb
Dr. M. Al hater
Eng. Meqdad
Eng. Rehan Gamil
Eng. Noman Ashraf

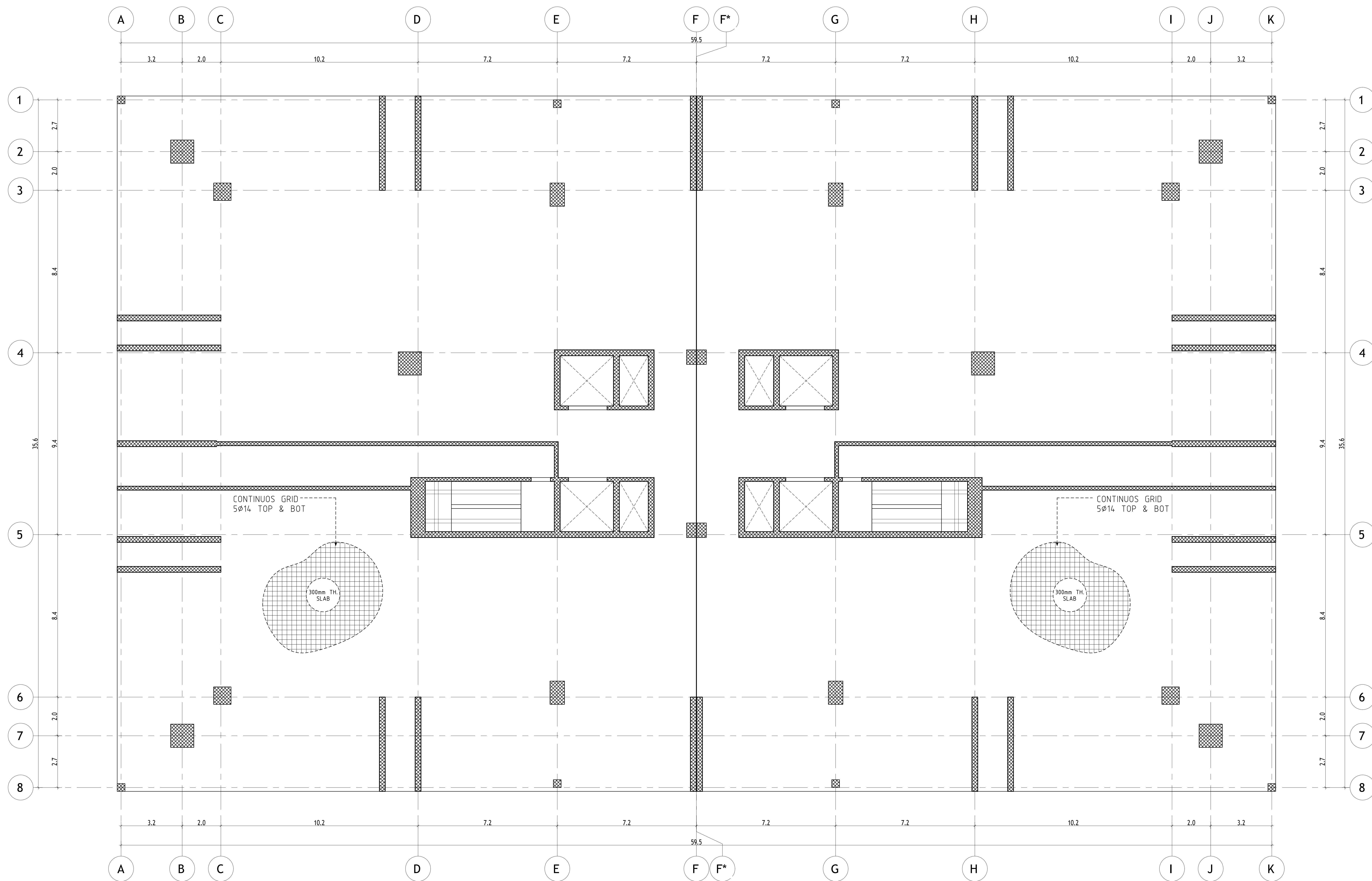
DONE BY:
ABDULELAH ALQABBANI

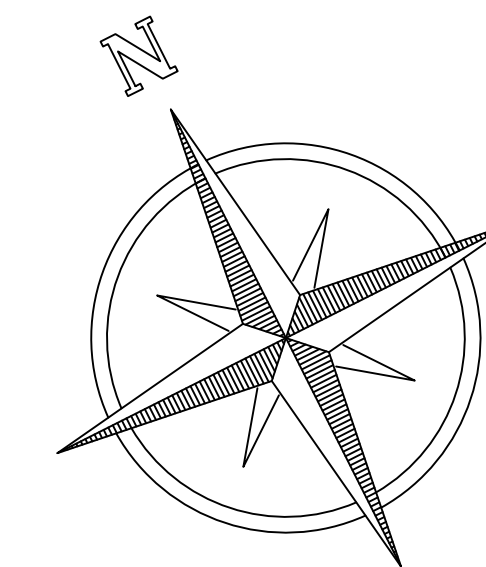
PROJECT:
Apartments Building

DRAWING NAME:
GF Framing

SCALE:
1:75

SHEET NUMBER:
A15





ALL UNITS IN METER

Design Studio (8), Technical
Project, 2024/2025

INSTRUCTORS:

Prof. M. Essam Shaawat Co.
Dr. Faris Almaziad
Dr. M. Farooqi
Dr. M. Al Mulhim
Dr. AL Tayeb
Dr. M. Al khater
Eng. Meqdad
Eng. Rehan Gamil
Eng. Noman Ashraf

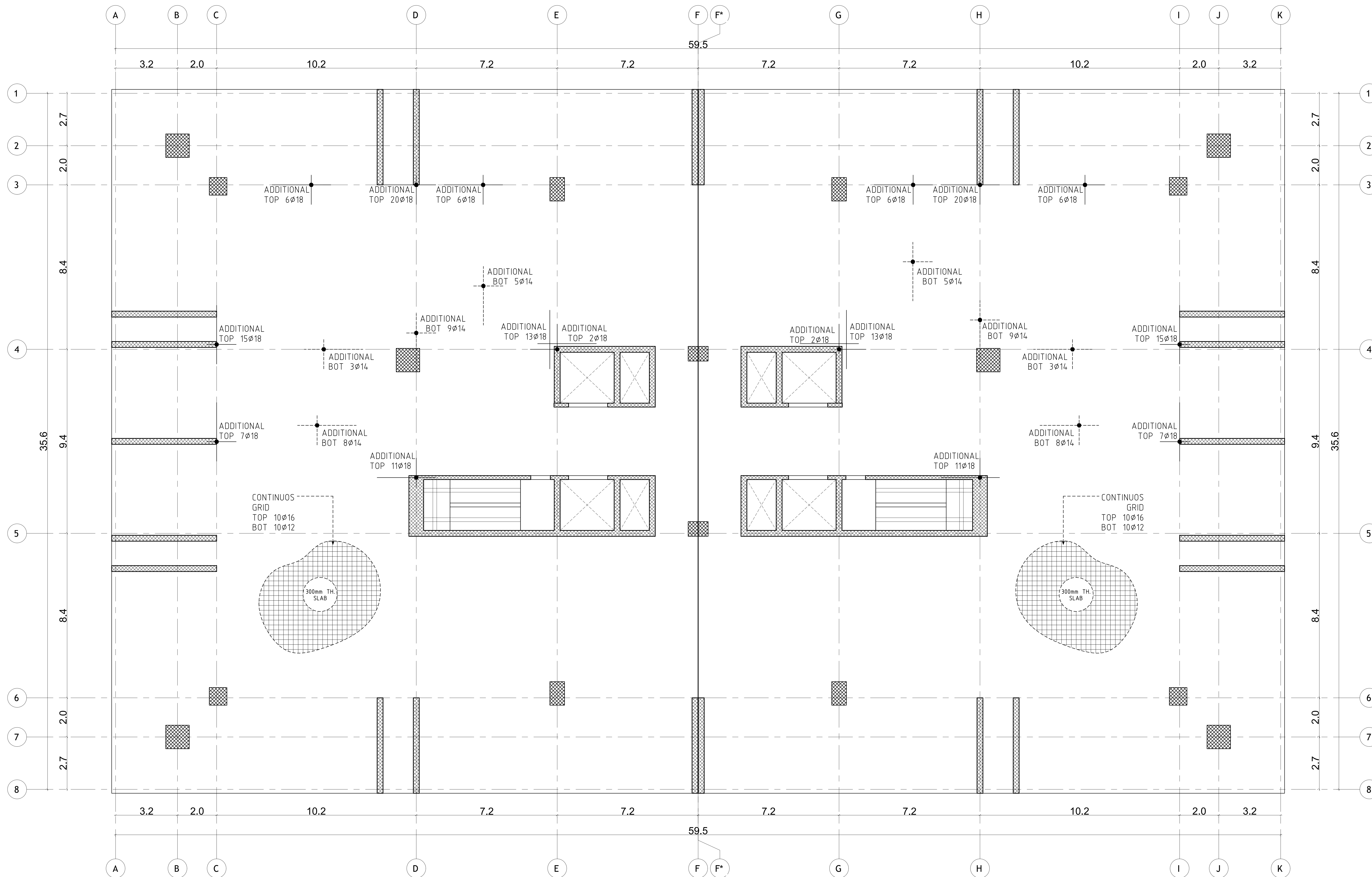
DONE BY:
ABDULELAH ALQABBANI

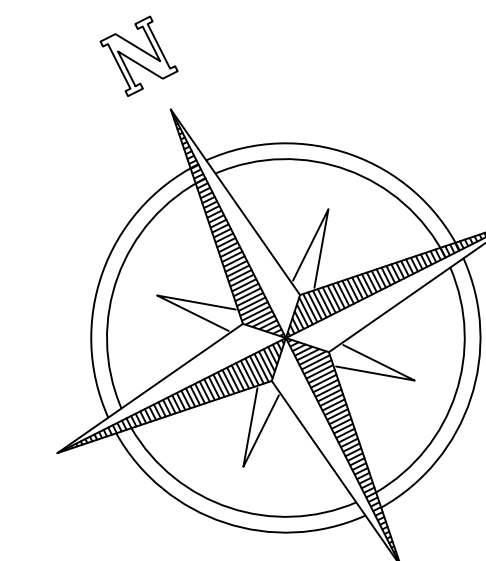
PROJECT:
Apartments Building

DRAWING NAME:
Typical 2nd-20th(2nd floor)
Framing plan

SCALE:
1:75

SHEET NUMBER:
A16





ALL UNITS IN METER

Design Studio (8), Technical
Project, 2024/2025

INSTRUCTORS:

Prof. M. Essam Shaawat Co.
Dr. Faris Almaziad
Dr. M. Farooqi
Dr. M. Al Mulhim
Dr. AL Tayeb
Dr. M. Al khater
Eng. Meqdad
Eng. Rehan Gamil
Eng. Noman Ashraf

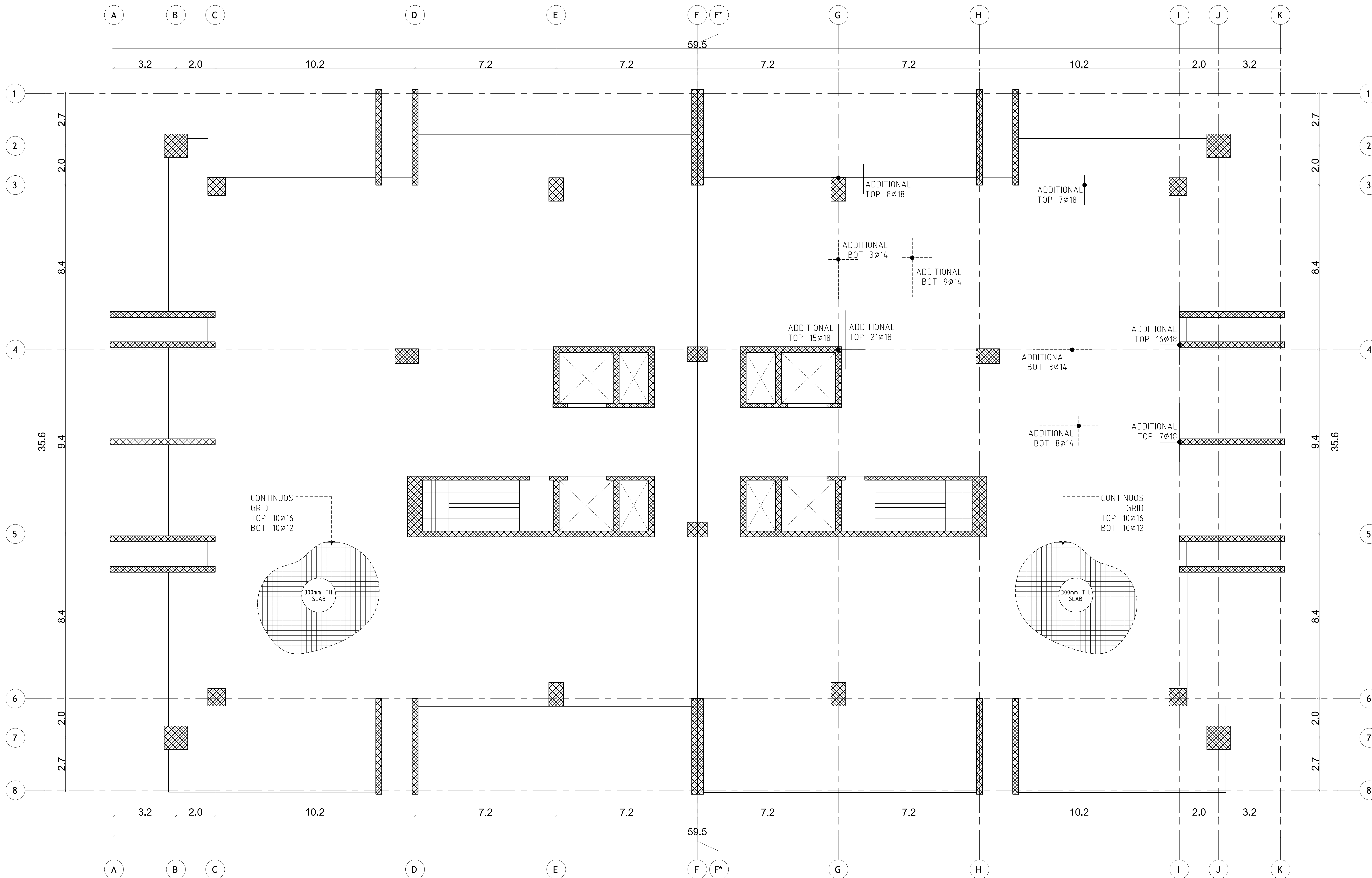
DONE BY:
ABDULELAH ALQABBANI

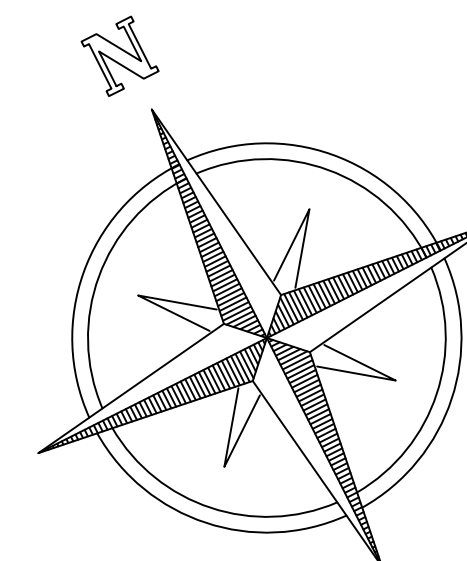
PROJECT:
Apartments Building

DRAWING NAME:
Typical 2nd-20(8th floor)
Framing plan

SCALE:
1:75

SHEET NUMBER:
A17





ALL UNITS IN METER

Design Studio (8), Technical
Project, 2024/2025

INSTRUCTORS:

Prof. M. Essam Shaawat Co.
Dr. Faris Almaziad
Dr. M. Farooqi
Dr. M. Al Mulhim
Dr. AL Tayeb
Dr. M. Al khater
Eng. Meqdad
Eng. Rehan Gamil
Eng. Noman Ashraf

DONE BY:

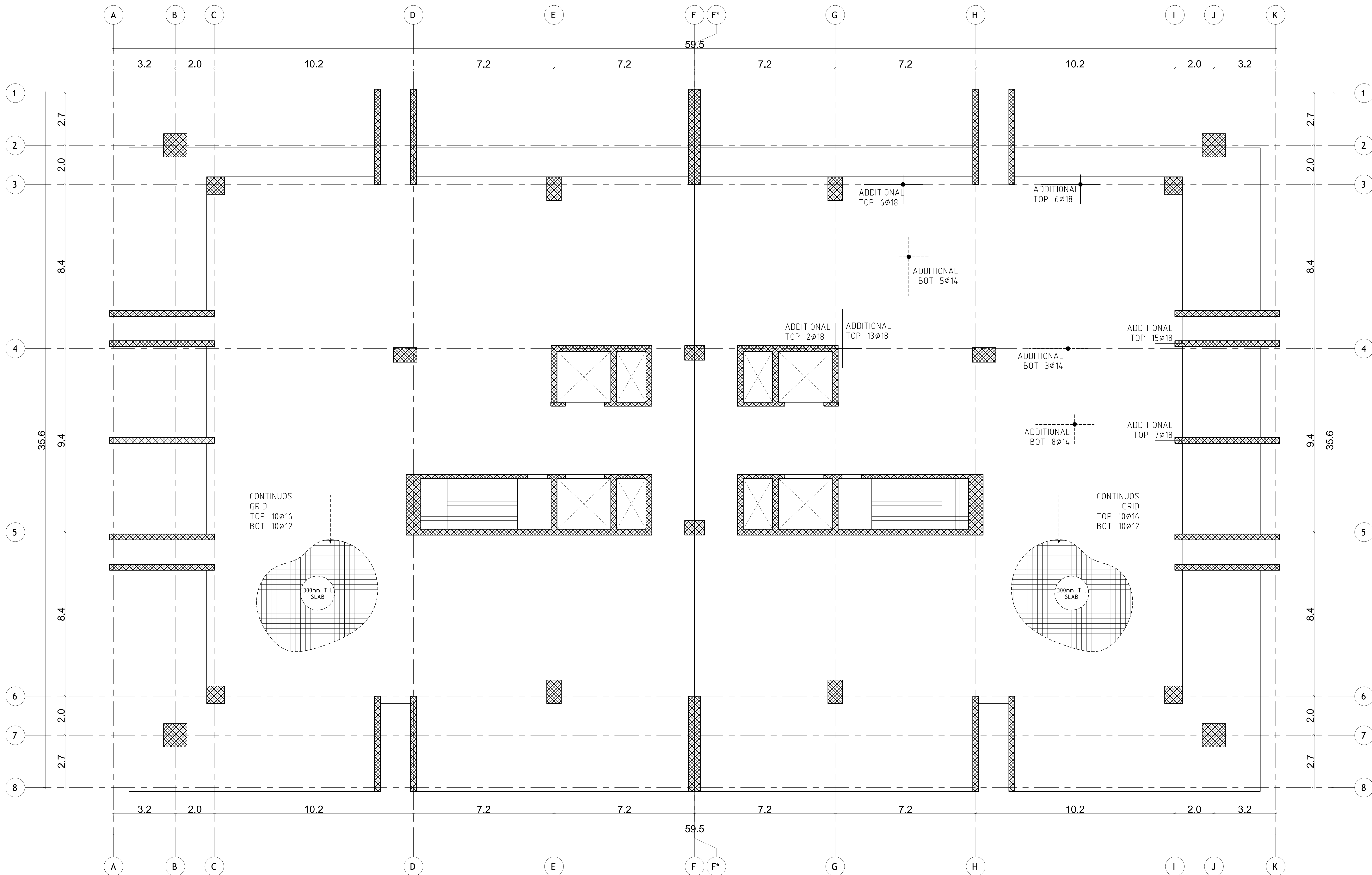
ABDULELAH ALQABBANI

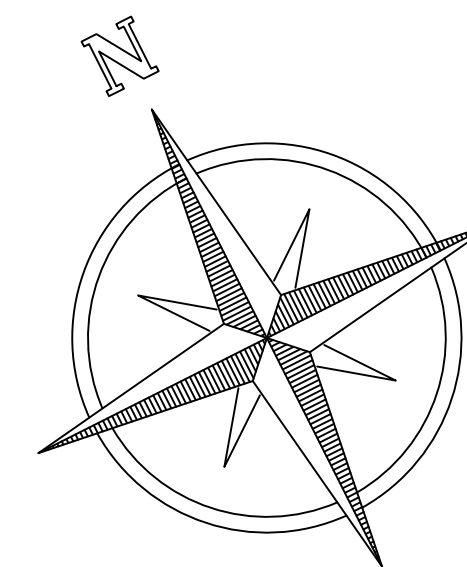
PROJECT:
Apartments Building

DRAWING NAME:
Typical 2nd-20th(12th floor)
Framing plan

SCALE:
1:75

SHEET NUMBER:
A18





ALL UNITS IN METER

Design Studio (8), Technical
Project, 2024/2025

INSTRUCTORS:

Prof. M. Essam Shaawat Co.
Dr. Faris Almaziad
Dr. M. Farooqi
Dr. M. Al Mulhim
Dr. AL Tayeb
Dr. M. Al khater
Eng. Meqdad
Eng. Rehan Gamil
Eng. Noman Ashraf

DONE BY:

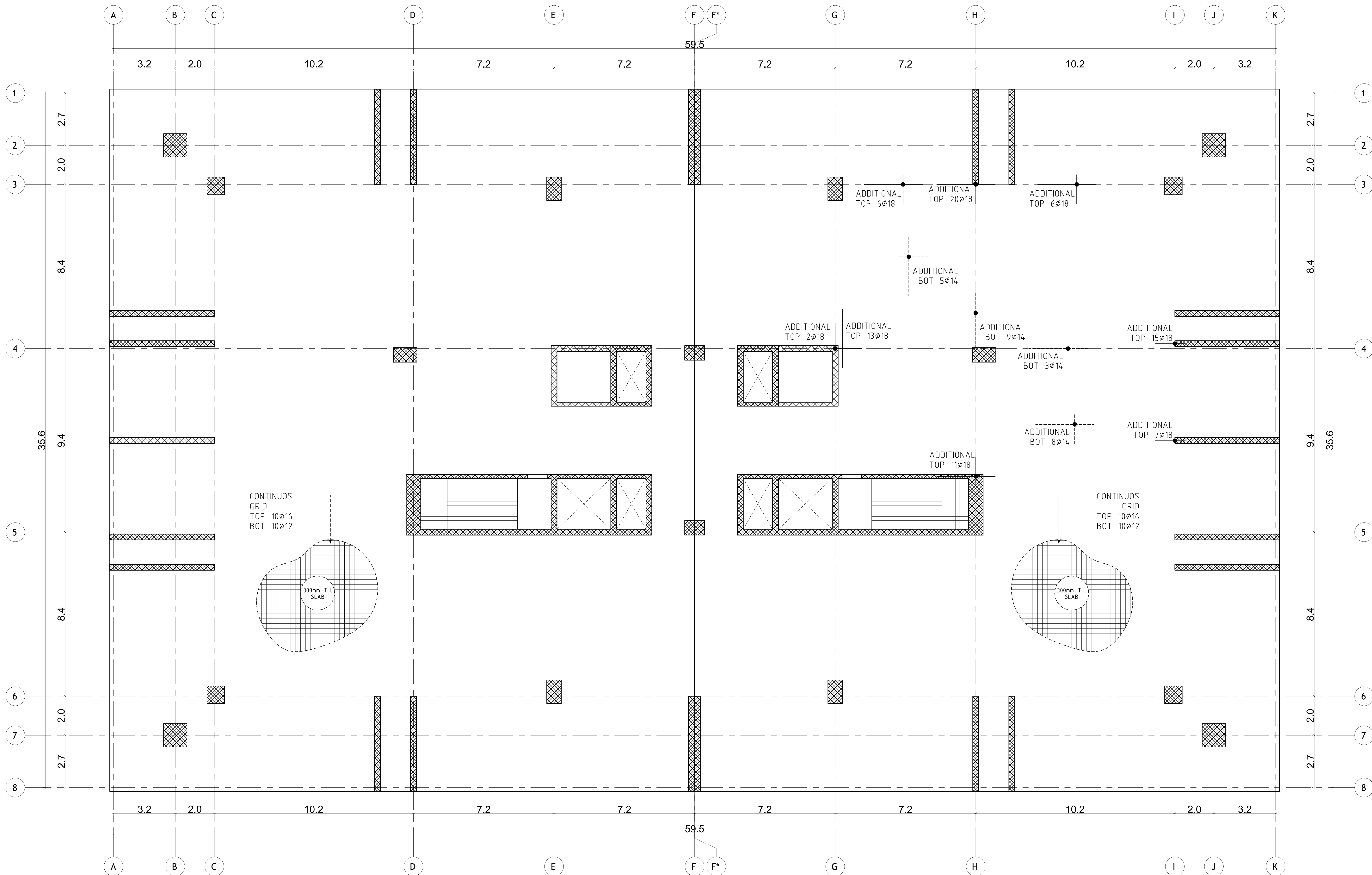
ABDULELAH ALQABBANI

PROJECT:
Apartments Building

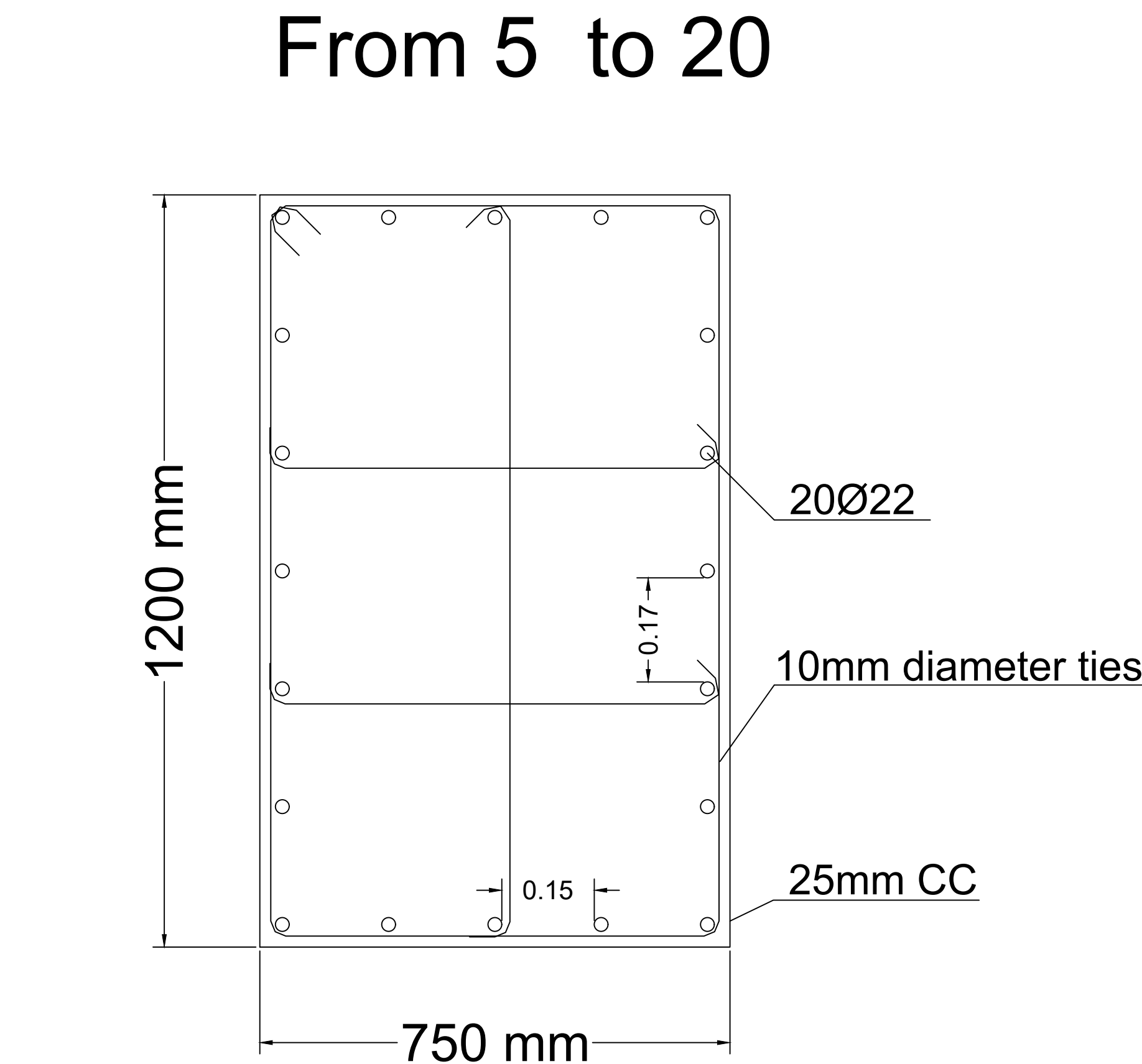
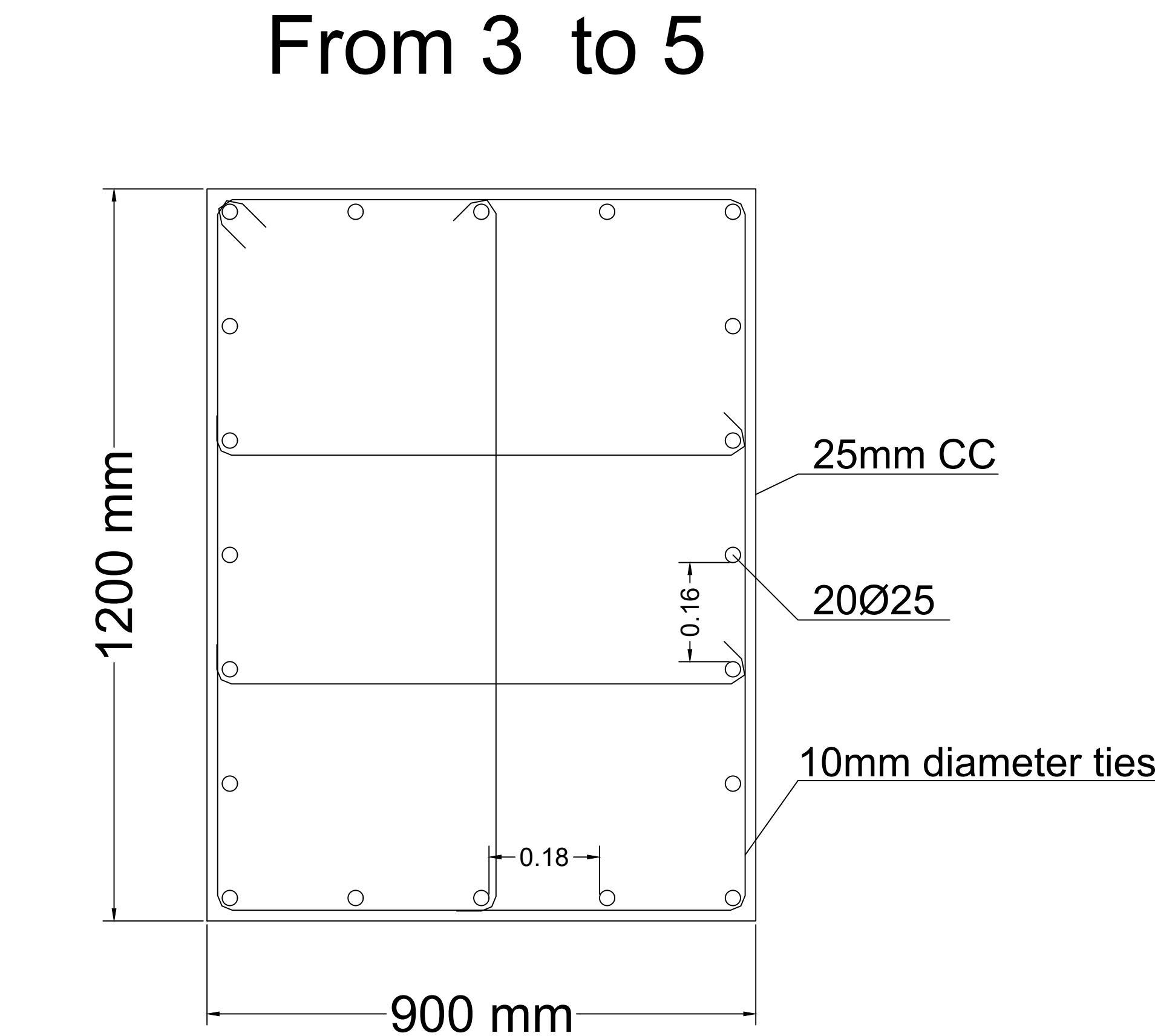
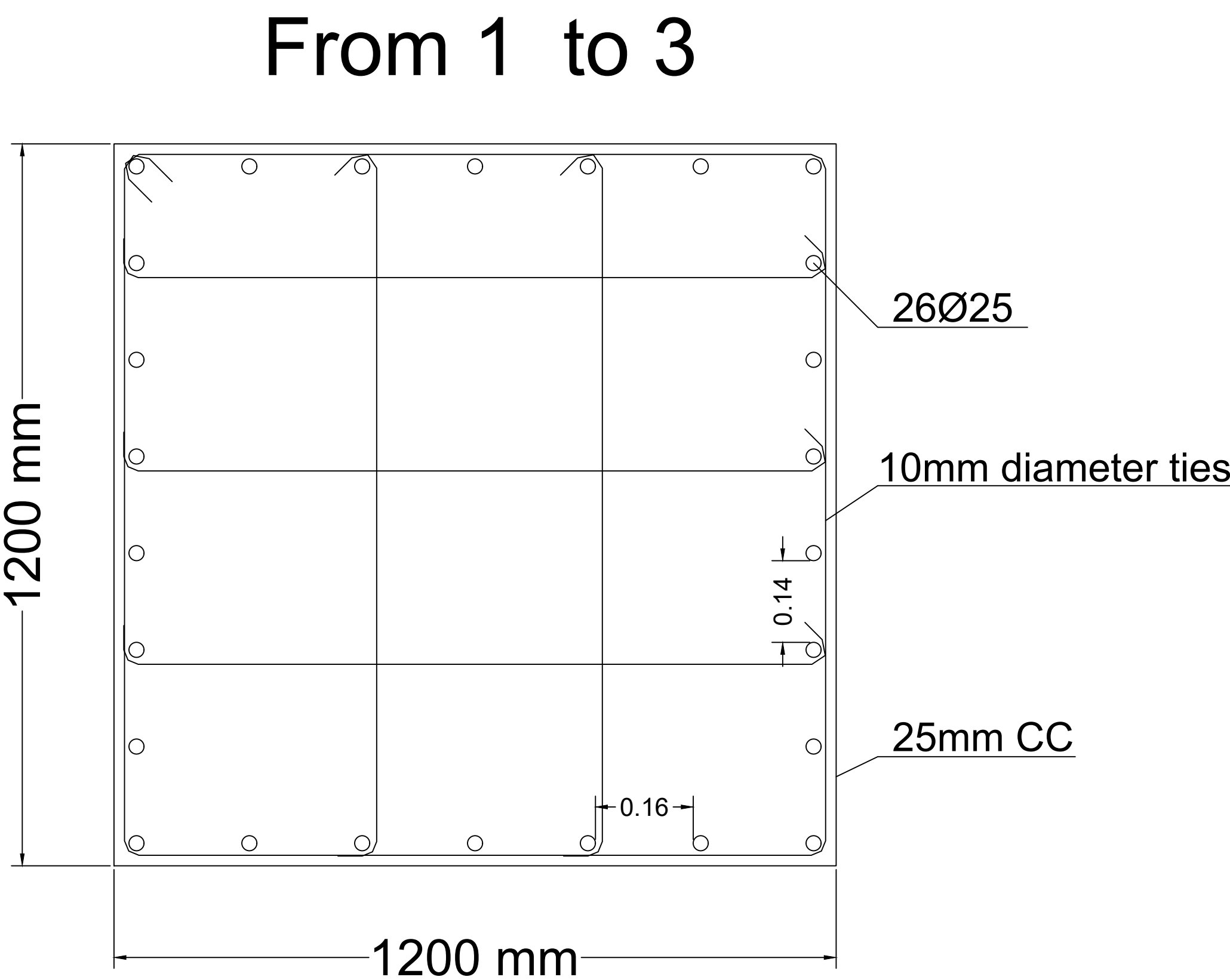
DRAWING NAME:
Roof(20th floor)
Framing plan

SCALE:
1:75

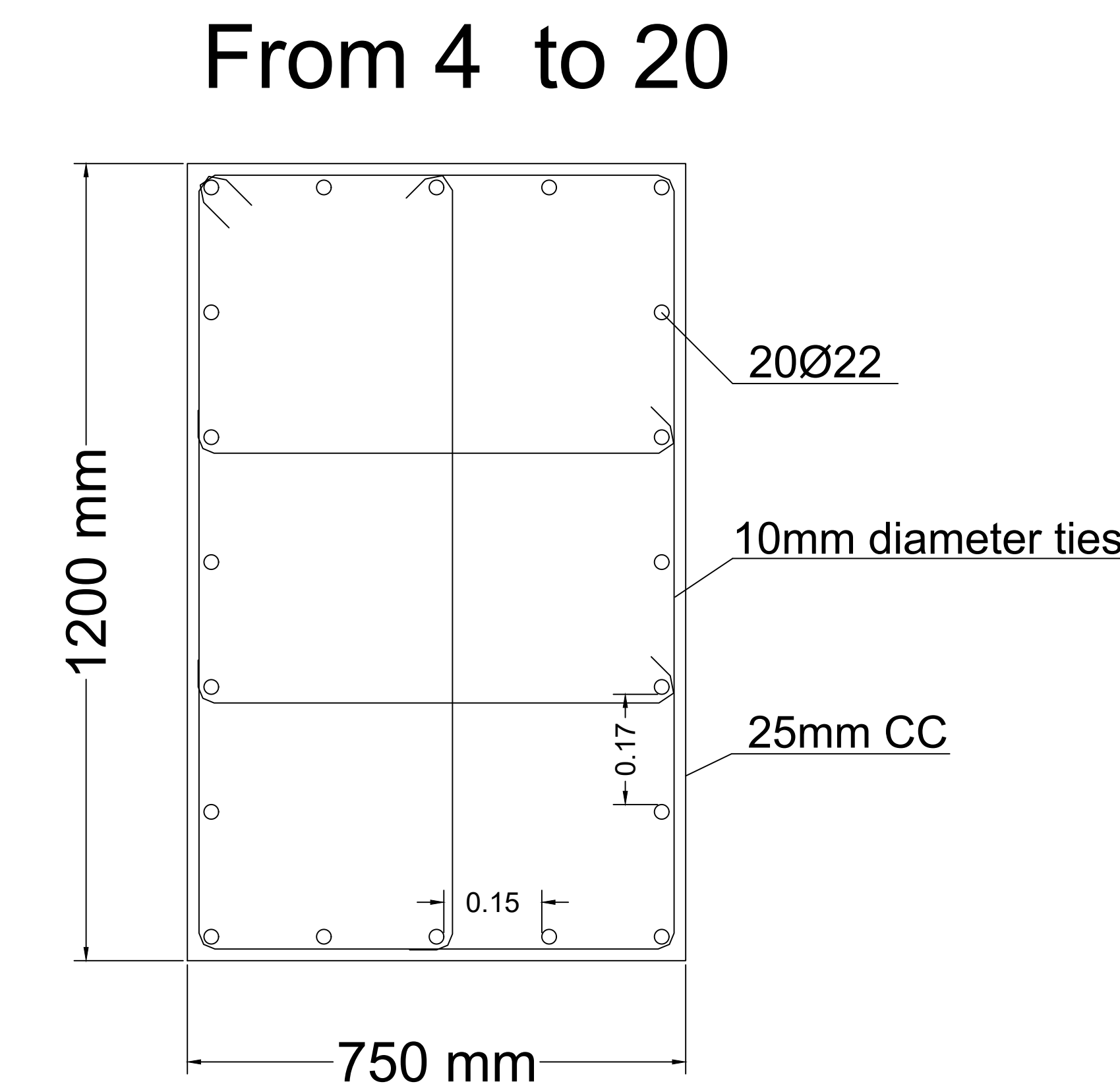
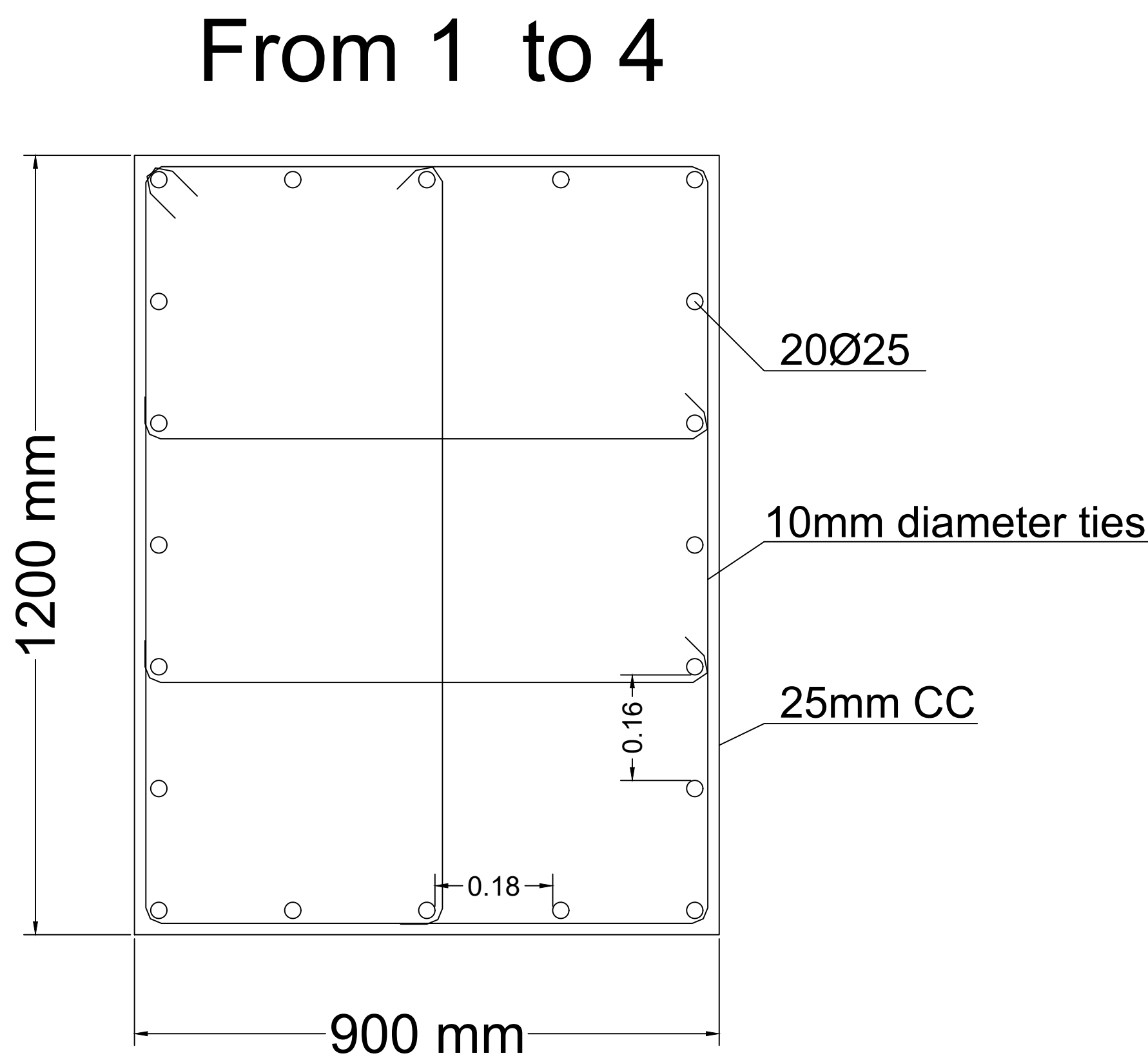
SHEET NUMBER:
A19



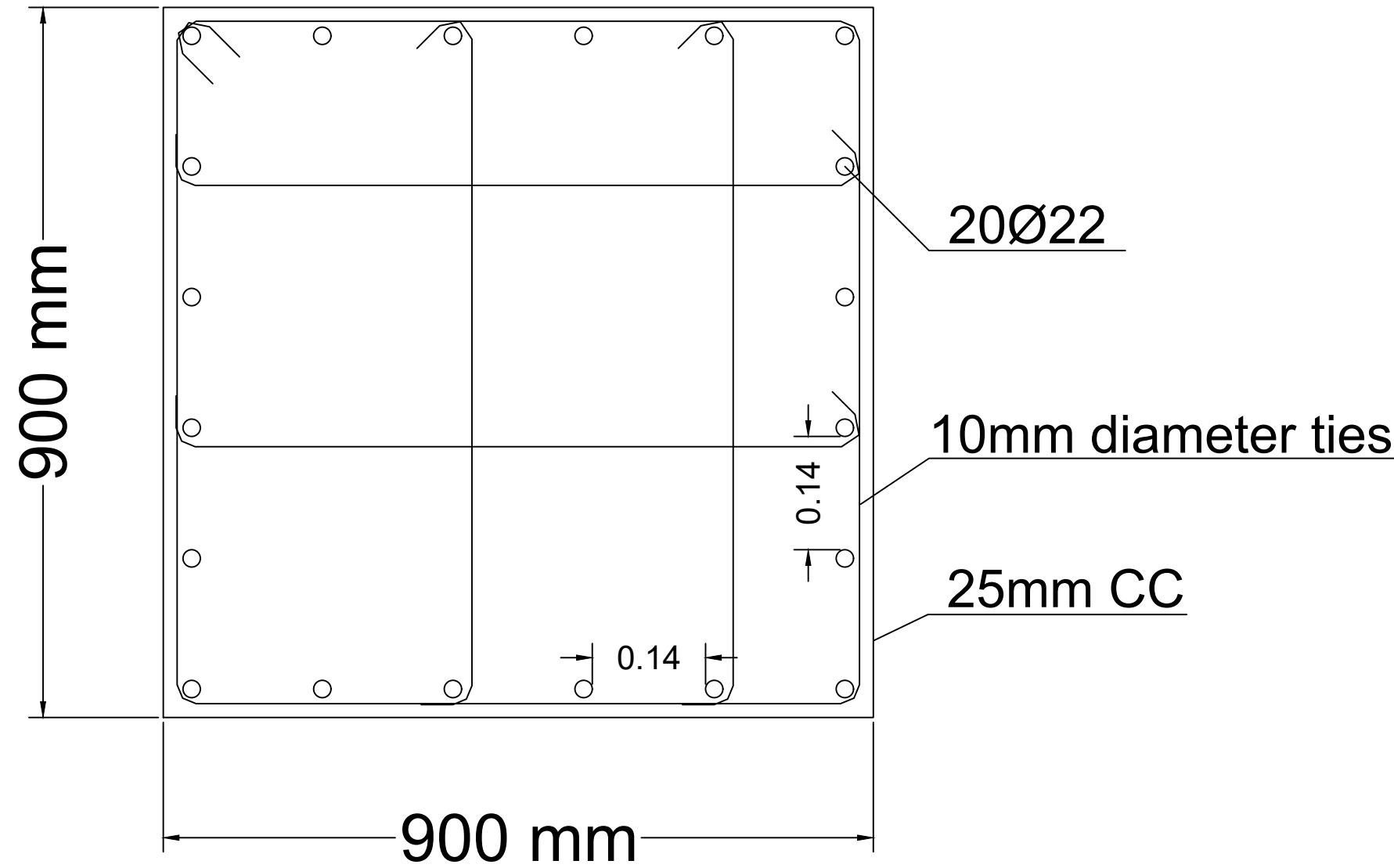
(C1 Internal)



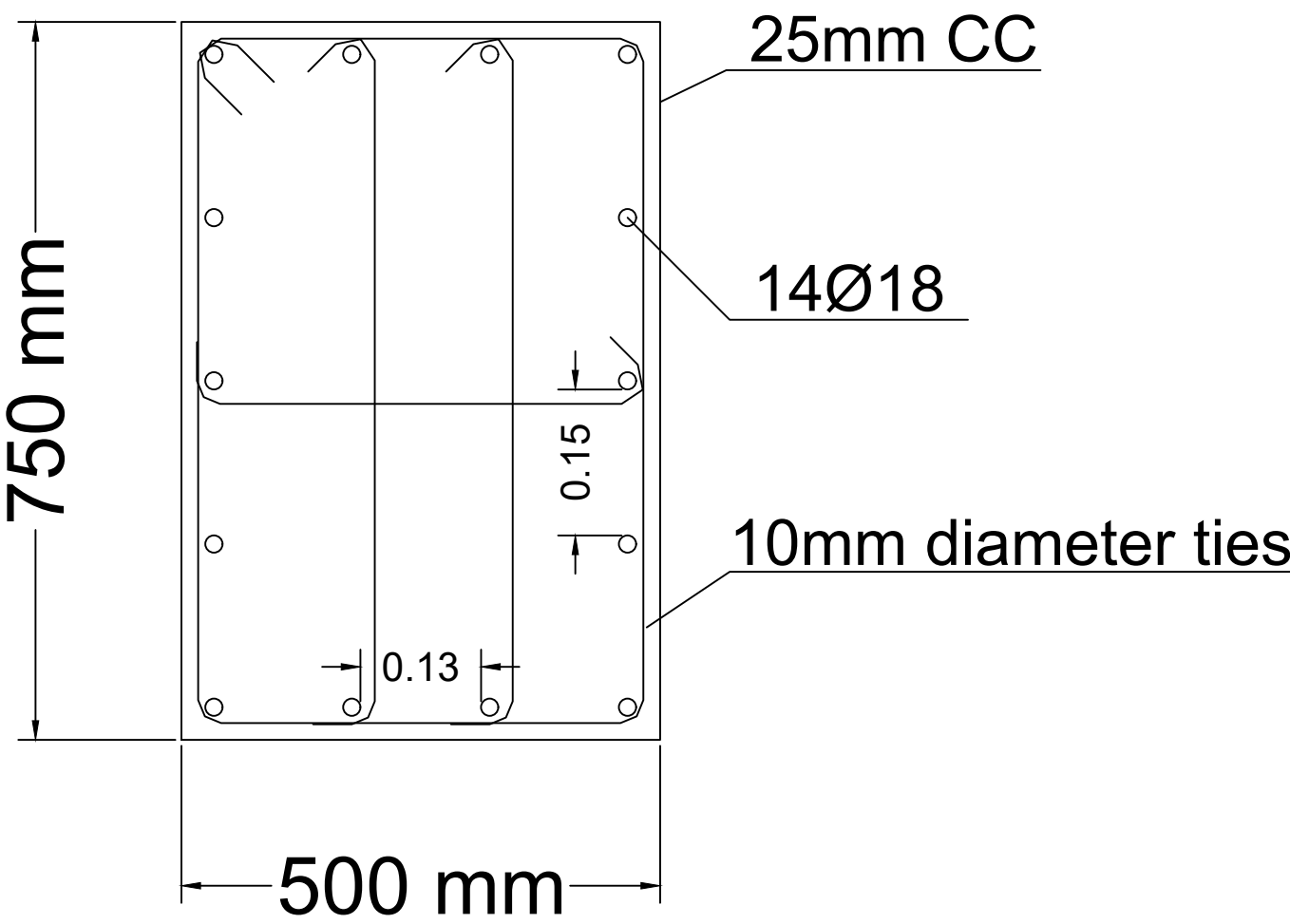
(C2 Edge)



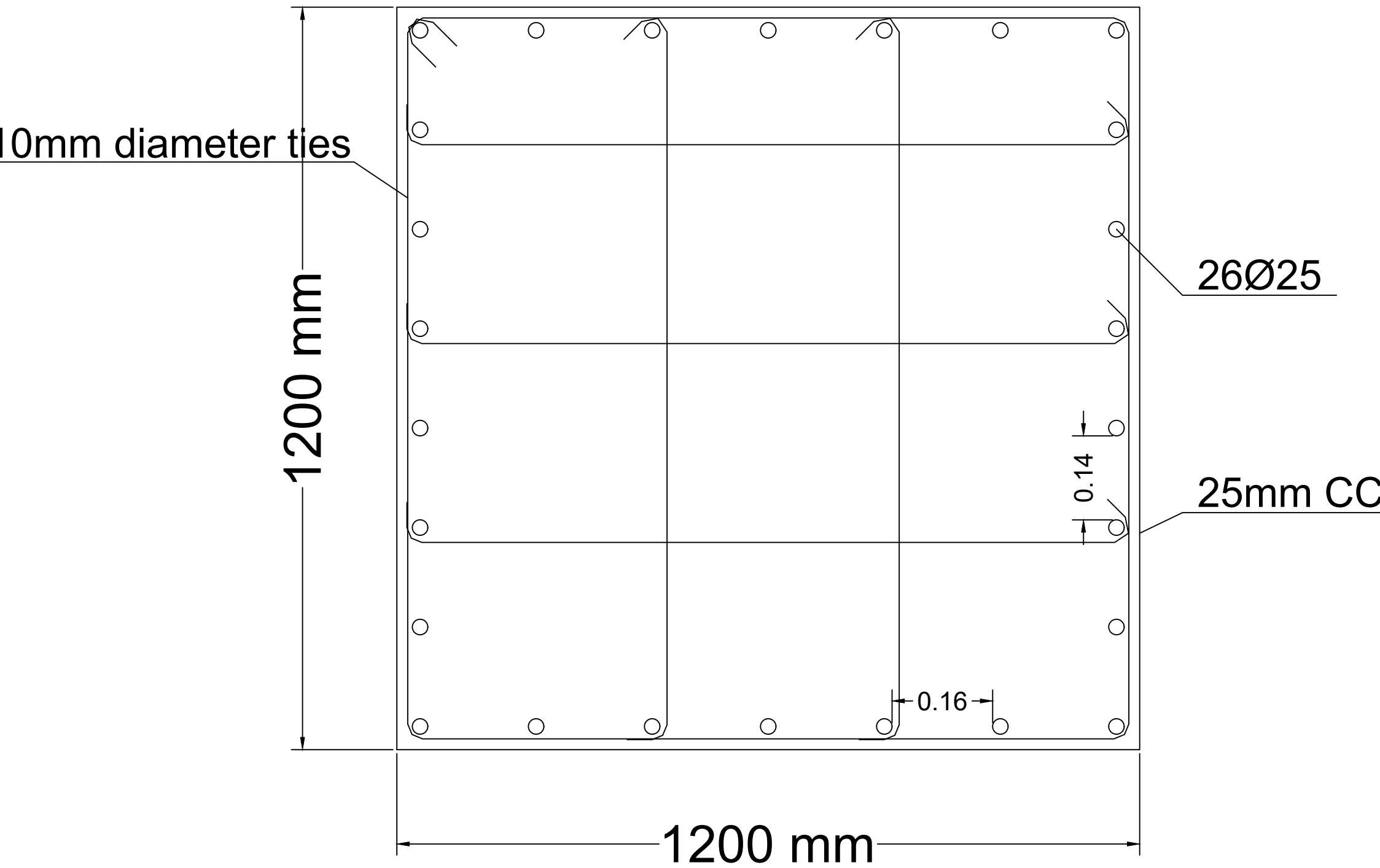
(C3 Corner)



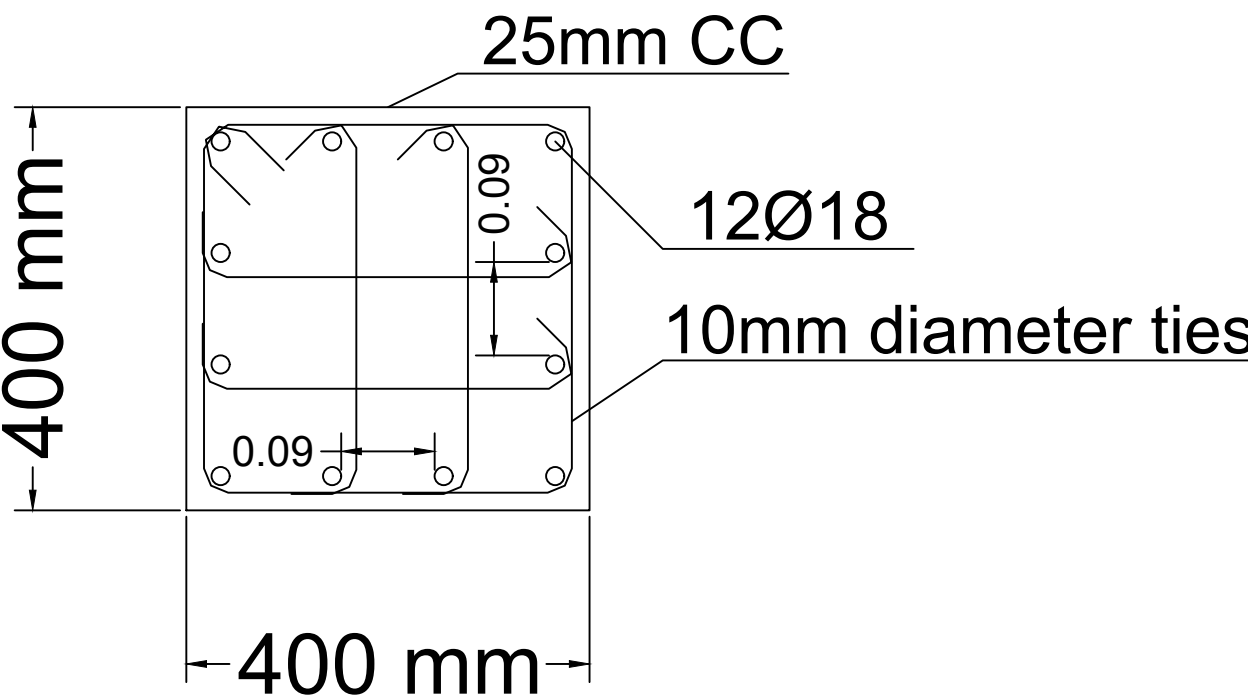
(C4 Edge)



(C5 External corner)



(C6 Edge & Corner)



Schedule of Columns

Column Type	Floor Levels	Dimensions (mm)	Vertical Reinforcement	Tie Bars	Tie Spacing	Clear Cover (mm)
C1(Internal)	1 - 3	1200 × 1200	26Ø25 @ 140mm	Ø10mm	150mm (ends), 450mm (mid)	25mm
---	3 - 5	1200 × 900	20Ø25 @ 160mm	Ø10mm	150mm (ends), 450mm (mid)	25mm
---	5 - 20	1200 × 750	20Ø22 @ 150mm	Ø10mm	150mm (ends), 450mm (mid)	25mm
C2 (Edge)	1 - 4	1200 × 900	20Ø25 @ 160mm	Ø10mm	150mm (ends), 450mm (mid)	25mm
---	4 - 20	1200 × 750	20Ø22 @ 150mm	Ø10mm	150mm (ends), 450mm (mid)	25mm
C3 (Corner)	1 - 20	900 × 900	20Ø22 @ 140mm	Ø10mm	150mm (ends), 450mm (mid)	25mm
C4 (Edge)	1 - 20	750 × 500	14Ø18 @ 130mm	Ø10mm	150mm (ends), 450mm (mid)	25mm
C5 (External Corner)	1 - 20	1200 × 1200	26Ø25 @ 140mm	Ø10mm	150mm (ends), 450mm (mid)	25mm
C6 (Edge & Corner)	1	400 × 400	12Ø18 @ 90mm	Ø10mm	150mm (ends), 450mm (mid)	25mm

INSTRUCTORS:

Prof. M. Essam Shaaawat Co.
Dr. Faris Almaziad
Dr. M. Farooqi
Dr. M. Al Mulhim
Dr. AL Tayeb
Dr. M. Al khater
Eng. Meqdad
Eng. Rehan Gamil
Eng. Noman Ashraf

DONE BY:
ABDULELAH ALQABBANI

PROJECT:
Apartments Building

DRAWING NAME:
Coloumns Details

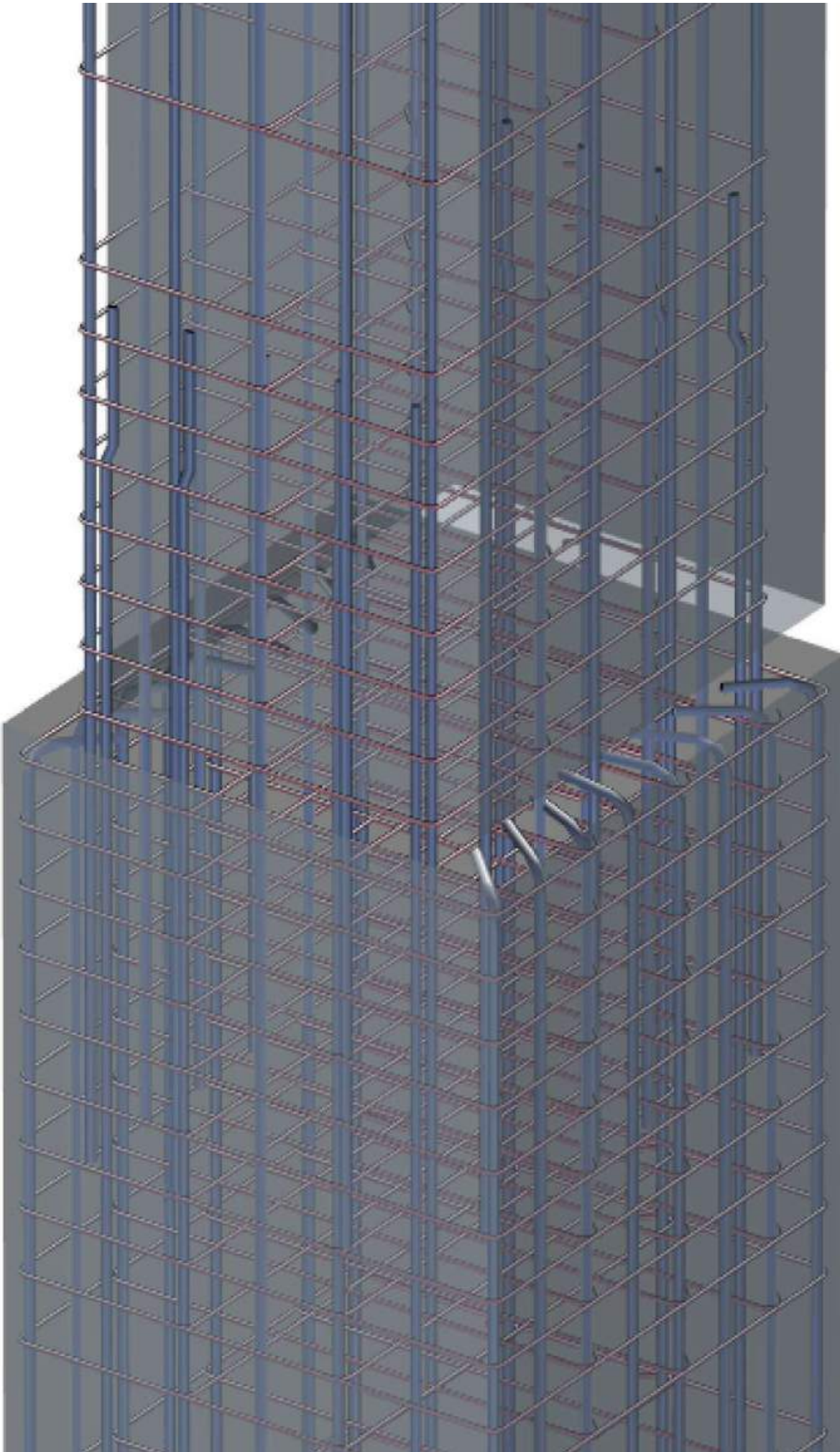
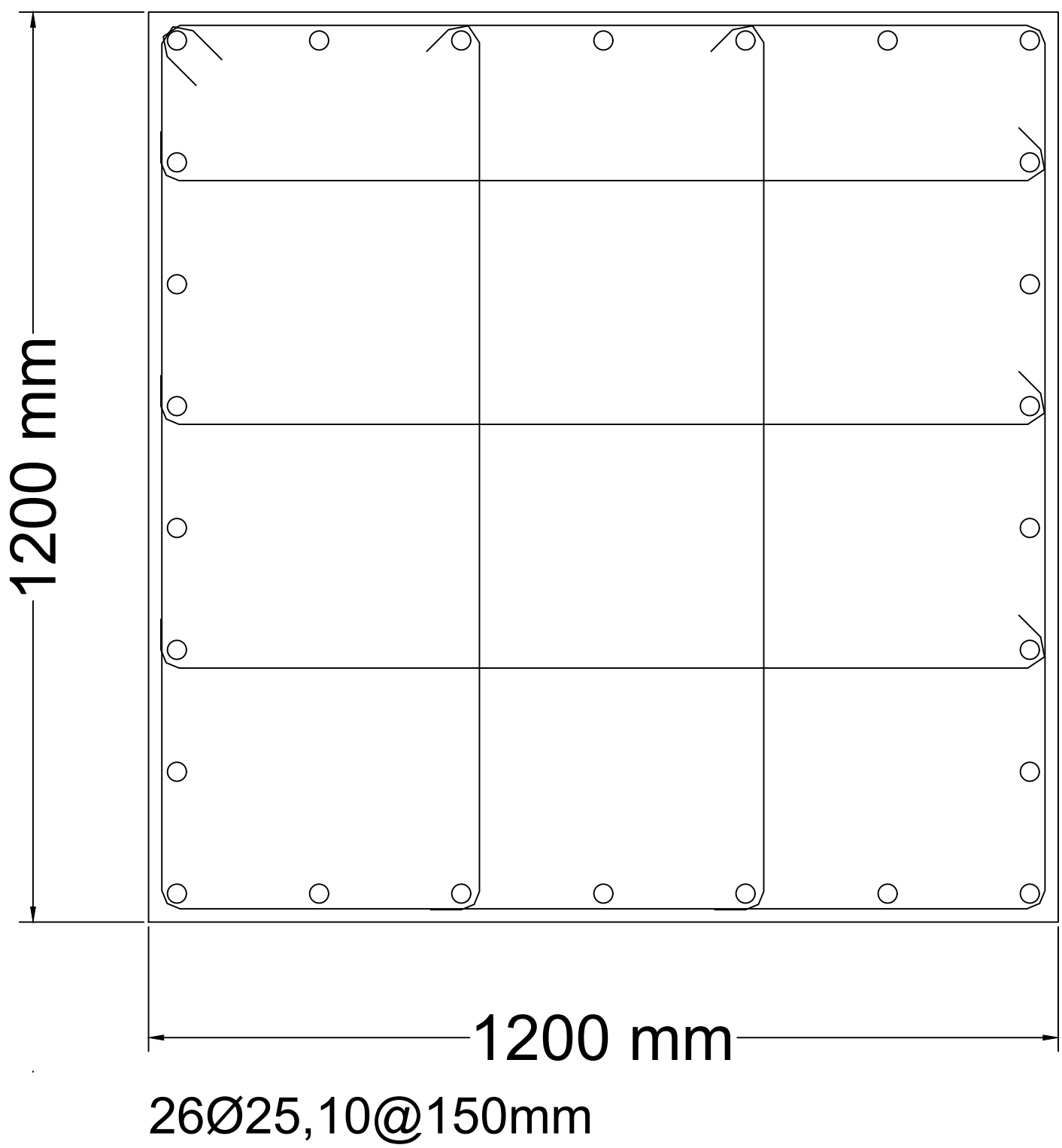
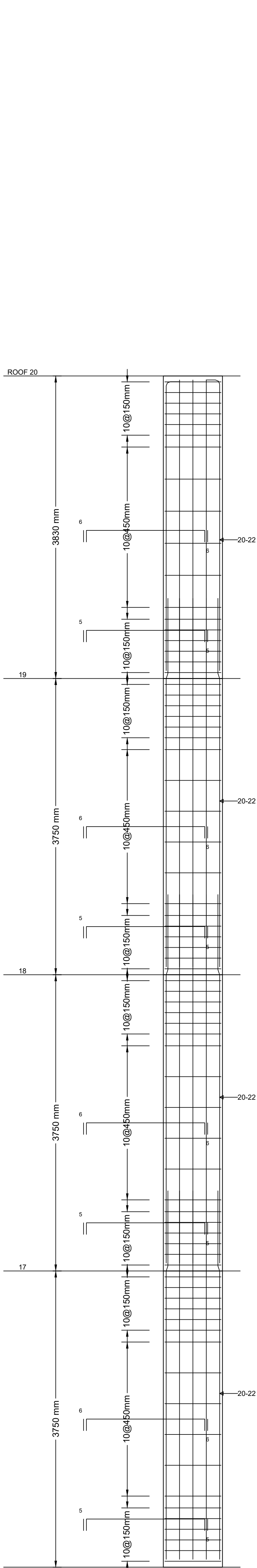
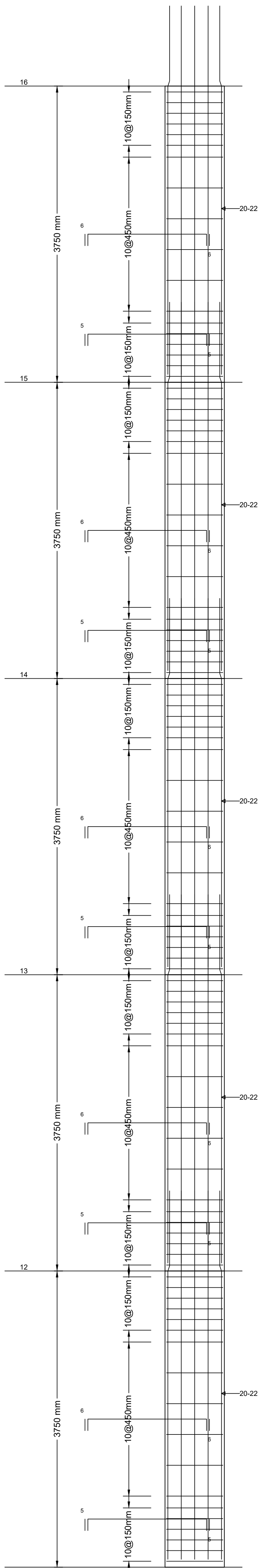
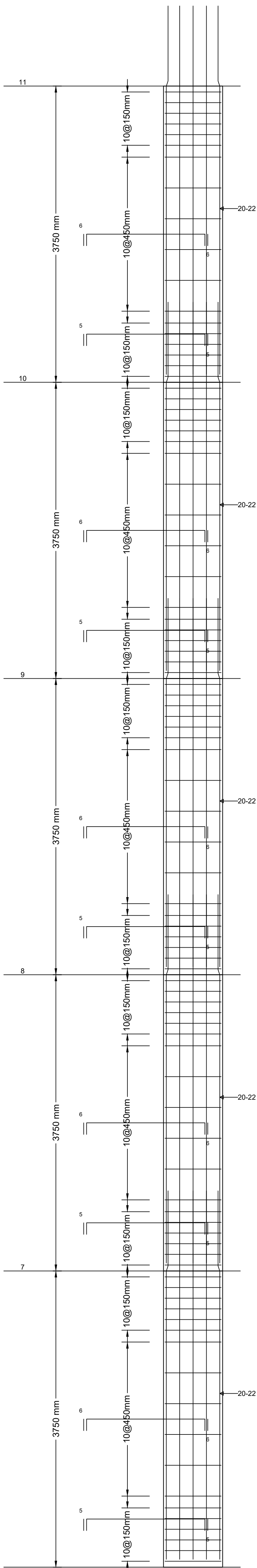
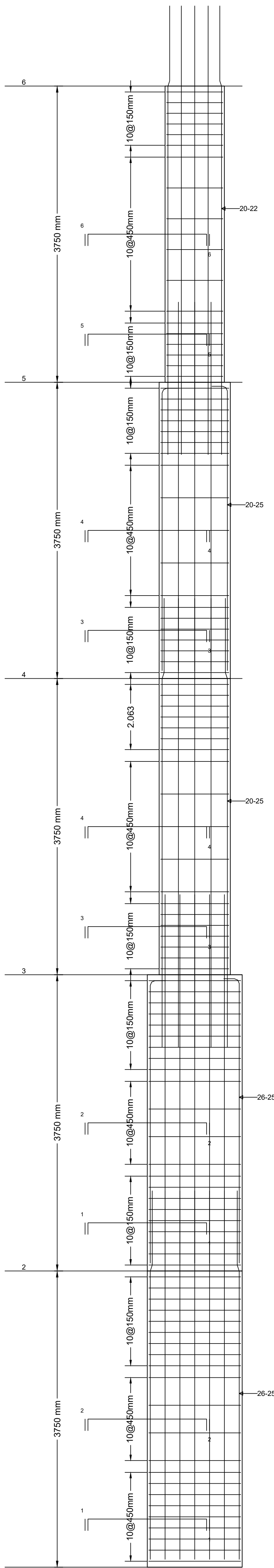
SCALE:
1:10

SHEET NUMBER:
A20

Coloumn Full Stack Detail (C1 Internal)

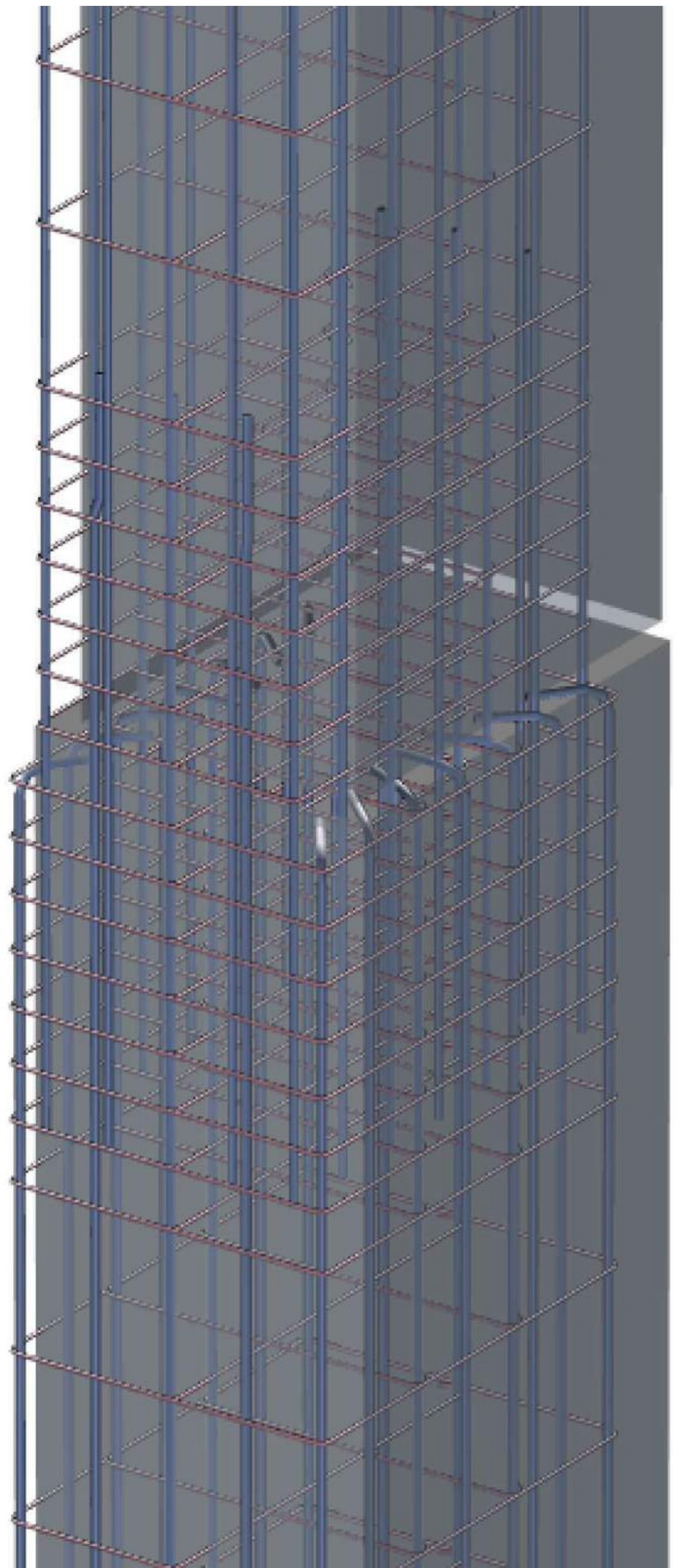
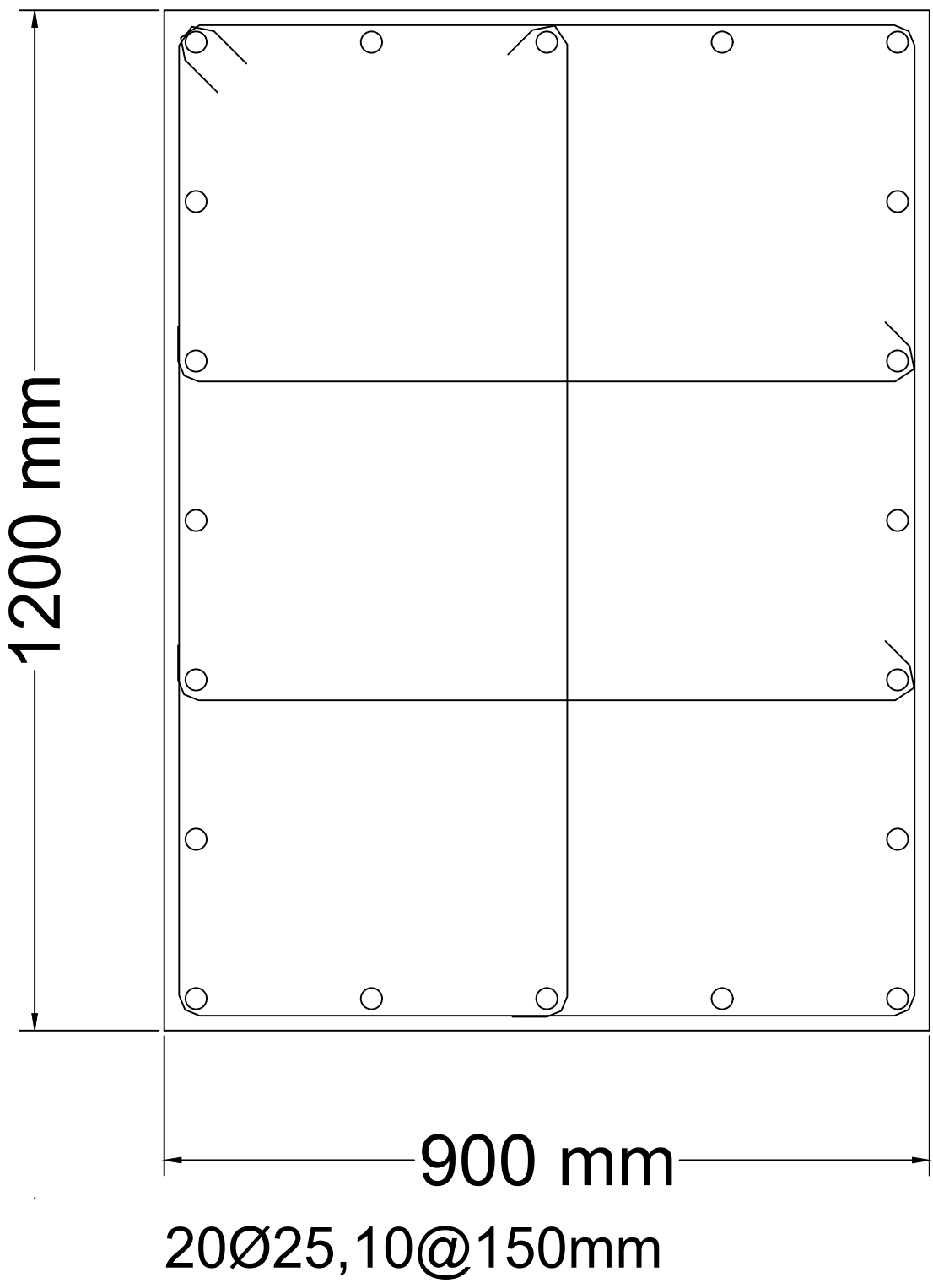
From 1 to 3

CONNECTION AT 3

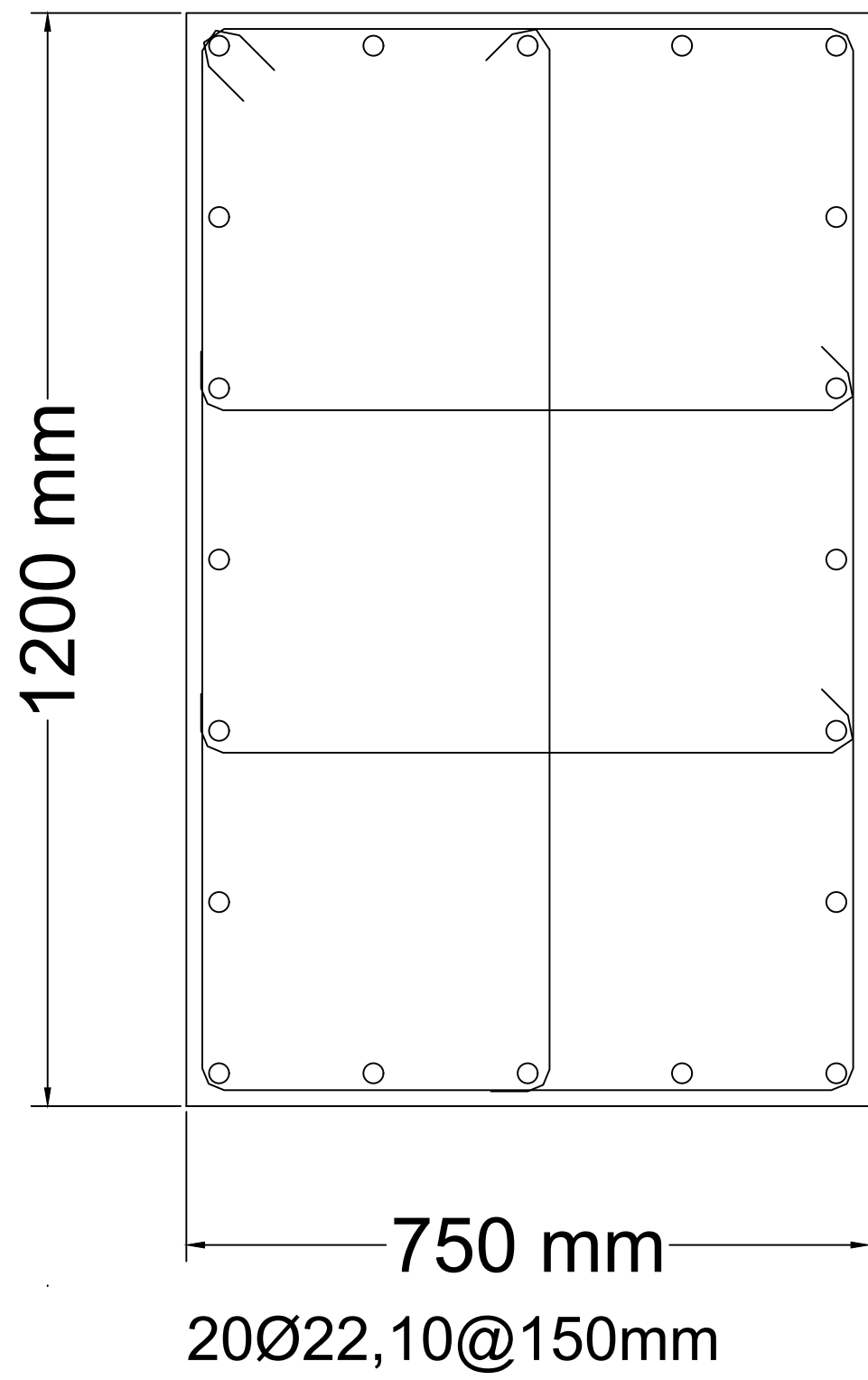


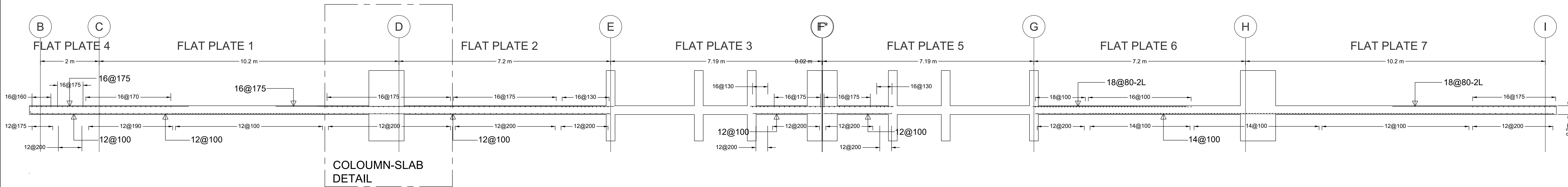
From 3 to 5

CONNECTION AT 5



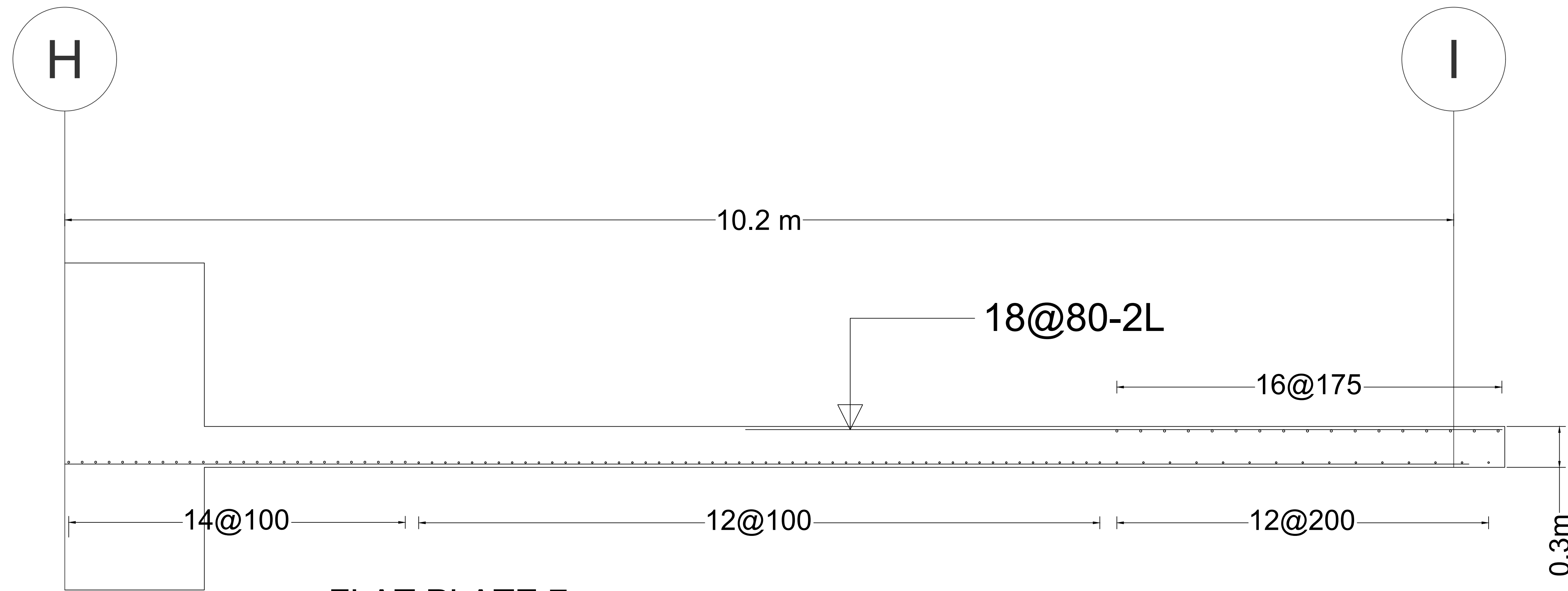
From 5 to 20



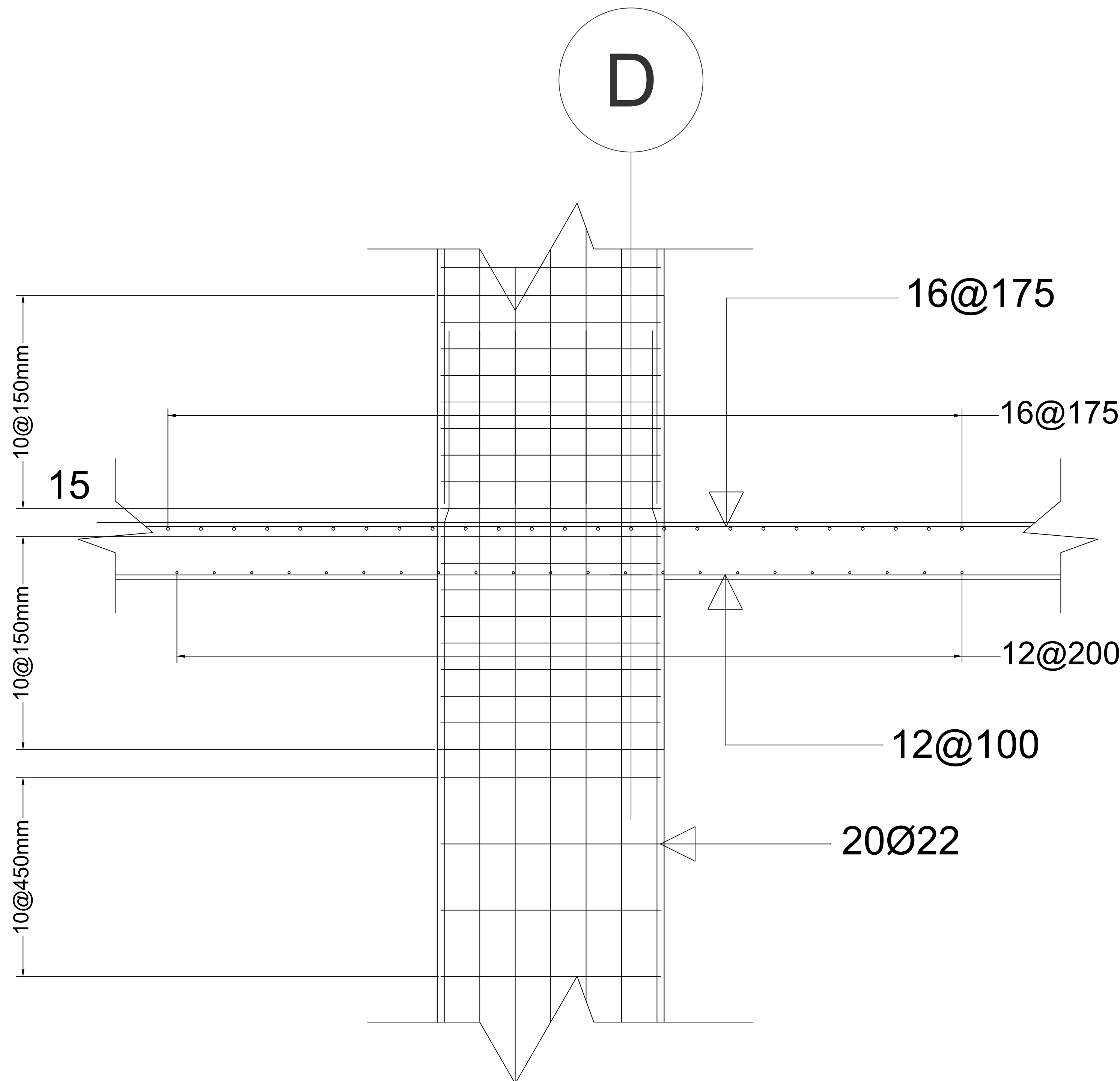


SECTION S-S
NOT TO SCALE

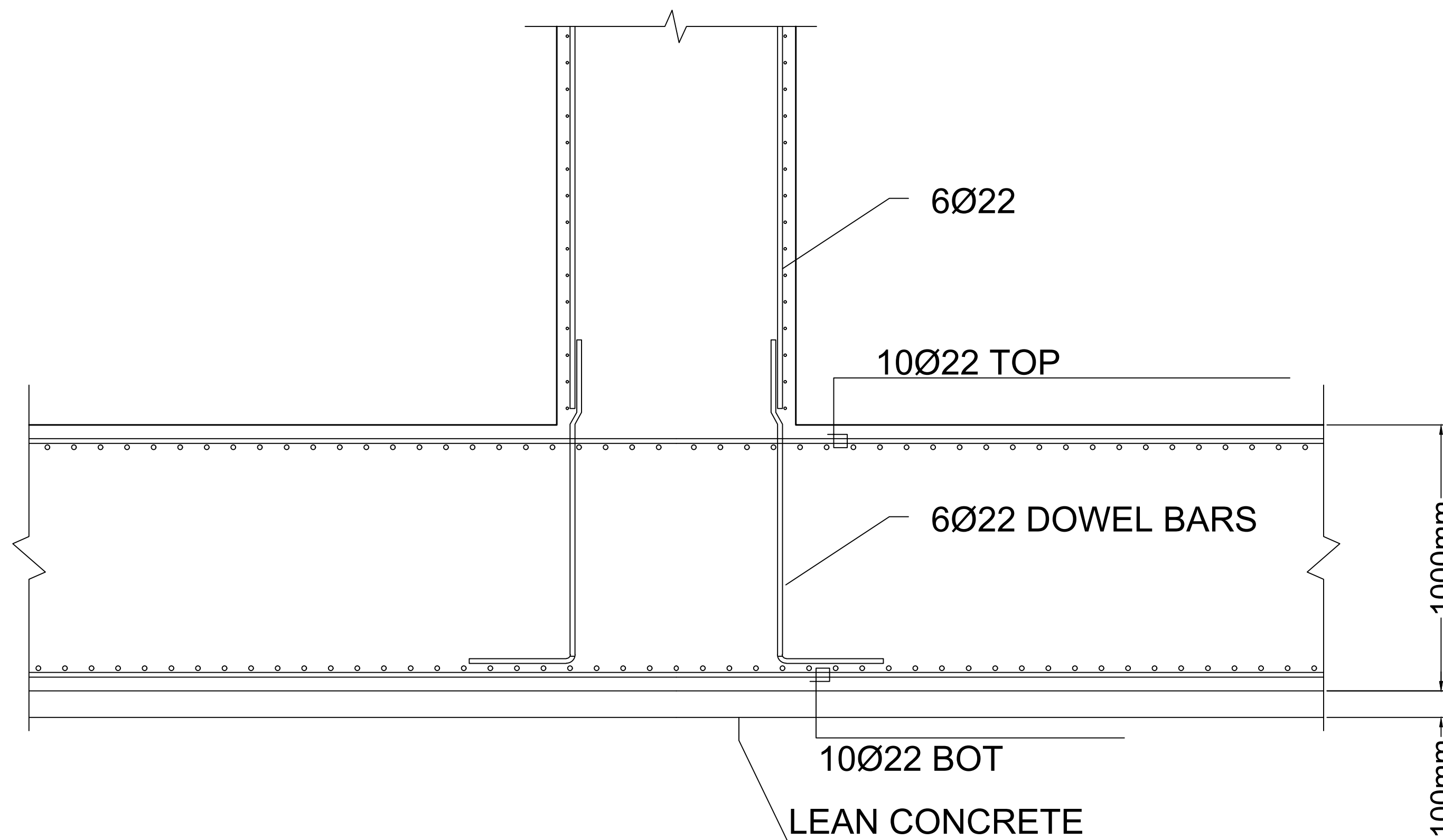
Schedule of Slabs	Slab Type	Span (m) (X-Axis)	Span (m) (Y-Axis)	General Reinforcement	Additional Reinforcement (X-Axis)	Additional Reinforcement (Y-Axis)	Clear Cover (mm)
	Flat Plate 1	10.2	8.4	TOP: 10Ø16 BOT: 10Ø12	---	---	40mm
	Flat Plate 2	7.2	8.4	TOP: 10Ø16 BOT: 10Ø12	---	---	40mm
	Flat Plate 3	7.19	8.4	TOP: 10Ø16 BOT: 10Ø12	---	---	40mm
	Flat Plate 4	2.4	8.4	TOP: 10Ø16 BOT: 10Ø12	---	---	40mm
	Flat Plate 5	7.19	8.4	TOP: 10Ø16 BOT: 10Ø12	---	---	40mm
	Flat Plate 6	7.2	8.4	TOP: 10Ø16 BOT: 10Ø12	ADDITIONAL TOP 2Ø18	ADDITIONAL TOP 13Ø18 BOT: 6Ø14	40mm
	Flat Plate 7	10.2	8.4	TOP: 10Ø16 BOT: 10Ø12	ADDITIONAL BOT 4Ø14	ADDITIONAL TOP 15Ø18 BOT: 9Ø14	40mm



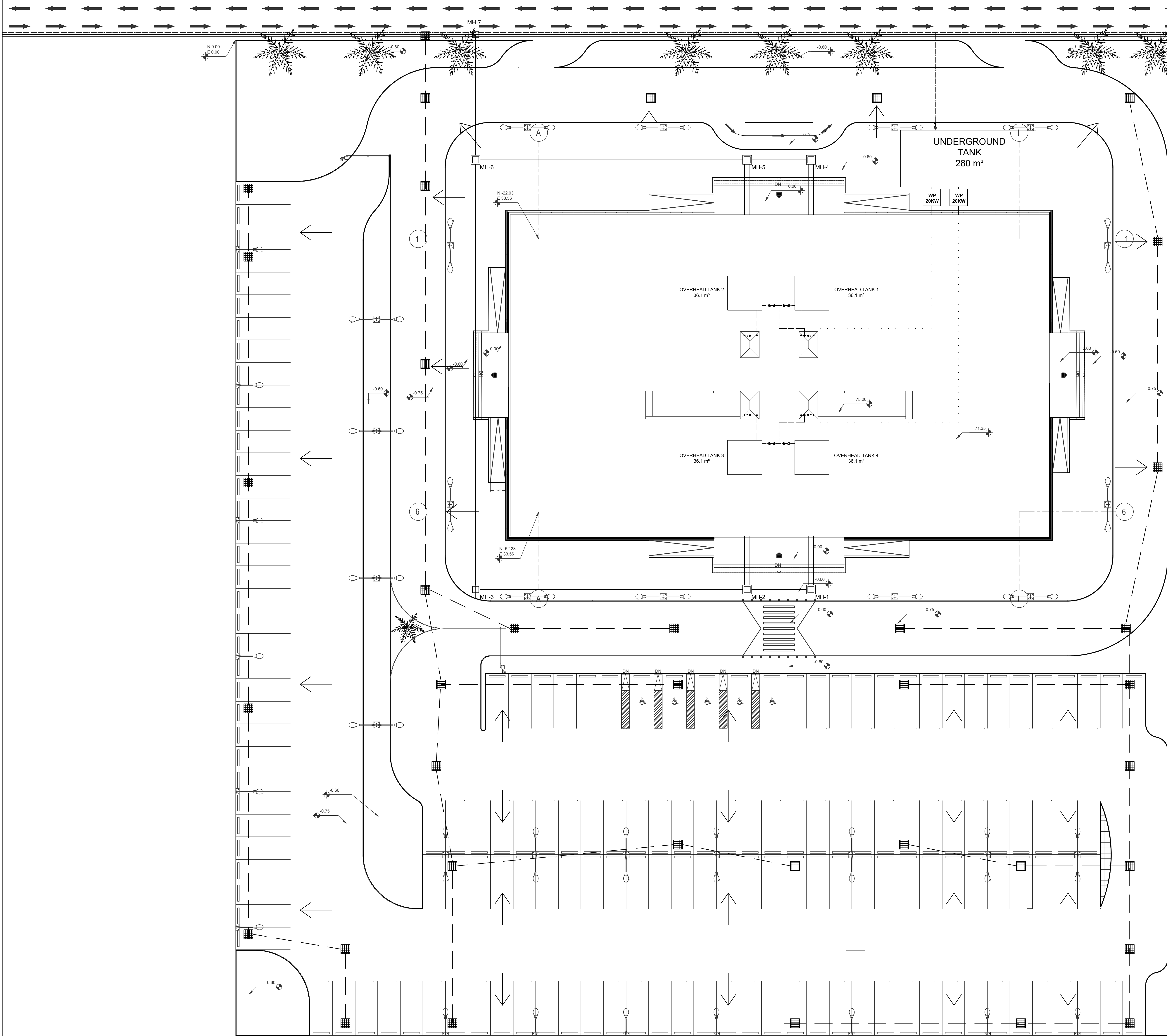
FLAT PLATE 7
SCALE 1:25



COLOUMN-SLAB DETAIL
1:20

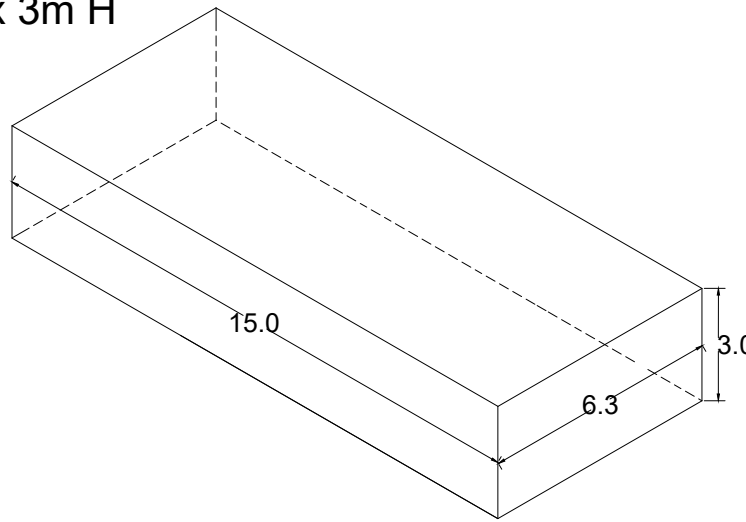


RAFT SECTION DETAIL
SCALE 1:25



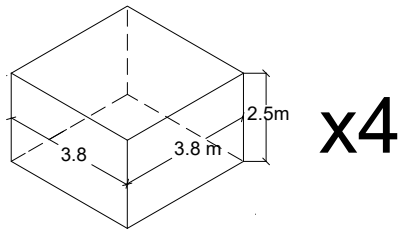
MH Sr.No.	Invert Level	Length from Previous pipe
MH-1	-1.00m	-
MH-2	-1.03m	6.06m
MH-3	-1.175m	29m
MH-4	-1.00m	-
MH-5	-1.03m	6.06m
MH-6	-1.40m & -1.175	45.55m & 29m
MH-7	-1.46m	12.8m

Occupants of the building = 608 persons
GPCD = 60
Water demand for the building = 608x60=36480 gallons/day
36480 * 0.00378 = 137.9 m³/day
Required storage for underground storage = 2 days
Volume required for underground square tank
137.9 * 2 = 275.8 m³, use 280 m³
Assumed water depth = 3 m
Area required = 280/3 = 93.33 m² use 94 m²
15m W x 6.3m D x 3m H



Rectangular under ground water tank
Water depth : 3m
Free board : 0.25m
Total depth : 3.25m
Volume of water : 280 m³

Required storage for Roof storage = 1 days
Volume required for underground square tank
137.9 * 1 = 137.9 m³, use 140 m³
Assumed water depth = 2.5 m
Area required = 140/2.5 = 56 m²
4x (3.8m W x 3.8m D x 2.5m H)



4 square Roof water tank
Water depth : 2.5m
Volume of water : 36. m³ each

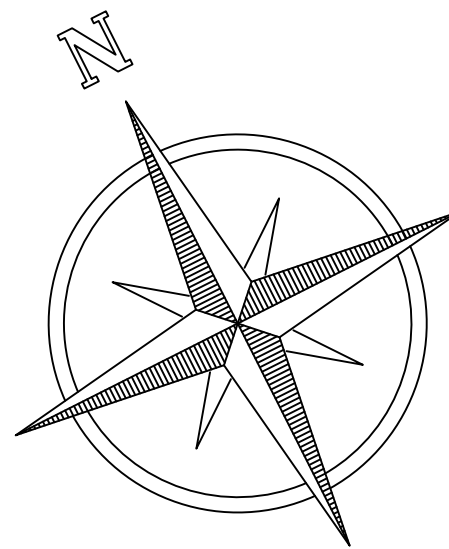
Calculations for Water Pump design :

$$P = \frac{Q H}{1200 \eta}$$

$$P = (4 \times 134.34)(71.25 + 7.125 + 5.4) / 1200 (0.7)$$
$$P = 53.6 \text{ hp} = 39.96 \text{ kW}$$



COLLEGE OF ARCH. & PLAN
DEPT. OF BUILDING ENGINEERING



ALL UNITS IN METER

SYMBOLS	SERVICES
	WATER PIPE Ø50mm
	DRAINAGE PIPE Ø250mm
	SEWERAGE PIPE Ø200mm
	ELECTRIC CABLE
	GATE VALVE
	MANHOLE
	CATCHPIT
	WATER PUMP

Design Studio (8), Technical
Project, 2024/2025

INSTRUCTORS:

Prof. M. Essam Shaawat Co.
Dr. Faris Almaziad
Dr. M. Farooqi
Dr. M. Al Mulhim
Dr. AL Tayeb
Dr. M. Al hater
Eng. Meqdad
Eng. Rehan Gamil
Eng. Noman Ashraf

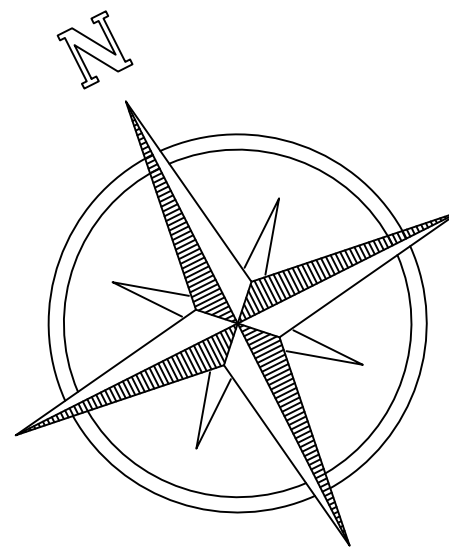
DONE BY:
ABDULELAH ALQABBANI

PROJECT:
Apartments Building

DRAWING NAME:
Site Plan

SCALE:
1:150

SHEET NUMBER:
A23



ALL UNITS IN METER

LEGENDES:

- CW RISER
- HW RISER
- CW GATE VALV
- HW SUPPLY
- CW SUPPLY

Design Studio (8), Technical
Project, 2024/2025

INSTRUCTORS:

Prof. M. Essam Shaawat Co.
Dr. Faris Almaziad
Dr. M. Farooqi
Dr. M. Al Mulhim
Dr. AL Tayeb
Dr. M. Al hater
Eng. Meqdad
Eng. Rehan Gamil
Eng. Noman Ashraf

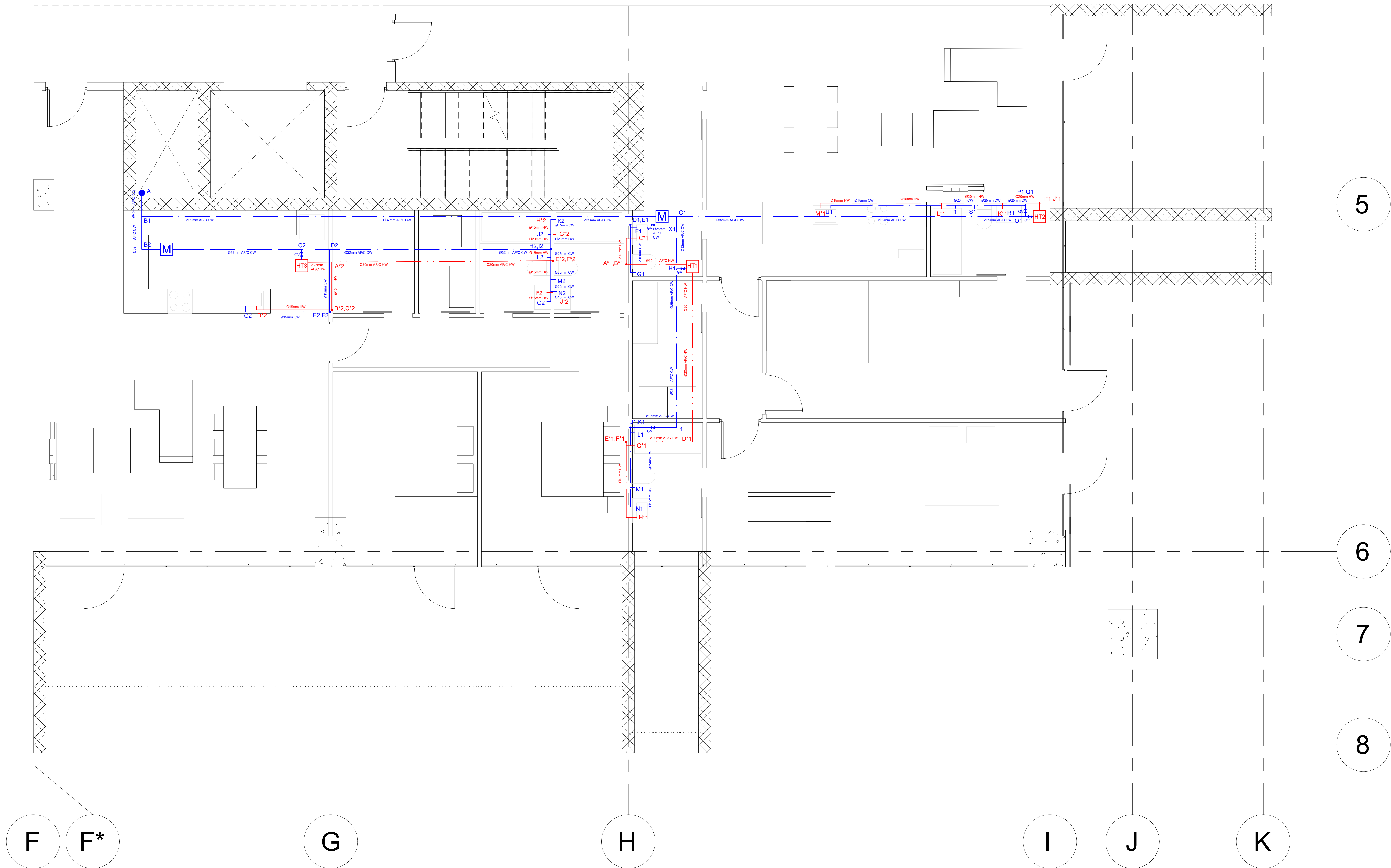
DONE BY:
ABDULELAH ALQABBANI

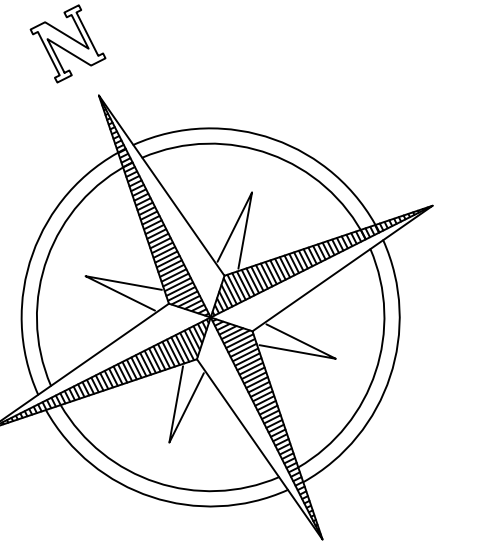
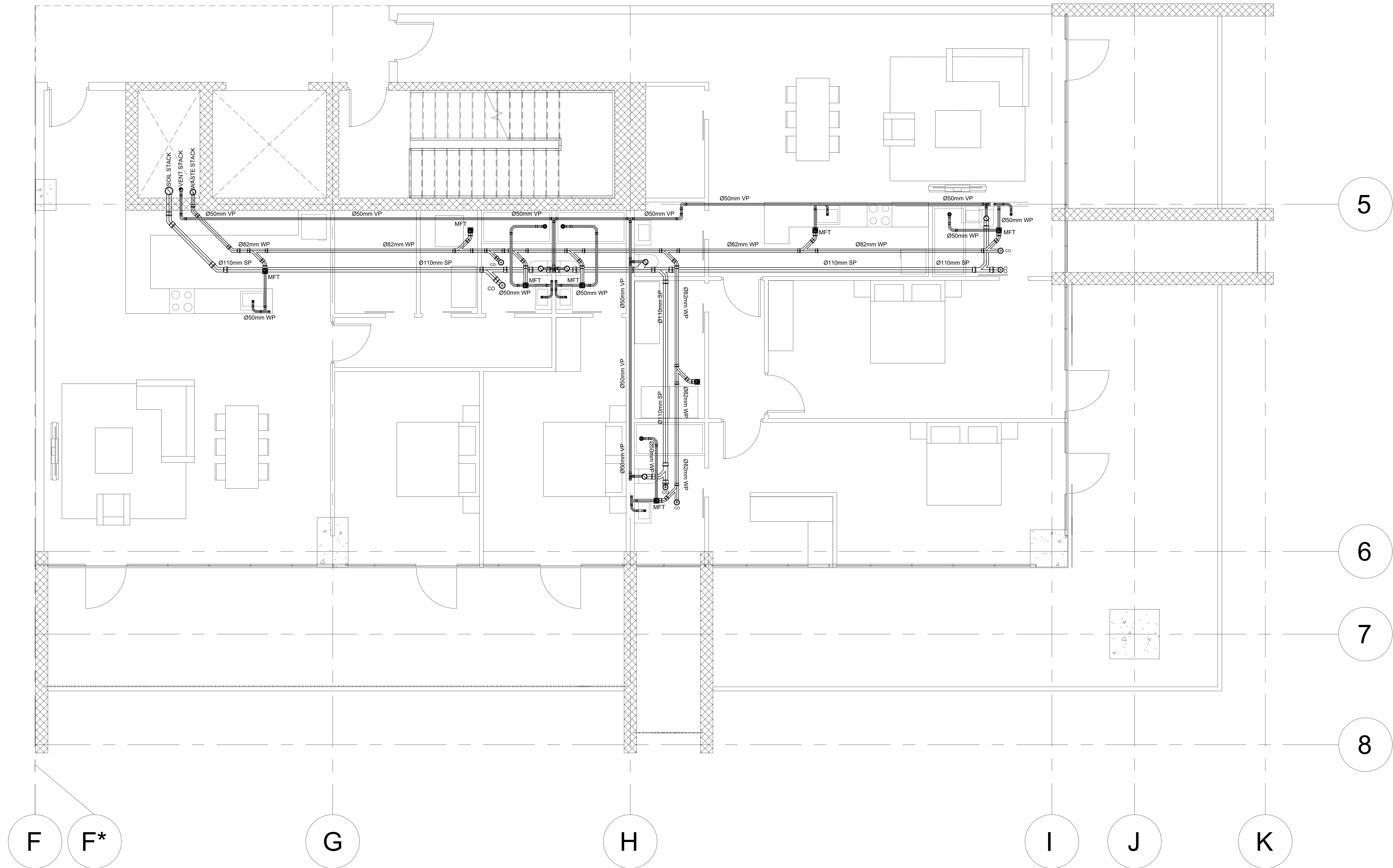
PROJECT:
Apartments Building

DRAWING NAME:
Water Supply Layout

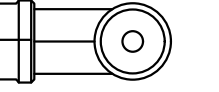
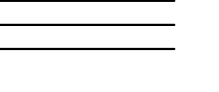
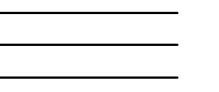
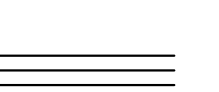
SCALE:
1:37.5

SHEET NUMBER:
A24





ALL UNITS IN METER

- MFT 
- CO 
- Ø82mm
WASTE PIPE 
- Ø110mm
SOIL PIPE 
- Ø50mm
VENT PIPE 

Design Studio (8), Technical
Project, 2024/2025

INSTRUCTORS:

Prof. M. Essam Shaawat Co.
Dr. Faris Almaziad
Dr. M. Farooqi
Dr. M. Al Mulhim
Dr. AL Tayeb
Dr. M. Al hater
Eng. Meqdad
Eng. Rehan Gamil
Eng. Noman Ashraf

DONE BY:
ABDULELAH ALQABBANI

PROJECT:
Apartments Building

DRAWING NAME:
Sewage System Layout

SCALE:
1:37.5

SHEET NUMBER:
A25

CALCULATION FOR PIPE SIZE

1 gpm = 0.000063 m³/s

pipe	Fixtures Attached	Total Fixture Units	flow Q (gpm)	ow Q (m3/s)	Calculated Diameter	Selected Diameter
GF(1)	3LAV+6WC	16.2	18.08	0.00114	31.102	15

CALCULATION FOR PIPE SIZE (COLD WATER)

1 gpm = 0.000063 m³/s

pipe	Fixtures Attached	Total Fixture Units	flow Q (gpm)	flow Q (m3/s)	Calculated Diameter	Selected Diameter
U1-T1	SINK	1	3	0.00019	12.669	15
T1-S1	SINK+SHW	1+1 = 2	5	0.00032	16.356	20
S1-R1	SINK+SHW+WC	2+2 = 4.2	8.28	0.00052	21.048	25
R1-Q1	SINK+SHW+WC+LAV	4.2+0.5 = 4.7	8.98	0.00057	21.919	25
P1-O1	SINK+SHW+WC+LAV	4.2+0.5 = 4.7	8.98	0.00057	21.919	25
HT2-O1	(HT2)	2.5	7.25	0.00046	19.695	20
O1-C1	SINK+SHW+WC+LAV+(HT2)	4.7+2.5 = 7.2	12	0.00076	25.338	32
N1-M1	LAV	0.5	3	0.00019	12.669	15
M1-L1	LAV+WC	0.5+2.2 = 2.7	7.55	0.00048	20.098	25
L1-K1	LAV+WC+SHW	2.7+3 = 5.7	10.31	0.00065	23.487	25
K1-J1	LAV+WC+SHW	2.7+3 = 5.7	10.31	0.00065	23.487	25
J1-I1	LAV+WC+SHW	2.7+3 = 5.7	10.31	0.00065	23.487	25
I1-H1	LAV+WC+SHW	2.7+3 = 5.7	10.31	0.00065	23.487	25
HT1-H1	HT1	2	5	0.00032	16.356	20
H1-X1	LAV+WC+SHW+(HT1)	5.7+2 = 7.7	12.5	0.00079	25.861	32
G1-F1	WC	2.2	5.3	0.00033	16.839	20
F1-E1	WC+LAV	2.2+0.5 = 2.7	7.55	0.00048	20.098	25
E1-D1	WC+LAV	2.2+0.5 = 2.7	7.55	0.00048	20.098	25
D1-X1	WC+LAV	2.2+0.5 = 2.7	7.55	0.00048	20.098	25
X1-C1	2LAV+2WC+SHW+(HT1)	7.7+2.7 = 10.4	14.92	0.00094	28.254	32
C1-B1	SINK+3LAV+3WC+2SHW+(HT1)+(HT2)	10.4+7.2 = 17.6	18.64	0.00117	31.580	32
K2-J2	SHW	1	3	0.00019	12.669	15
J2-I2	2SHW	2	5	0.00032	16.356	20
O2-N2	LAV	0.5	3	0.00019	12.669	15
N2-M2	2LAV	1	5	0.00032	16.356	20
M2-L2	2LAV+WC	1+2.2 = 3	6.5	0.00041	18.649	20
L2-I2	2LAV+2WC	1+4.4 = 5.4	9.92	0.00062	23.038	25
I2-H2	2LAV+2WC+2SHW	1+4.4+2 = 7.4	12.2	0.00077	25.549	32
H2-D2	2LAV+2WC+2SHW	1+4.4+2 = 7.4	12.2	0.00077	25.549	32
G2-F2	SINK	1	3	0.00019	12.669	15
F2-E2	SINK	1	3	0.00019	12.669	15
E2-D2	SINK	1	3	0.00019	12.669	15
D2-C2	2LAV+2WC+2SHW+SINK	7.4+1 = 8.4	13.16	0.00083	26.535	32
HT3-C2	(HT3)	4	8	0.00050	20.689	25
C2-B2	2LAV+2WC+2SHW+SINK+(HT3)	8.4+4 = 12.4	16.2	0.00102	29.441	32
B2-B1	2LAV+2WC+2SHW+SINK+(HT3)	8.4+4 = 12.4	16.2	0.00102	29.441	32
B1-A	5LAV+5WC+4SHW+2SINK+(HT1,2,3)	12.4+17.6 = 30	23.3	0.00147	35.308	40

CALCULATION FOR PIPE SIZE (HOT WATER)

1 gpm = 0.000063 m³/s

pipe	Fixtures Attached	Total Fixture Units	flow Q (gpm)	flow Q (m3/s)	Calculated Diameter	Calculated Diameter
M*1-L*1	SINK	1	3	0.00019	12.67	15
L*1-K*1	SINK+SHW	1+1 = 2	5	0.00032	16.36	20
K*1-J*1	SINK+SHW+LAV	1+1+0.5 = 2.5	7.25	0.00046	19.70	20
J*1-I*1	SINK+SHW+LAV	2.5	7.25	0.00046	19.70	20
I*1-HT1	SINK+SHW+LAV	2.5	7.25	0.00046	19.70	20
H*1-G*1	LAV	0.5	3	0.00019	12.67	20
G*1-F*1	LAV+SHW	0.5+1 = 1.5	6	0.00038	17.92	20
F*1-E*1	LAV+SHW	1.5	6	0.00038	17.92	20
E*1-D*1	LAV+SHW	1.5	6	0.00038	17.92	20
D*1-HT1	LAV+SHW	1.5	6	0.00038	17.92	20
C*1-B*1	LAV	0.5	3	0.00019	12.67	15
B*1-A*1	LAV	0.5	3	0.00019	12.67	15
A*1-HT1	LAV	0.5	3	0.00019	12.67	15
H*2-G*2	SHW	1	3	0.00019	12.67	15
G*2-F*2	2SHW	1+1 = 2	5	0.00032	16.36	20
J*2-I*2	LAV	0.5	3	0.00019	12.67	15
I*2-F*2	2LAV	0.5+0.5 = 1	3	0.00019	12.67	15
F*2-E*2	2SHW+2LAV	2+1 = 3	6.5	0.00041	18.65	20
E*2-A*2	2SHW+2LAV	2+1 = 3	6.5	0.0004095	18.65	20
D*2-C*2	SINK	1	3	0.000189	12.67	15
C*2-B*2	SINK	1	3	0.000189	12.67	15
B*2-A*2	SINK	1	3	0.000189	12.67	15
A*2-HT3	2SHW+2LAV+SINK	2+1+1 = 4	8	0.000504	20.69	25

Cold Water Supply System - Head Loss Calculations

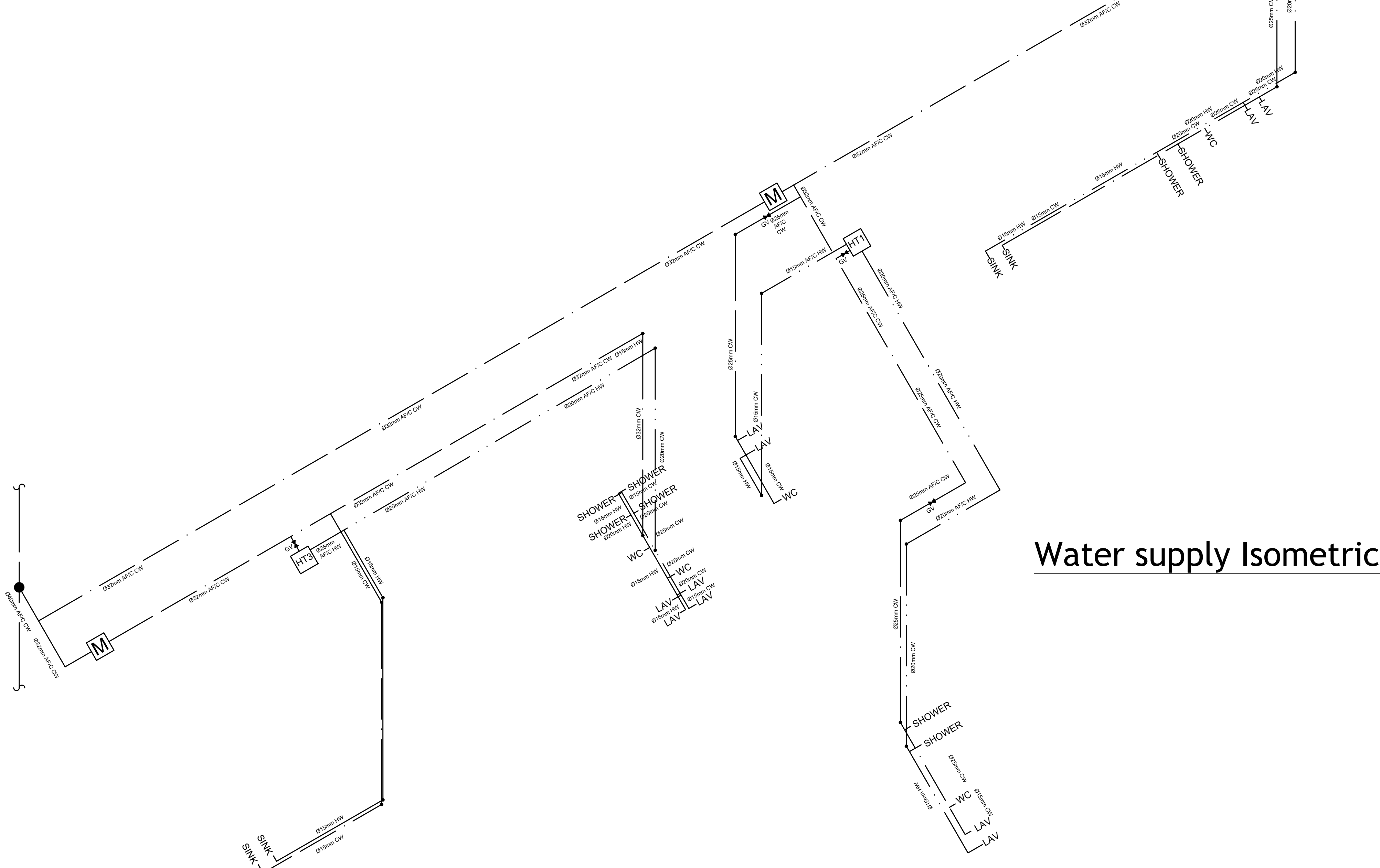
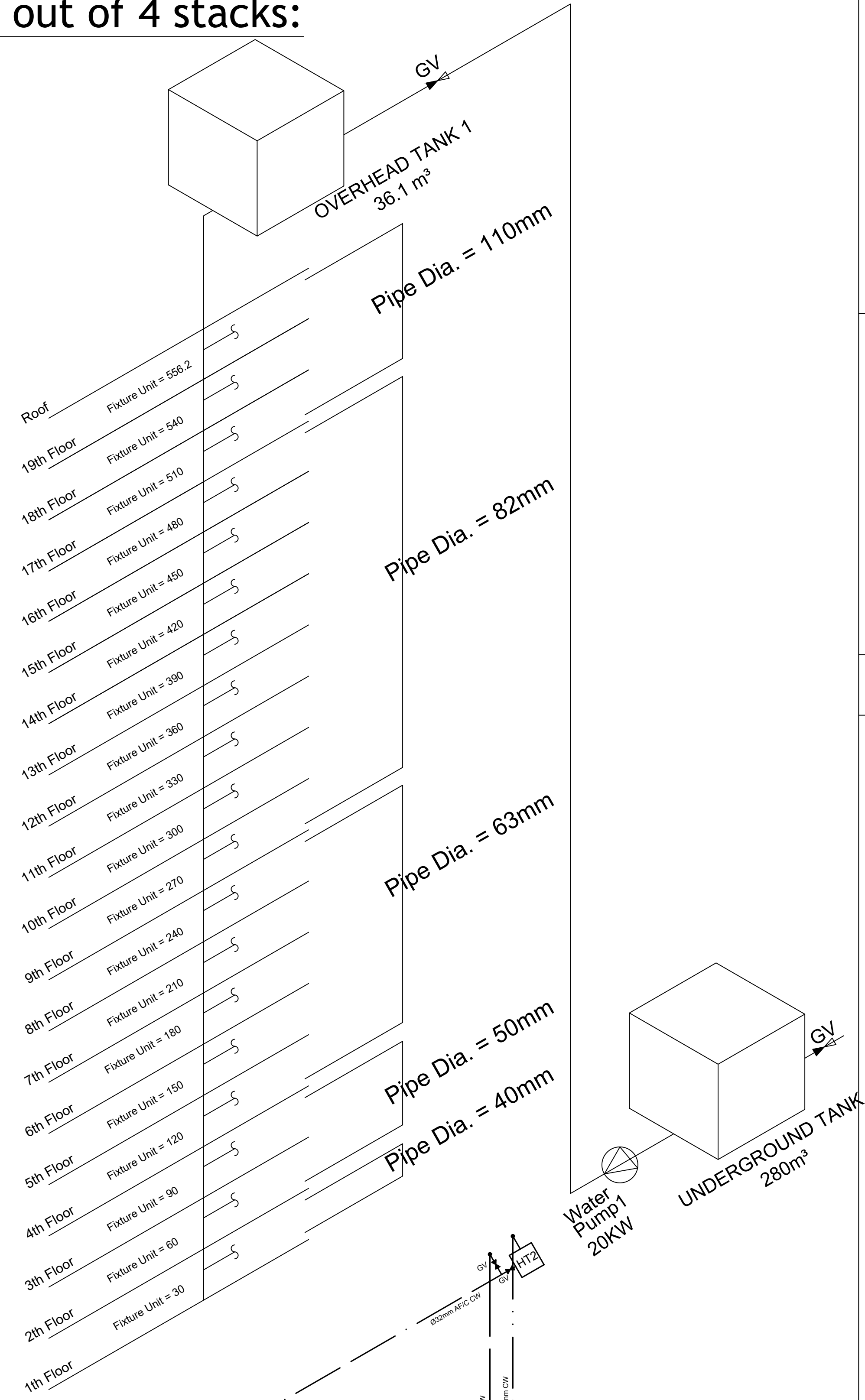
Pipe Section	Pipe Length	Selected D (mm)	Major HL - h _f (m)	HL Check h _f /L (m/m)	Fittings Attached	Minor HL - h _m (m)	Total HL - h _f +h _m (m)
U1-T1	3	15	0.2163	0.0721	90 Elbow k=1.5	0.1720	0.388
T1-S1	0.46	20	0.0249	0.0541	Line T k=0.9	0.0619	0.087
S1-R1	0.93	25	0.0402	0.0433	Line T k=0.9	0.0619	0.102
R1-Q1	0.3	25	0.0130	0.0433	Line T k=0.9	0.0619	0.075
Q1-P1	3	25	0.1298	0.0433	90 Elbow k=1.5	0.0619	0.192
P1-O1	2.8	25	0.1211	0.0433	90 Elbow k=1.5, GV k=0.15	0.0619	0.183
O1-C1	8.4	32	0.2839	0.0338	90 Elbow k=1.5	0.1720	0.456
N1-M1	0.46	15	0.0332	0.0721	90 Elbow k=1.5	0.1720	0.205
M1-L1	1.3	25	0.0562	0.0433	Line T k=0.9	0.0619	0.118
L1-K1	0.12	25	0.0052	0.0433	Line T k=0.9	0.1720	0.177
K1-J1	3	25	0.1298	0.0433	90 Elbow k=1.5	0.4044	0.534
J1-I1	1.1	25	0.0476	0.0433	90 Elbow k=1.5, GV k=0.15	0.1720	0.220
I1-H1	3.8	25	0.1644	0.0433	90 Elbow k=1.5	0.3058	0.470
H1-X1	1.5	32	0.0507	0.0338	Line T k=0.9	0.1720	0.223
G1-F1	1.1	15	0.0793	0.0721	90 Elbow k=1.5	0.0619	0.141
F1-E1	0.05	25	0.0022	0.0433	Line T k=0.9	0.0619	0.064
E1-D1	3	25	0.1298	0.0433	90 Elbow k=1.5	0.0619	0.192
D1-X1	1.1	25	0.0476	0.0433	90 Elbow k=1.5, GV k=0.15	0.0619	0.110
X1-C1	0.2	32	0.0068	0.0338	Line T k=0.9	0.0619	0.069
C1-B1	13	32	0.4393	0.0338	Line T k=0.9, Water Meter k=7	0.0619	0.501
K2-J2	0.37	15	0.0267	0.0721	90 Elbow k=1.5	0.1720	0.199
J2-I2	35	20	1.8925	0.0541	Line T k=0.9	0.6881	2.581
O2-N2	0.25	15	0.0180	0.0721	90 Elbow k=1.5	0.1720	0.190
N2-M2	0.29	20	0.0157	0.0541	Line T k=0.9	0.0619	0.078
M2-L2	0.53	20	0.0287	0.0541	Line T k=0.9	0.0619	0.091
L2-I2	0.2	25	0.0087	0.0433	Line T k=0.9	0.0619	0.071
I2-H2	3	32	0.1014	0.0338	Line T k=0.9	0.0619	0.163
H2-D2	5.4	32	0.1825	0.0338	90 Elbow k=1.5, GV k=0.15	0.0619	0.244
G2-F2	2	32	0.0676	0.0338	90 Elbow k=1.5	0.1720	0.240
F2-E2	3	32	0.1014	0.0338	90 Elbow k=1.5	0.1720	0.273
E2-D2	1.5	32	0.0507	0.0338	90 Elbow k=1.5	0.0619	0.113
D2-C2	0.7	32	0.0237	0.0338	Line T k=0.9	0.1720	0.196
C2-B2	3.9	32	0.1318	0.0338	Line T k=0.9, Water Meter k=7	0.4044	0.536
B2-B1	0.8	32	0.0270	0.0338	90 Elbow k=1.5	0.1720	0.199
B1-A	0.5	40	0.0135	0.0270	Line T k=0.9	0.3058	0.319
							10.015m

CALCULATION FOR PIPE SIZE

1 gpm = 0.000063 m³/s

pipe	Total Fixture Units	ow Q (gpm)	flow Q (m3/s)	Calculated Diameter	Selected Diameter
19	556.2	134.34	0.00846	84.780	110
18	540	131.36	0.00828	83.834	110
17	510	125.84	0.00793	82.054	110
16	480	120.2	0.00757	80.194	82
15	450	114.5	0.00721	78.270	82
14	420	108.8	0.00685	76.296	82
13	390	103	0.00649	74.235	82
12	360	97	0.00611	72.040	82
11	330	91	0.00573	69.777	82
10	300	85	0.00536	67.437	82
9	270	79	0.00498	65.013	82
8	240	73	0.00460	62.496	63
7	210	67	0.00422	59.872	63
6	180	61	0.00384	57.129	63
5	150	54.75	0.00345	54.123	63
4	120	48	0.00302	50.677	63
3	90	41	0.00258	46.836	50
2	60	32	0.00202	41.378	50
1	30	23.3	0.00147	35.308	40

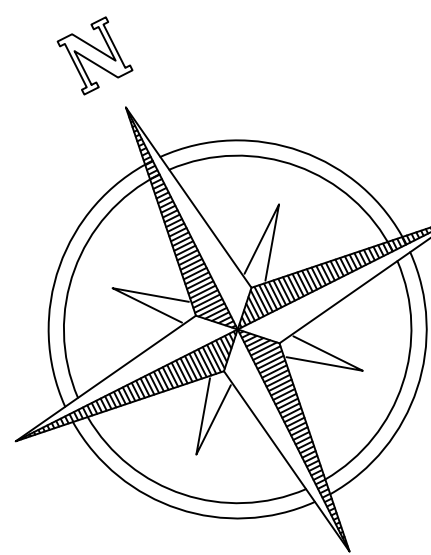
1 Riser out of 4 stacks:



Water supply Isometric



COLLEGE OF ARCH. & PLAN
DEPT. OF BUILDING ENGINEERING



ALL UNITS IN METER

LEGENDES:

- CW RISER
- HW RISER
- ▽ GATE VALVE
- HW SUPPLY
- CW SUPPLY

Design Studio (8), Technical
Project, 2024/2025

INSTRUCTORS:

Prof. M. Essam Shaawat Co.
Dr. Faris Almaziad
Dr. M. Farooqi
Dr. M. Al Mulhim
Dr. AL Tayeb
Dr. M. Al hater
Eng. Meqdad
Eng. Rehan Gamil
Eng. Noman Ashraf

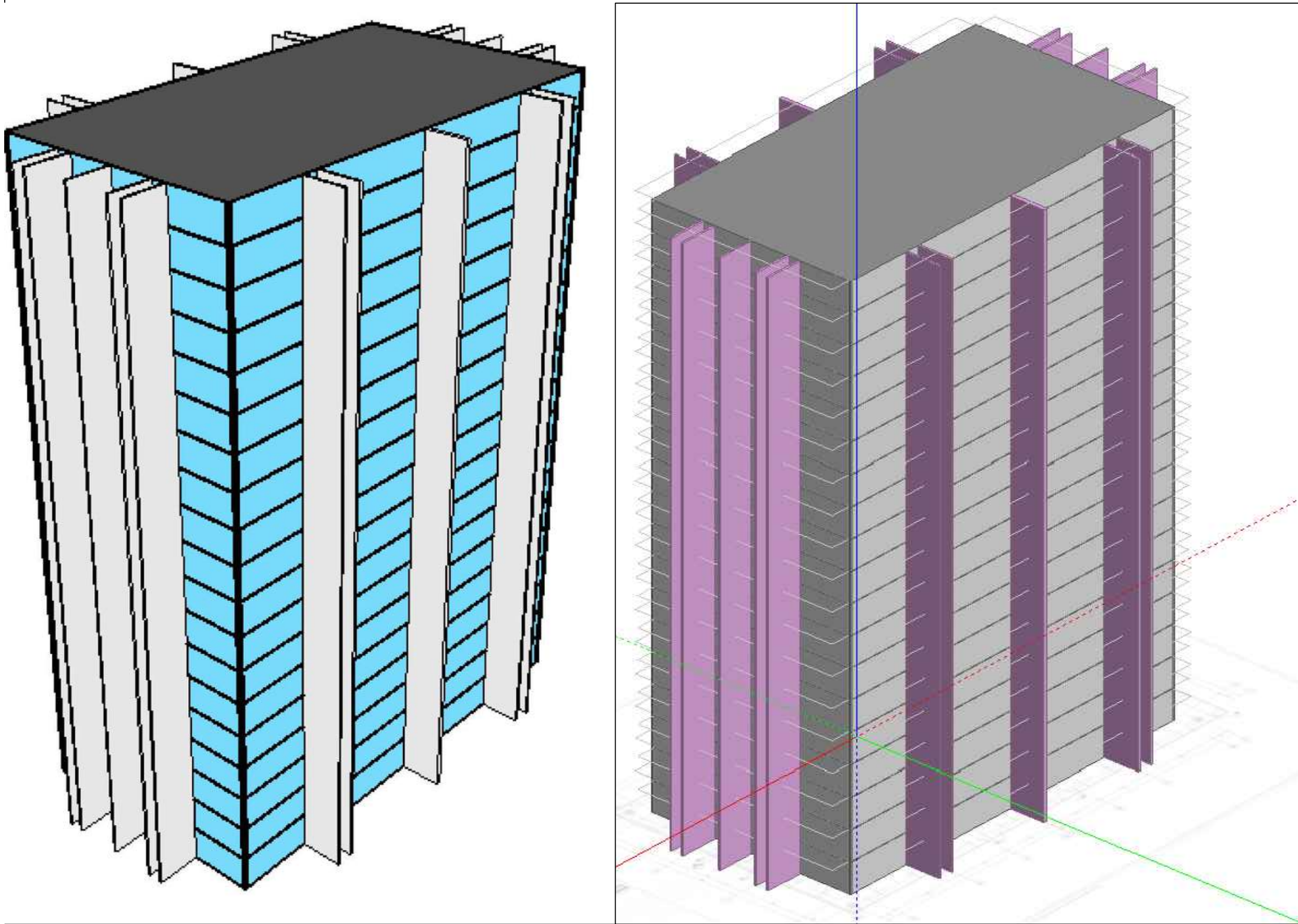
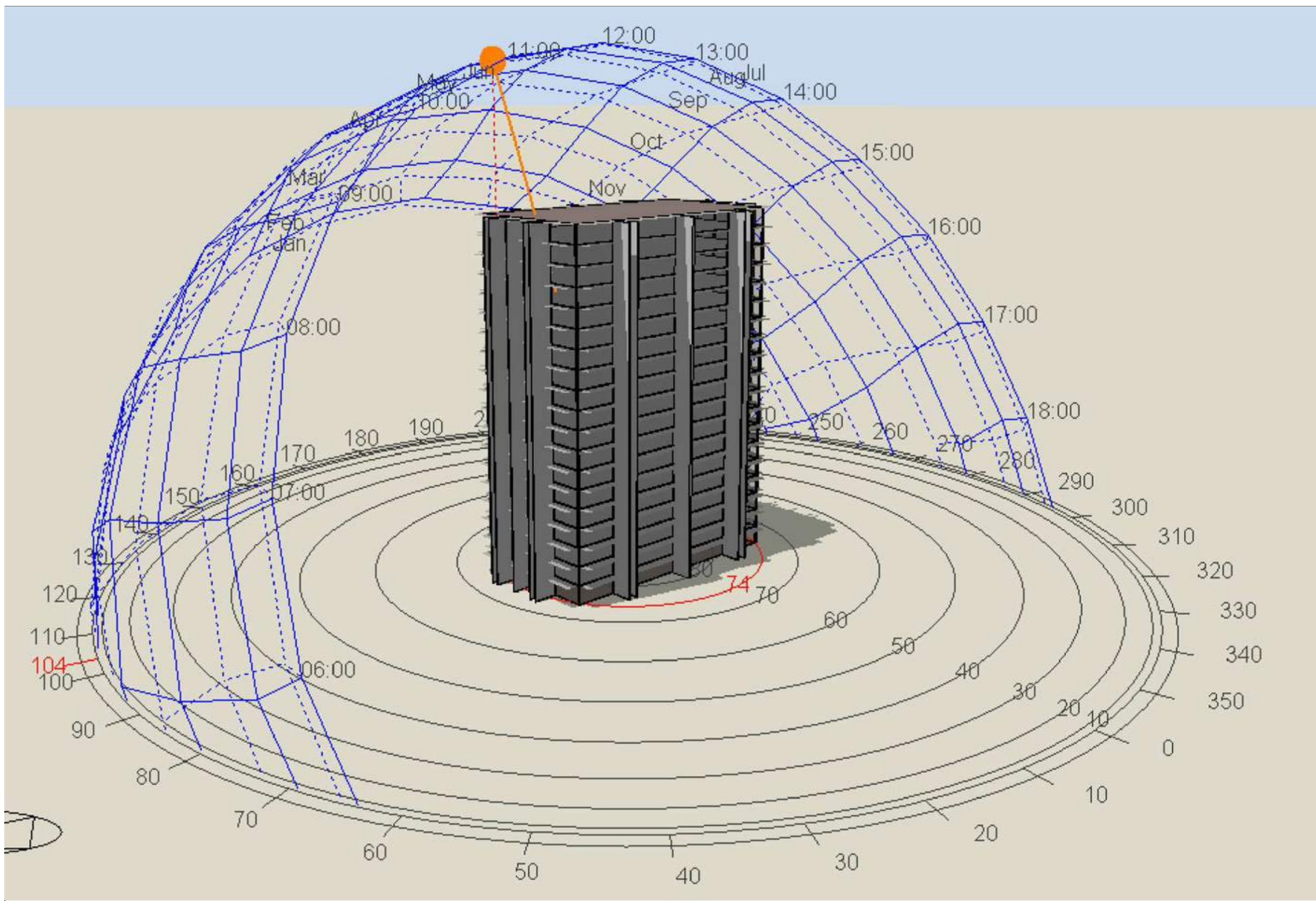
DONE BY:
ABDULELAH ALQABBANI

PROJECT:
Apartments Building

DRAWING NAME:
Plumbing Calculations

SCALE:

SHEET NUMBER:
A26



BASE CASE ROOF

Outer surface

50.00mm Gravel (not to scale)

50.00mm MW Glass Wool (rolls)

50.00mm Bitumen, pure

300.00mm Cast Concrete

Inner surface

U-Value (W/m2-K) 0.479

GLAZING

Layers

Number layers 1

Outermost pane Generic Clear 6mm glass

Pane type

Flip layer

Radiance Daylighting

U-Value (W/m2-K) 5.801

Total solar transmission (SHGC) 0.828

LIGHTING

Lighting Template

Template Fluorescent (no details)

General Lighting

On

Normalised power density (W/m2-100 lux) 5.0000

Schedule occupants APARTEMENTS SCHEDULE

Luminaire type 3-Recessed

Return air fraction 0.000

Radiant fraction 0.370

Visible fraction 0.180

Convective fraction 0.450

COOLING

Cooling

Cooled

Cooling system Default

Fuel 1-Electricity from grid

Cooling system seasonal CoP 1.800

PASSIVE SOLUTIONS CONSTRUCTION & OPENINGS CASE

SHADING

Shading types used:

- 1-Overhangs
- 2-Fins
- 3-Double Skin

Shading

Window shading

Type MASHRABYA

Position 3-Outside

Control type 1-Always on

Slat Angle Control

Local shading

Type 2.5m Overhang

Edit window shade - MASHRABYA

Window shading

General Slat data Cost and Carbon

Slat Properties

Blind-to-glass distance (m) 0.0500

Slat orientation Horizontal

Slat width (m) 0.04000

Slat separation (m) 0.10000

Slat thickness (m) 0.04000

Slat conductivity (W/m-K) 16.000

Slat angle (°) 0.0

Minimum slat angle (°) 0

Maximum slat angle (°) 0

Slat Solar Properties

Slat Visible Properties

Slat visible transmittance 0.000

Slat visible reflectance, front side 0.700

Slat visible reflectance, back side 0.700

Slat IR (Thermal) Properties

Openings

Blind top opening multiplier 0.500

Blind bottom opening multiplier 0.500

Blind left-side opening multiplier 0.500

Blind right-side opening multiplier 0.500

ROOF

Outer surface

50.00mm Gravel (not to scale)

1.50mm Insuwrap PVC 1500 W/MR(not to scale)

100.00mm Foam Board Thermal insulation

1.50mm Polyethylene / Polythene, high density(not to scale)

1.50mm Insuwrap PVC 1500 W/MR(not to scale)

100.00mm Cast Concrete (Lightweight)

5.00mm Bitumen, felt/sheet(not to scale)

2.00mm Bitumen, pure(not to scale)

300.00mm Concrete, Reinforced (with 1% steel)

Inner surface

U-Value (W/m2-K) 0.197

GLAZING

DOUBLE GLAZED WITH LOW-E
COATING ON BOTH FILLED WITH
XENON FOR OPTIMAL INSULATION

Layers

Number layers 2

Outermost pane Generic ECREF-2 COLORED 6M

Pane type

Flip layer

Window gas 1

Window gas type XENON 13MM

Innermost pane Generic ECREF-2 COLORED 6M

Pane type

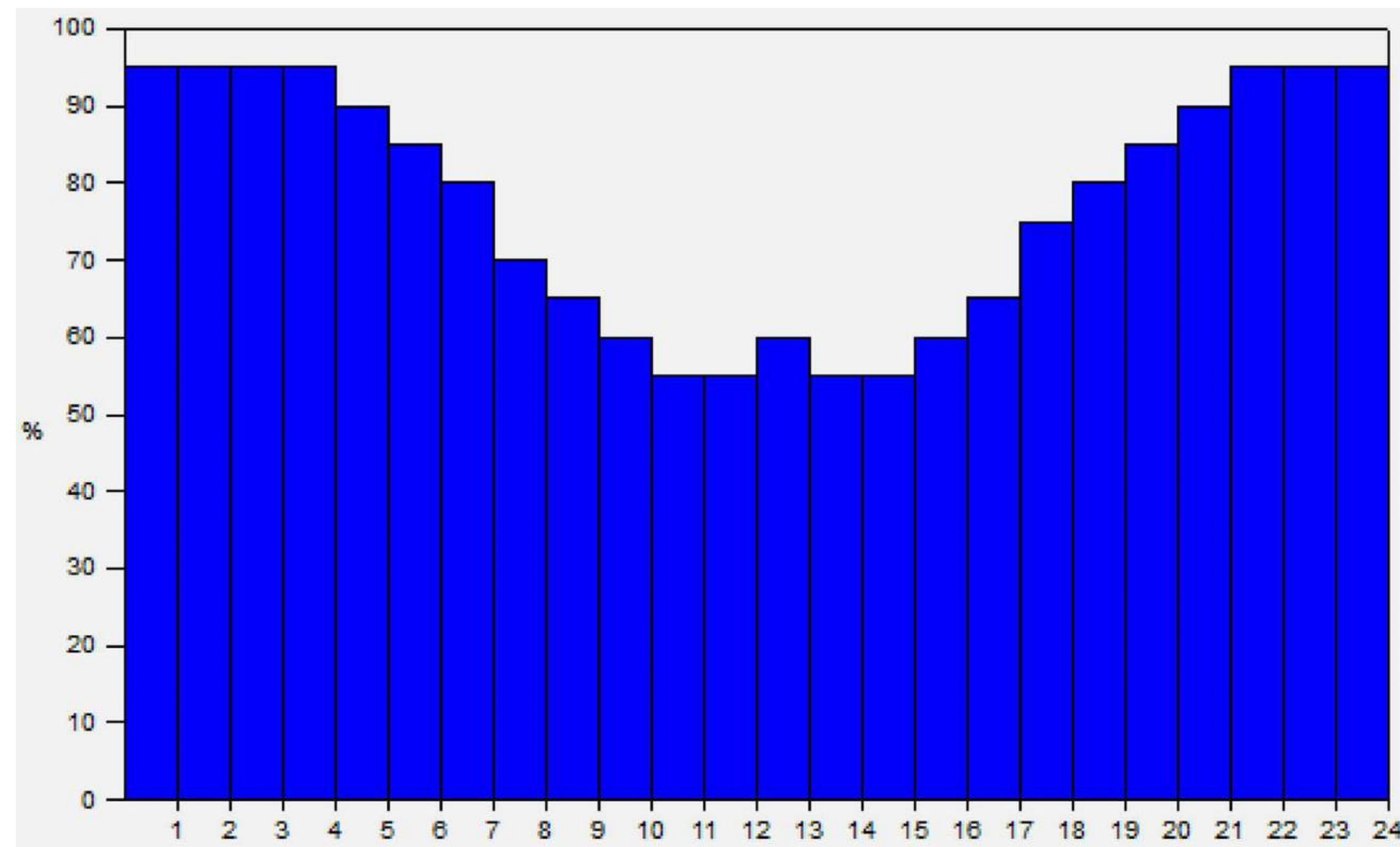
Flip layer

Radiance Daylighting

U-Value (W/m2-K) 1.046

Total solar transmission (SHGC) 0.098

ACTIVITY IS CONSTANT



Occupancy

Occupancy density (people/m2) 0.1000

Schedule occupants APARTEMENTS SCHEDULE

Metabolic

Activity Bedroom

Factor (Men=1.00, Women=0.85, Children=...

CO2 generation rate (m3/s-W) 0.0000000382

Clothing

Clothing schedule definition 1-Generic summer and winter clothing

Winter clothing (clo) 1.00

Summer clothing (clo) 0.50

Comfort Radiant Temperature Weighting

Contaminant Generation and Removal

Holidays

DHW

Consumption rate (l/m2-day) 0.100

Environmental Control

Heating Setpoint Temperatures

Heating (°C) 20.0

Heating set back (°C) 18.0

Cooling Setpoint Temperatures

Cooling (°C) 24.0

Cooling set back (°C) 28.0

Humidity Control

RH Humidification setpoint (%) 40.0

RH Dehumidification setpoint (%) 60.0

Ventilation Setpoint Temperatures

Minimum Fresh Air

Fresh air (l/s-person) 2.500

Mech vent per area (l/s-m2) 0.300

Lighting

Computers

Office Equipment

Miscellaneous

On

Power density (W/m2) 4.00

Schedule MISC APARTEMENTS SCHEDULE

Fuel 1-Electricity from grid

Fraction lost 0.100000

Latent fraction 0.000000

Radiant fraction 0.200000

ACTIVE SOLUTIONS HVAC & LIGHTING CASE

LIGHTING

Lighting Template

Template LED

General Lighting

On

Normalised power density (W/m2-100 lux) 2.5000

Schedule occupants APARTEMENTS SCHEDULE

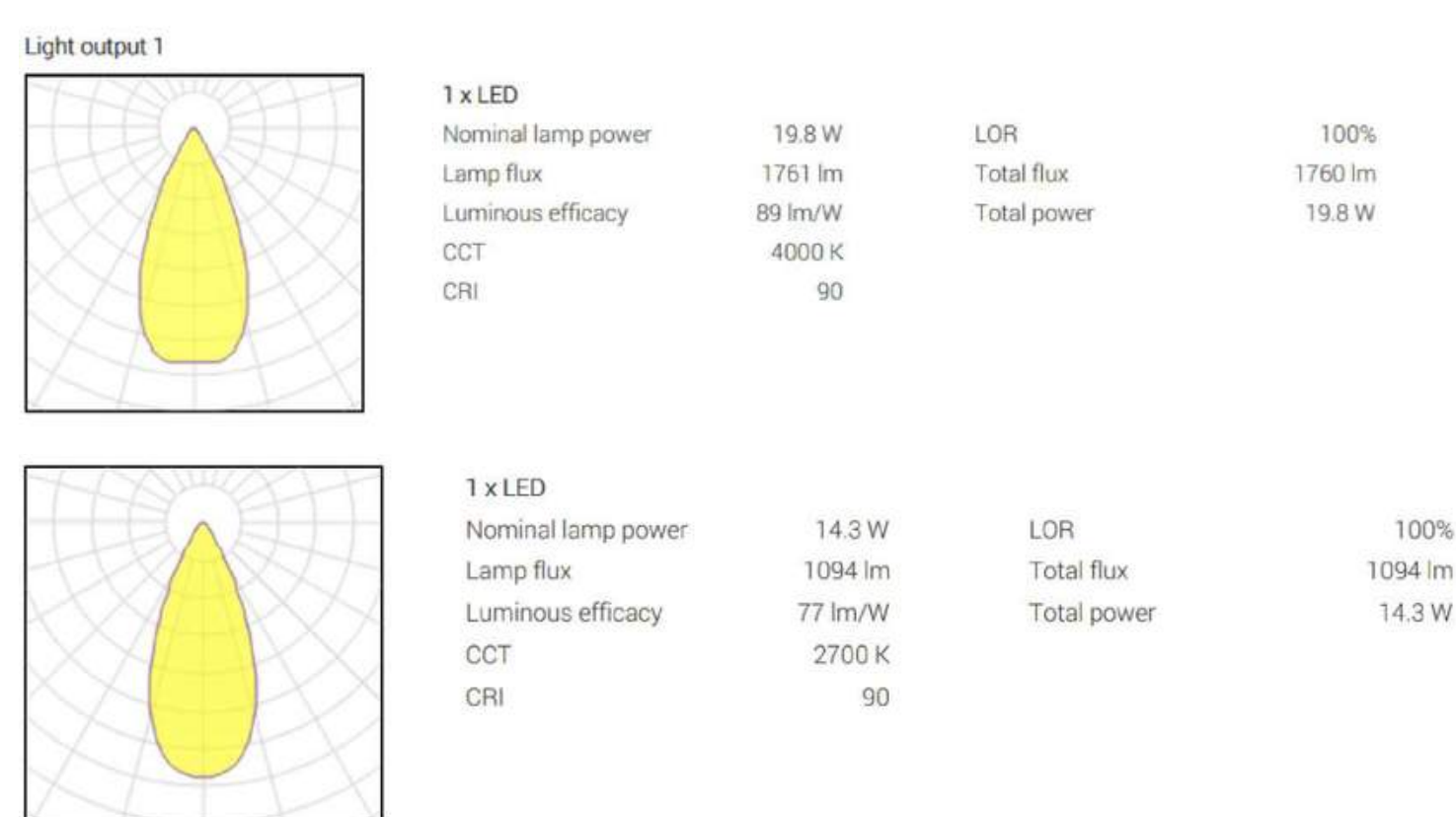
Luminaire type 3-Recessed

Return air fraction 0.000

Radiant fraction 0.370

Visible fraction 0.180

Convective fraction 0.450



COOLING

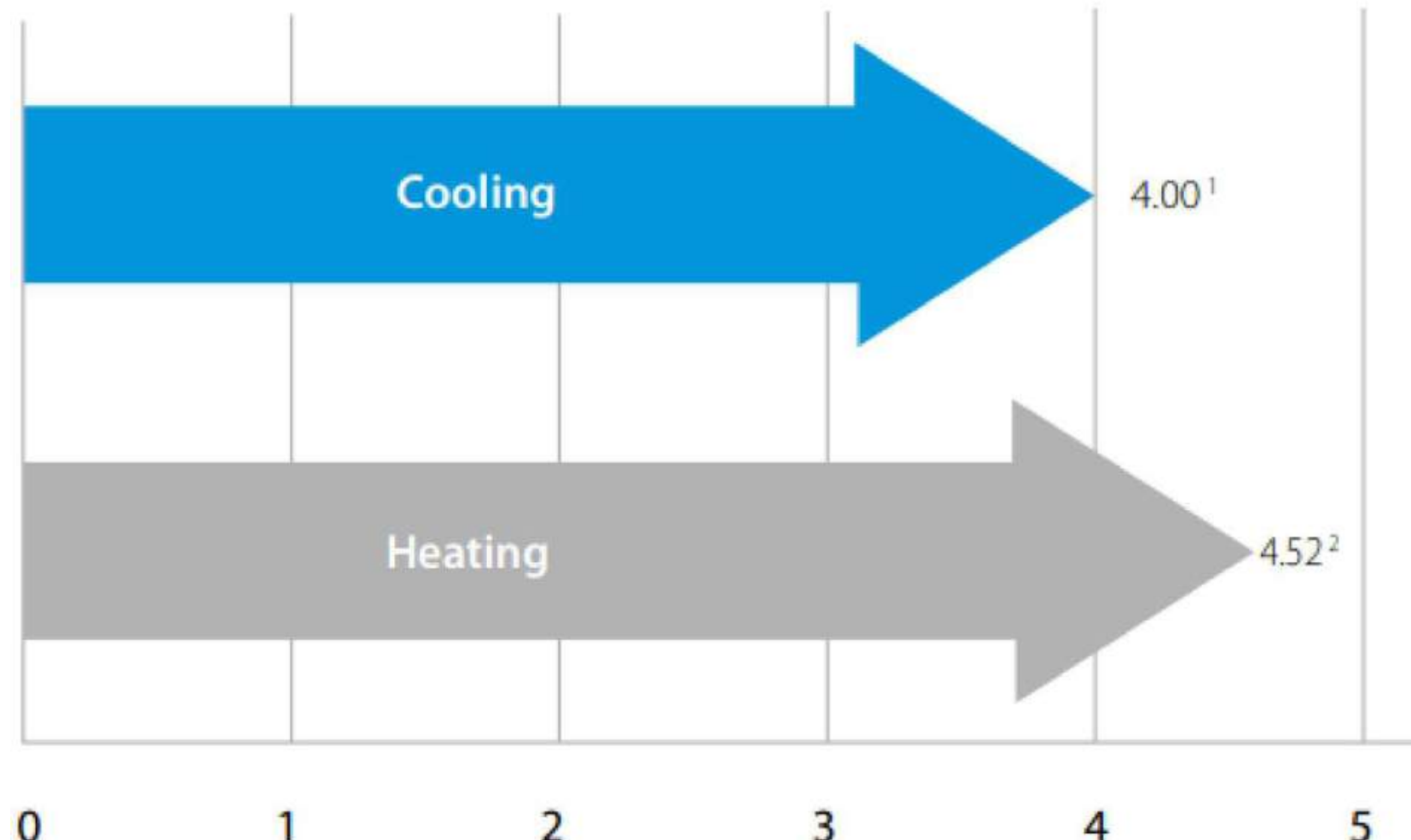
Cooling

Cooled

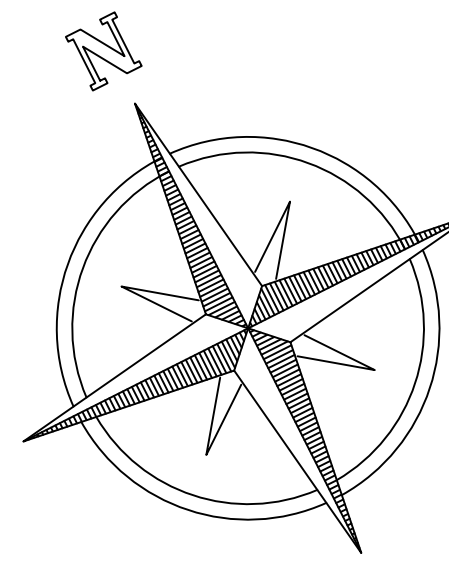
Cooling system Default

Fuel 1-Electricity from grid

Cooling system seasonal CoP 4.000



COLLEGE OF ARCH. & PLAN
DEPT. OF BUILDING ENGINEERING



ALL UNITS IN METER



كود البناء السعودي
Saudi Building Code

SBC 601 & 501

Design Studio (8), Technical
Project, 2024/2025

INSTRUCTORS:

Prof. M. Essam Shaawat Co.
Dr. Faris Almaziad
Dr. M. Farooqi
Dr. M. Al Mulhim
Dr. AL Tayeb
Dr. M. Al hater
Eng. Meqdad
Eng. Rehan Gamil
Eng. Noman Ashraf

DONE BY:
ABDULELAH ALQABBANI

PROJECT:
Apartments Building

DRAWING NAME:
DESIGN BUILDER
INPUTS

SCALE:

--

SHEET NUMBER:
A27

BASE CASE

PASSIVE CASE

ACTIVE CASE

BEST CASE

BASE CASE

BEST CASE

COOLING LOADS (kW)

HEAT BALANCE (kWh)

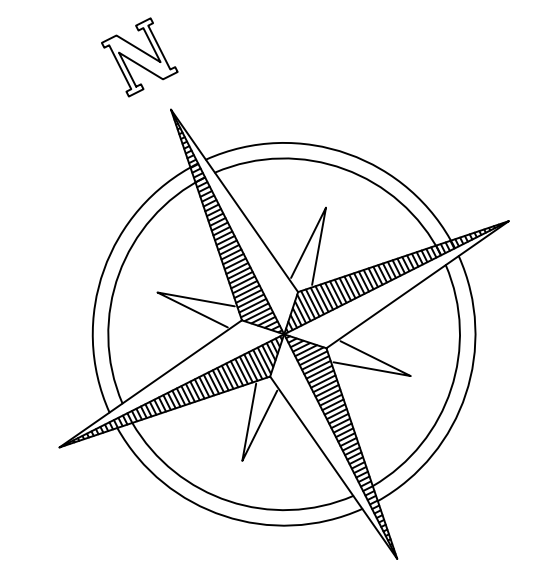
FUEL (kWh)

CASES COMPARISON

IMPROVEMENT



COLLEGE OF ARCH. & PLAN
DEPT. OF BUILDING ENGINEERING



ALL UNITS IN METER



كود البناء السعودي
Saudi Building Code

SBC 601 & 501

Design Studio (8), Technical
Project, 2024/2025

INSTRUCTORS:

Prof. M. Essam Shaawat Co.
Dr. Faris Almaziad
Dr. M. Farooqi
Dr. M. Al Mulhim
Dr. AL Tayeb
Dr. M. Al hater
Eng. Meqdad
Eng. Rehan Gamil
Eng. Noman Ashraf

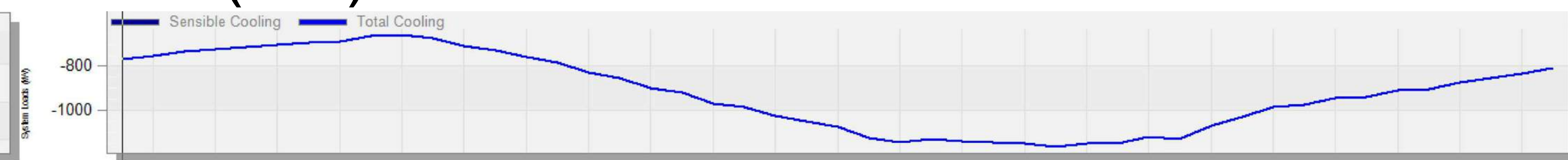
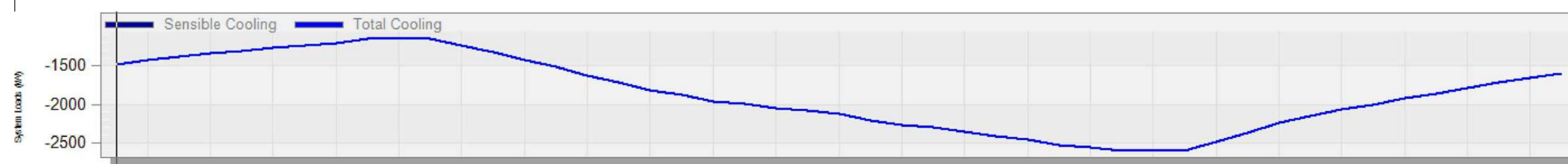
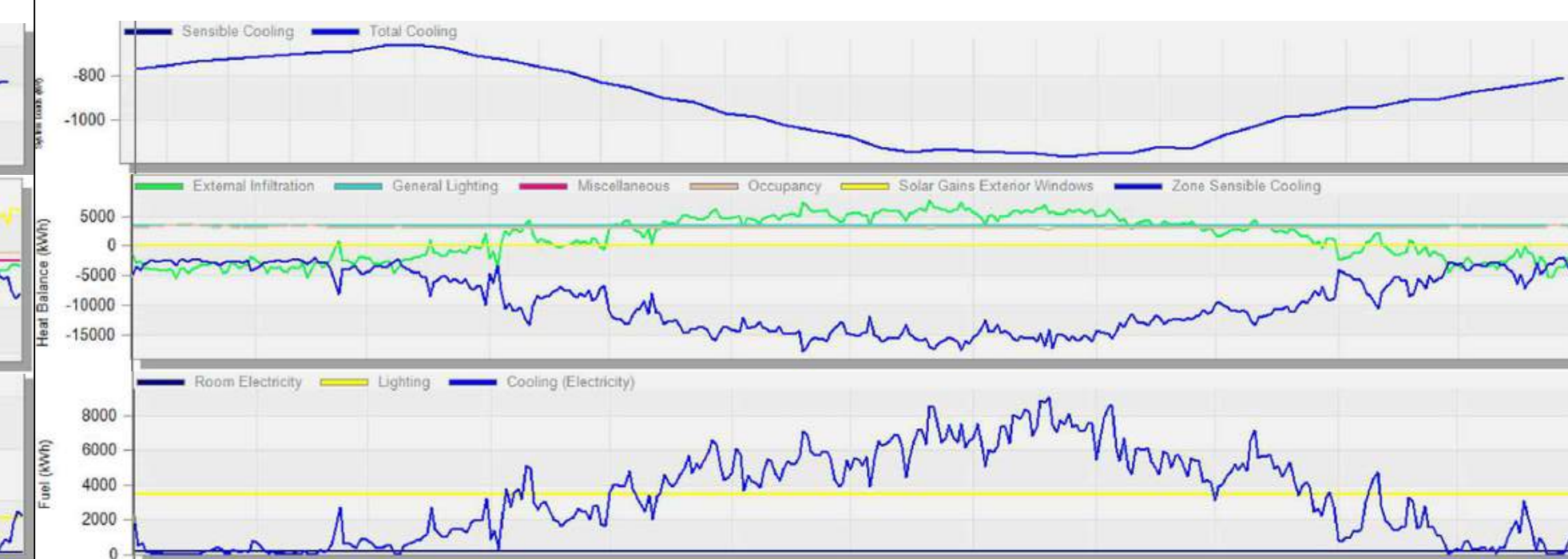
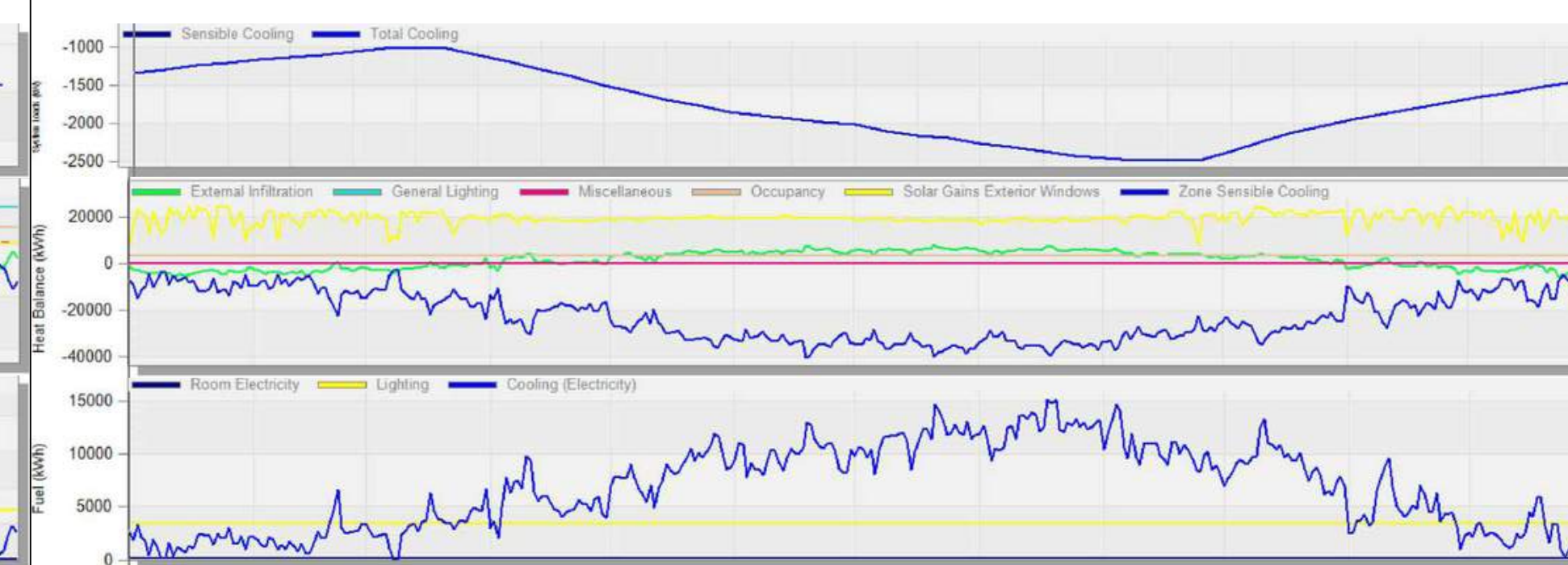
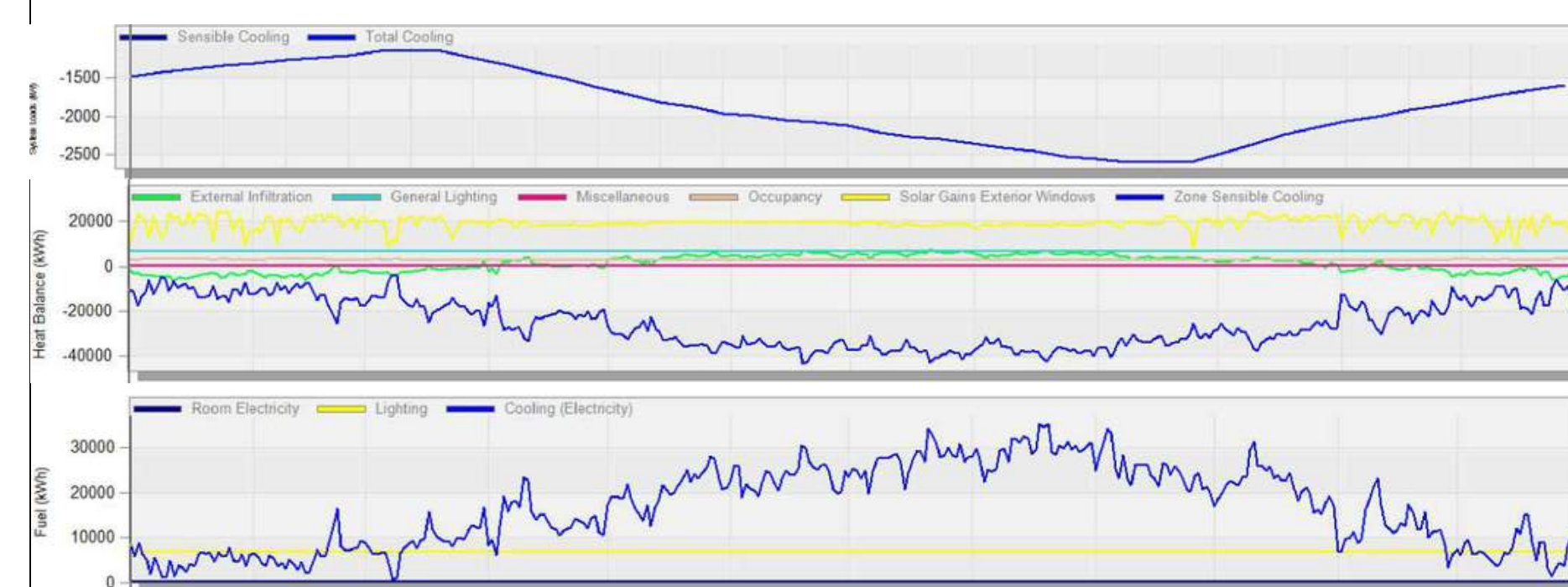
DONE BY:
ABDULELAH ALQABBANI

PROJECT:
Apartments Building

DRAWING NAME:
DESIGN BUILDER
OUTPUTS

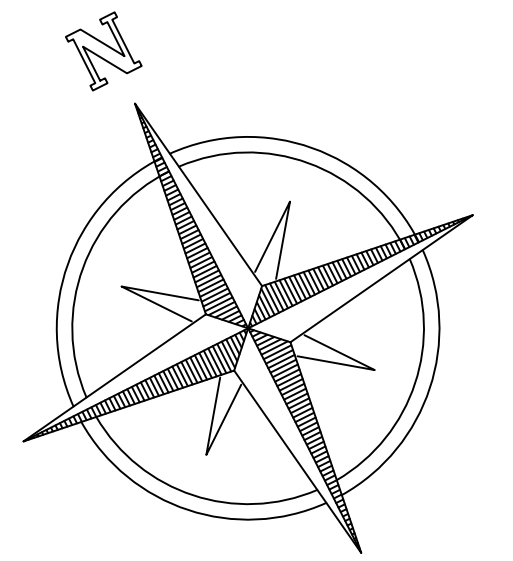
SCALE:
--

SHEET NUMBER:
A28



Case	Energy (kWh/m²)	Cooling Load (W/m²)	Flow Rate (L/s·m²)	Lighting P.D. (W/m²/100lux)	COP	Roof U-Value (W/m²·K)	Glazing U-Value (W/m²·K)	SHGC	Shading	Design Capacity (kW)
Base	795.71	125.9	8.41	5	1.8	0.429	5.801	0.828	None	3135.15
Passive	588.08	62.6	4.04	5	1.8	0.197	1.046	0.098	Yes	1557.98
Active	591.62	120.9	8.07	2.5	4.0	0.429	5.801	0.828	None	3135.15
Best (Merged)	374.34	57.0	3.66	2.5	4.0	0.197	1.046	0.098	Yes	1418.75

Parameter	Base Case	Best Case	SBC Requirement	%Improvement (Base → Best)	%Better Than SBC
Roof U-Value (W/m²·K)	0.429	0.197	0.272	Down 54.1%	27.6% Better
Glazing U-Value (W/m²·K)	5.801	1.046	1.870	Down 81.9%	44.1% Better
Glazing SHGC	0.828	0.098	0.25	Down 88.2%	60.8% Better
Energy Use (kWh/m²)	795.71	374.34	----	Down 53.0%	----
Cooling Load (W/m²)	125.9	57.0	----	Down 54.7%	----
Design Capacity (kW)	3135.15	1418.75	----	Down 54.7%	----
Flow Rate (L/s·m²)	8.41	3.66	----	Down 56.5%	----



ALL UNITS IN METER

LEGENDES:

SUPPLY GRILL	
RETURN GRILL	
FAHU-DUCT	
FLEXIBLE DUCT	
FAN COIL UNIT	
EXHAUST FAN	
DRAINAGE PIPE	
SUPPLY PIPE	
RETURN PIPE	

Design Studio (8), Technical
Project, 2024/2025

INSTRUCTORS:

Prof. M. Essam Shaawat Co.
Dr. Faris Almaziad
Dr. M. Farooqi
Dr. M. Al Mulhim
Dr. AL Tayeb
Dr. M. Al hater
Eng. Meqdad
Eng. Rehan Gamil
Eng. Noman Ashraf

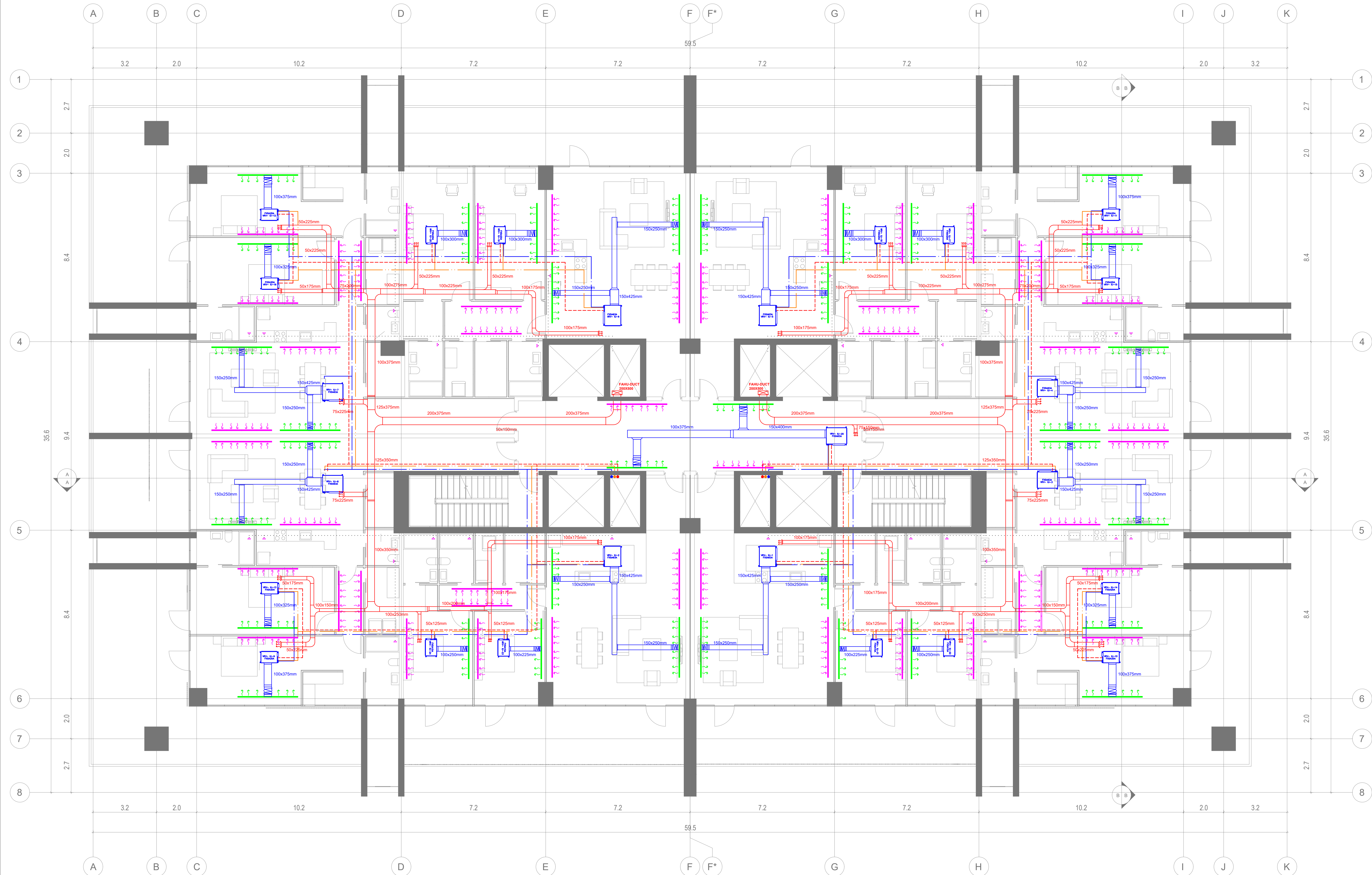
DONE BY:
ABDULELAH ALQABBANI

PROJECT:
Apartments Building

DRAWING NAME:
Typcal Plan

SCALE:
1:75

SHEET NUMBER:
A29



DesignTools DuctSizer...	
Exit Print Clear Units About	
20°C Air STP	
Fluid density	1.2014 kg/m³
Fluid viscosity	0.0643 kg/m·h
Specific Heat	1.0048 kJ/kg·°C
Energy factor	1.21 W/°C·L/s
☒ Flow rate	425 L/s
☒ Head loss	0.85 Pa/m
☐ Velocity	4.784 m/s
☐ Equivalent diameter	336.3 mm
Duct size	200 mm X 500 mm
Equivalent Diameter	337.03 mm
Flow Area	0.0888 m²
Fluid velocity	4.786 m/s
Reynolds Number	108.281
Friction factor	0.02067
Velocity Pressure	13.7632 Pa
Head Loss	0.847 Pa/m



DesignTools DuctSizer...	
Exit Print Clear Units About	
20°C Air STP	
Fluid density	1.2014 kg/m³
Fluid viscosity	0.0643 kg/m·h
Specific Heat	1.0048 kJ/kg·°C
Energy factor	1.21 W/°C·L/s
☒ Flow rate	237.9 L/s
☒ Head loss	0.85 Pa/m
☐ Velocity	4.143 m/s
☐ Equivalent diameter	270.4 mm
Duct size	150 mm X 425 mm
Equivalent Diameter	267.19 mm
Flow Area	0.0561 m²
Fluid velocity	4.241 m/s
Reynolds Number	76.226
Friction factor	0.02215
Velocity Pressure	10.8851 Pa
Head Loss	0.897 Pa/m



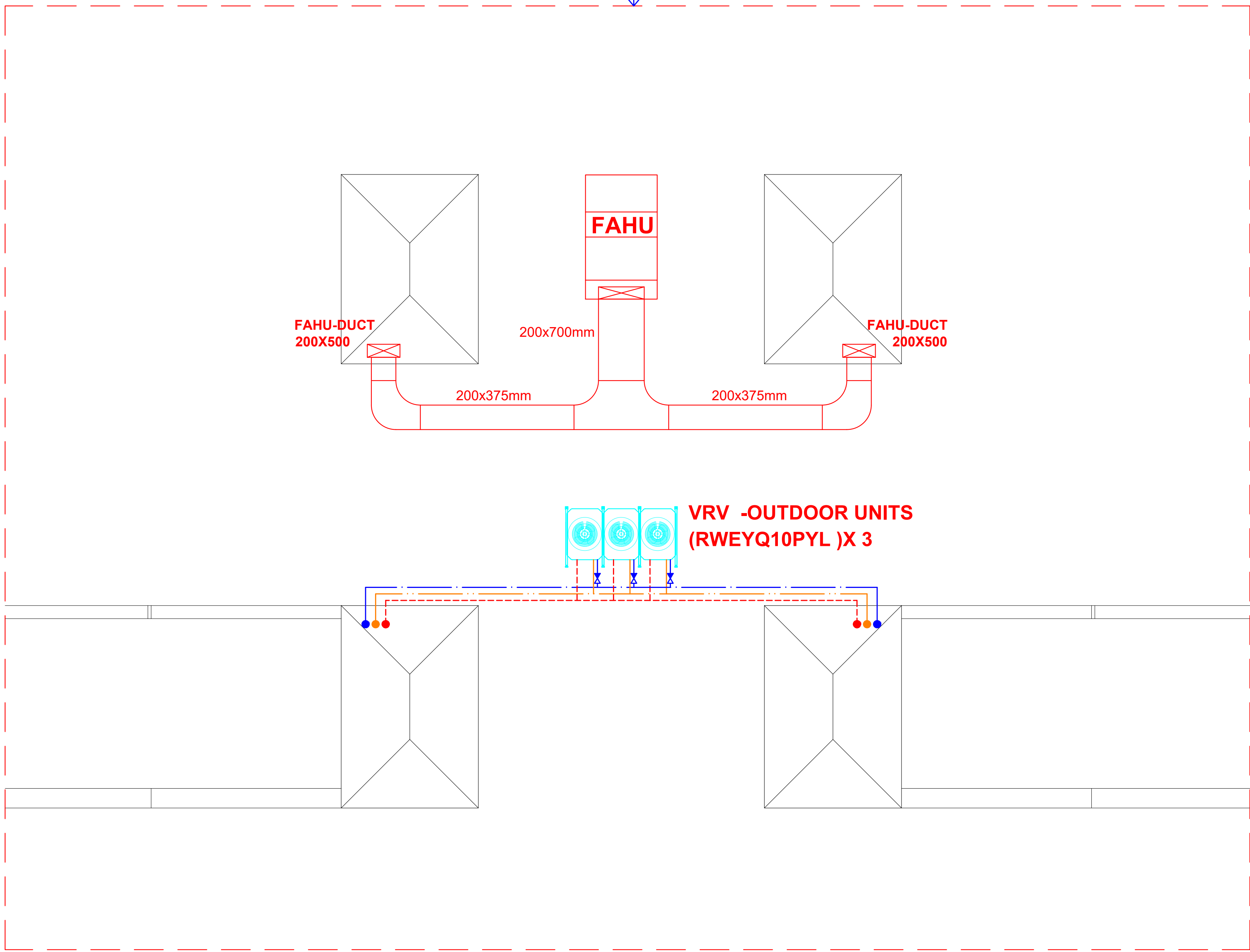
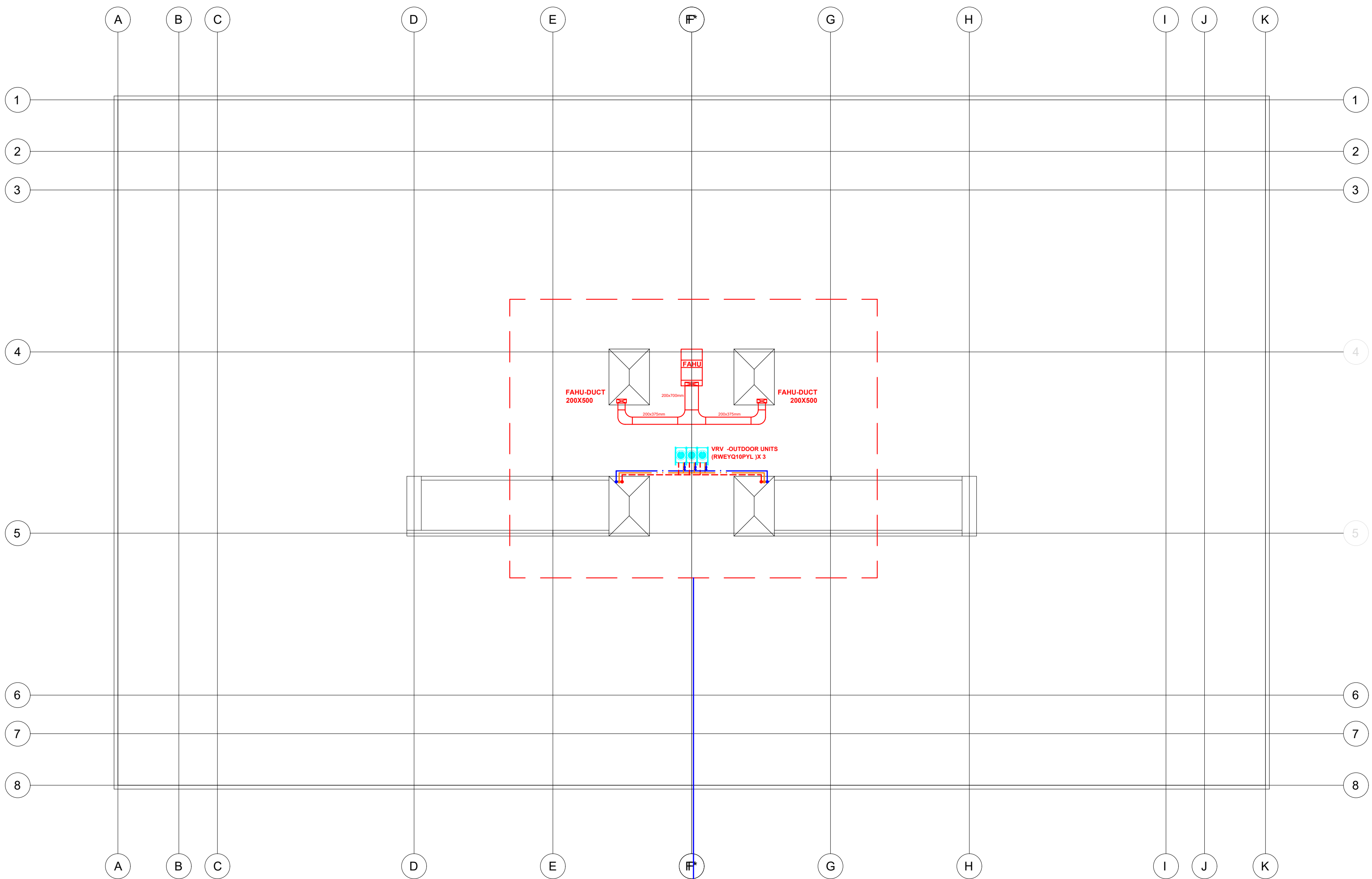
The design uses a cooling load of 57 W/m² and an air flow rate of 3.66 L/s·m², optimized through enhanced insulation, efficient glazing, and effective shading to reduce thermal gains. An additional 0.65 L/s·m² is included for fresh air supply, representing 17% of the total flow rate, ensuring adequate indoor air quality while maintaining energy efficiency.

Indoor Units			FXSA15A	FXSA20A	FXSA25A	FXSA32A	FXSA40A	FXSA50A	FXSA63A	FXSA80A	FXSA100A	FXSA125A	FXSA140A	
Capacity	UK Total Cooling	kW	1.10	1.50	2.00	2.50	3.10	3.90	4.80	6.20	7.80	10.20	11.20	
	UK Sensible Cooling	kW	0.90	1.20	1.50	2.00	2.50	3.20	3.90	4.90	6.30	7.80	9.00	
	Nominal Cooling	kW	1.7	2.2	2.8	3.6	4.5	5.6	7.1	9.0	11.2	14.0	16.0	
	Nominal Heating	kW	1.9	2.5	3.2	4.0	5.0	6.3	8.0	10.0	12.5	16.0	18.0	
Air Flow Rate	High	m3/sec	0.145	0.150	0.150	0.158	0.250	0.253	0.350	0.383	0.533	0.600	0.650	
	Nom	m3/sec	0.125	0.125	0.125	0.133	0.208	0.208	0.300	0.325	0.450	0.525	0.567	
	Low	m3/sec	0.108	0.108	0.108	0.117	0.183	0.183	0.250	0.267	0.383	0.433	0.467	
External Static Pressure	High	Pa	150	150	150	150	150	150	150	150	150	150	150	
	Low	Pa	30	30	30	30	30	30	30	40	40	50	50	
Dimensions	Height x Width x Depth		mm	245 x 550 x 800			245 x 700 x 800			245 x 1000 x 800		245 x 1400 x 800		245 x 1550 x 800
Weight			kg	23.5	23.5	23.5	24.0	28.5	29.0	35.5	36.5	46.0	47.0	51.0
Electrical Details	Running Current	amps	0.7	0.7	0.7	0.8	1.3	1.3	1.3	1.5	1.8	2.0	2.7	
	Power Supply	Phase / Hz / V						1 / 50 / 230						
	Max Fuse Amp	amps						6						
Sound Level	Sound Pressure	High	dBA	29.5	30.0	30.0	31.0	35.0	35.0	33.0	35.0	36.0	39.0	41.5
		Nom	dBA	28.0	28.0	28.0	29.0	32.0	32.0	30.0	32.0	34.0	36.0	38.0
	Sound Power	Low	dBA	25.0	25.0	25.0	26.0	29.0	29.0	27.0	29.0	31.0	33.0	34.0
			dBA	54.0	54.0	54.0	55.0	60.0	60.0	59.0	61.0	61.0	64.0	64.0
Piping Connections	Liquid	inches (mm)	1/4 (6.4)	1/4 (6.4)	1/4 (6.4)	1/4 (6.4)	1/4 (6.4)	1/4 (6.4)	1/4 (6.4)	1/4 (6.4)	3/8 (9.5)	3/8 (9.5)	3/8 (9.5)	
	Gas	inches (mm)	3/8 (9.5)	3/8 (9.5)	3/8 (9.5)	3/8 (9.5)	1/2 (12.7)	1/2 (12.7)	1/2 (12.7)	1/2 (12.7)	5/8 (15.9)	5/8 (15.9)	5/8 (15.9)	

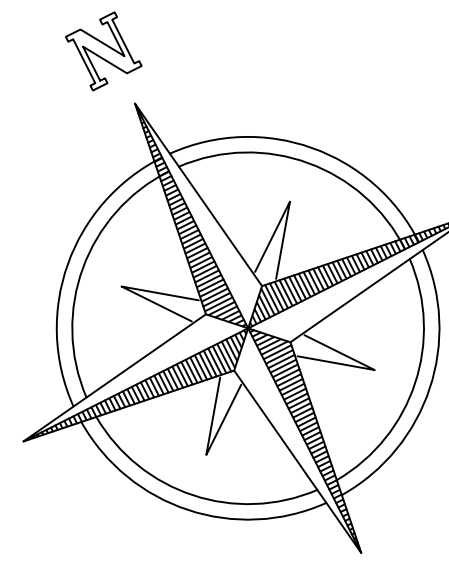
OUTSIDE UNITS			
MODEL		RWEYQ10PYL	RWEYQ20PYL
Combination units		—	(RWEYQ10PYL) X 2
Power supply		YL: 3-phase 4-wire system, 380 V, 60 Hz	
Cooling capacity (*1)(*2)	kcal/h(*1)	23,200	46,400
	Btu/h(*1)	92,100	184,000
	(*1)	27.0	54.0
	kW (*2)	26.7	53.4
Heating capacity	kcal/h	27,100	54,200
	Btu/h	107,000	215,000
		31.5	63.0
	kW	6.03	12.1
Power consumption	Cooling (*2)	6.05	12.1
	Heating		18.2
Casing colour		Ivory white (SY7.5/1)	
Dimensions (H x W x D)		1,000 x 780 x 550	(1,000 x 780 x 550) x 2
Compressor	Type	Hermetically sealed scroll type	
	Motor output	kW	4.2 x 2
Refrigerant piping connections	Liquid		ø9.5 (Flare)
			ø15.9 (Flare)
	Suction gas ★1		ø22.2 (Brazing)
			ø28.6 (Brazing)
Water piping connections	High and low pressure gas		ø19.1 ★2, ø22.2 ★3 (Brazing)
			ø22.2 ★2, ø28.6 ★3 (Brazing)
	Water inlet		PT1 1/4B internal thread
			(PT1 1/4B) x 2 internal thread
Machine weight (Operating weight)	Water outlet		PT1 1/4B internal thread
			(PT1 1/4B) x 2 internal thread
	Drain outlet		PS1/2B internal thread
			(PS1/2B) x 2 internal thread
Sound level		kg	150 (152)
		dB(A)	51
Operation range (Inlet water temp.)			10 to 45
			11-100
Capacity control		%	23-100
			R-410A
Refrigerant	Type		
	Charge	kg	4.2

COMBINATION MODEL			RAHU-VR1500	RAHU-VR2000
AHU	Model Name		740x900	750x1100
	Air Flow		CFM	1,500
	ESP		Pa	250
	Motor Power		KW	1.0
	Dimension	Width	mm	900
		Height	mm	840
		Length	mm	1780
	Prefilter		G4	G4
	Total Cooling Capacity (1)		KW	11.3
	Sensible Cooling Capacity		KW	8.9
	Fan Type		EC FAN	EC FAN

Number of slots A ₁ in m ²	Air flow rate per meter length		Face Velocity m/sec	Throw in meters	P _t in mm H ₂ O	Noise Criteria (NC)
	Cfm	m ³ /sec				
1	50	0.024	2.61	3.6 - 2.1 - 0.6	0.65	<15
	75	0.035	3.80	4.9 - 3.0 - 1.8	1.44	18
	100	0.047	5.11	6.1 - 4.3 - 2.7	2.55	30
	125	0.059	6.41	7.0 - 4.9 - 3.7	3.97	35
	150	0.071	7.72	7.6 - 5.8 - 4.6	4.37	38
0.0092	175	0.083	9.02	8.5 - 6.7 - 5.2	6.12	41
	100	0.047	2.61	4.3 - 2.4 - 0.6	0.97	<15
	125	0.059	3.28	5.2 - 3.0 - 1.5	1.47	20
	150	0.071	3.94	6.1 - 3.9 - 2.1	2.12	28
	175	0.083	4.61	6.7 - 4.6 - 3.0	2.87	30
0.016	200	0.094	5.22	7.0 - 5.2 - 4.3	3.73	33
	250	0.118	6.55	7.9 - 6.1 - 4.9	5.78	39
	125	0.059	2.11	4.2 - 2.2 - 0.7	0.86	<15
	150	0.071	2.53	4.9 - 2.9 - 1.4	1.21	18
	200	0.094	3.36	5.5 - 3.7 - 2.4	2.17	25
0.028	225	0.106	3.78	6.5 - 4.4 - 3.3	2.73	29
	250	0.118	4.21	7.6 - 5.3 - 4.0	3.34	34
	300	0.142	5.07	8.6 - 6.2 - 4.7	4.37	37
	150	0.071	1.91	4.6 - 2.7 - 1.1	0.72	<15
	200	0.094	2.53	5.4 - 3.6 - 2.0	1.39	21
0.0372	250	0.118	3.17	6.1 - 4.5 - 3.1	1.98	25
	300	0.142	3.82	6.7 - 5.2 - 3.9	2.85	30
	350	0.165	4.43	7.9 - 5.6 - 4.5	3.84	35
	400	0.189	5.08	8.8 - 6.7 - 5.2	4.51	38



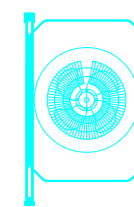
COLLEGE OF ARCH. & PLAN
DEPT. OF BUILDING ENGINEERING



ALL UNITS IN METER

LEGENDES:

VRF OUTDOOR UNIT



FRESH AIR HANDLING UNIT



DRAINAGE PIPE

SUPPLY PIPE

RETURN PIPE

GATE VALVE



Design Studio (8), Technical
Project, 2024/2025

INSTRUCTORS:

Prof. M. Essam Shaawat Co.
Dr. Faris Almaziad
Dr. M. Farooqi
Dr. M. Al Mulhim
Dr. AL Tayeb
Dr. M. Al hater
Eng. Meqdad
Eng. Rehan Gamil
Eng. Noman Ashraf

DONE BY:
ABDULELAH ALQABBANI

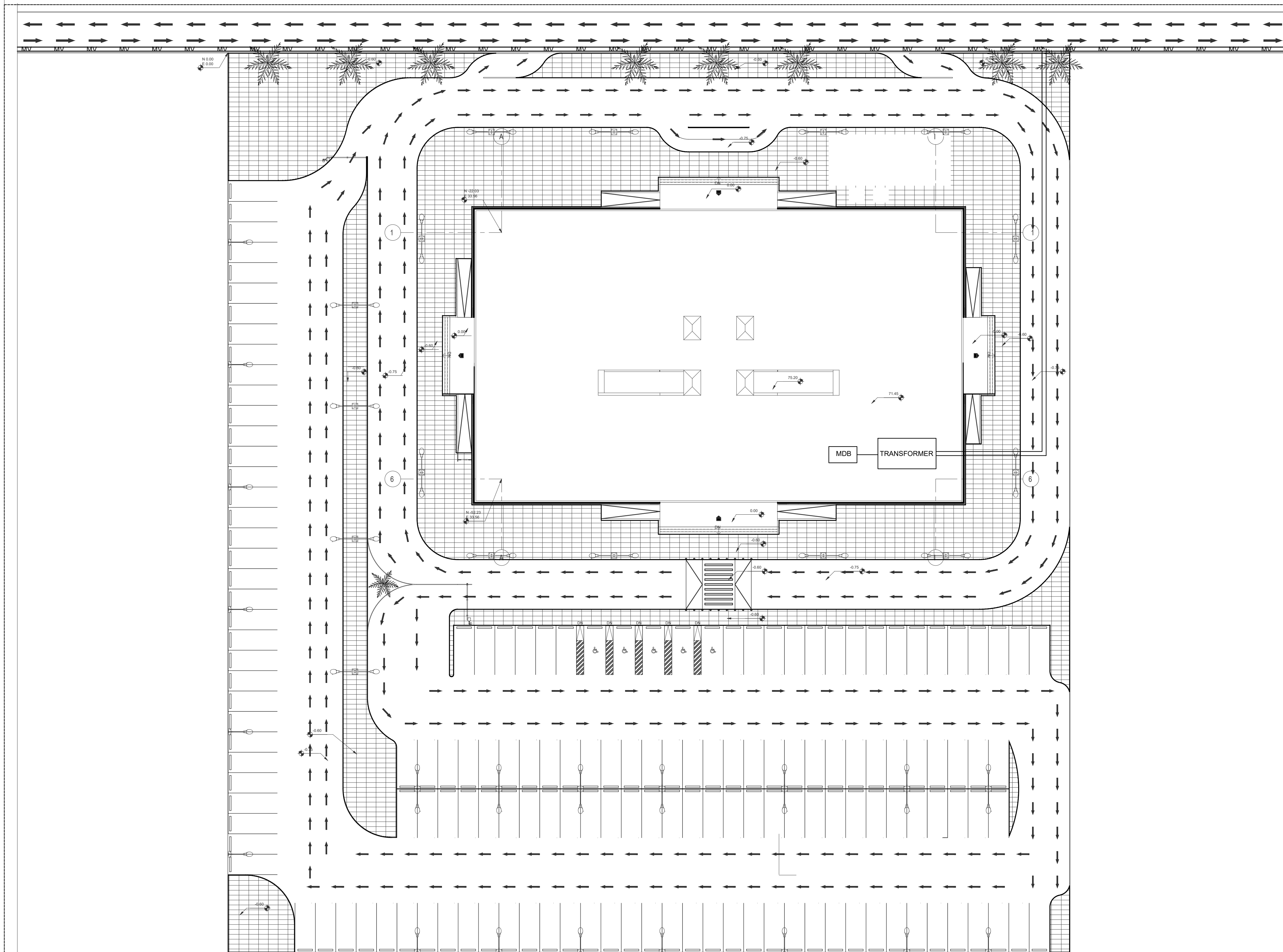
PROJECT:
Apartments Building

DRAWING NAME:
Roof Plan

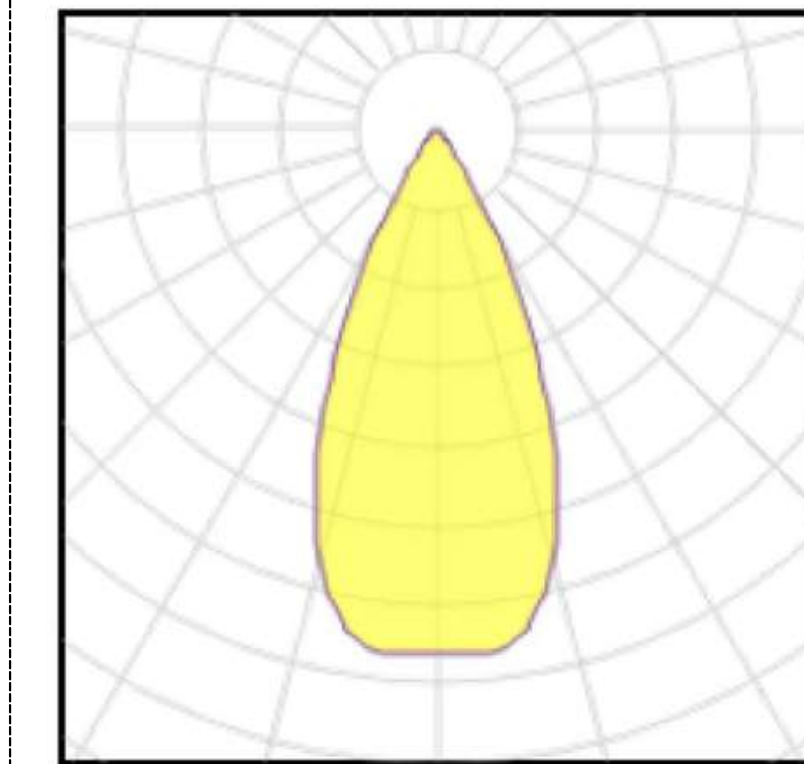
SCALE:

--

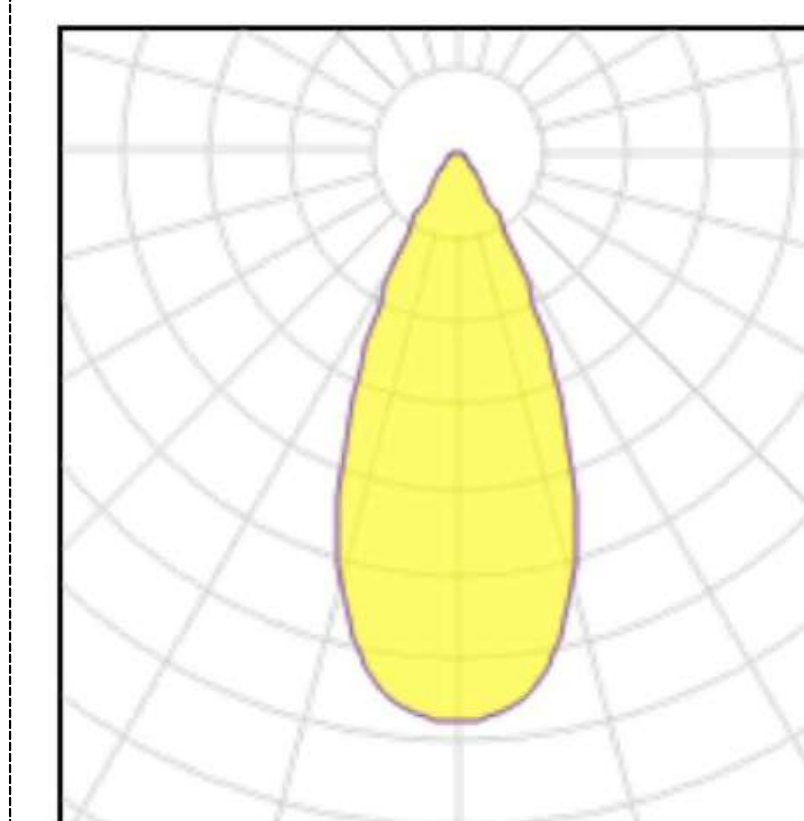
SHEET NUMBER:
A30



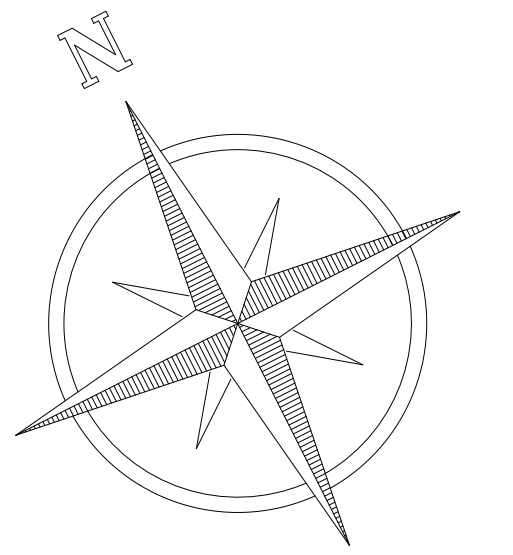
Light output 1



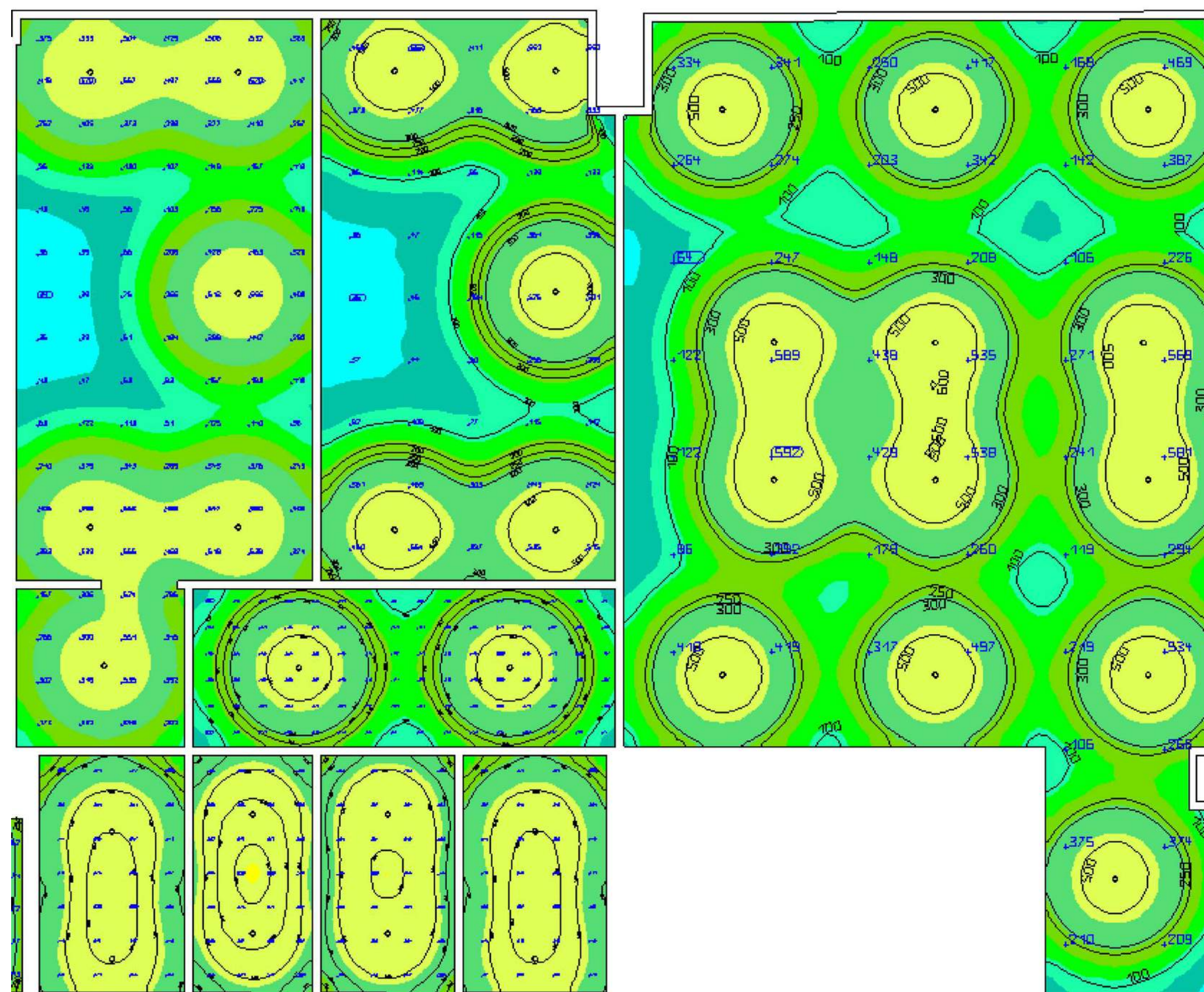
1 x LED			
Nominal lamp power	19.8 W	LOR	100%
Lamp flux	1761 lm	Total flux	1760 lm
Luminous efficacy	89 lm/W	Total power	19.8 W
CCT	4000 K		
CRI	90		



1 x LED			
Nominal lamp power	14.3 W	LOR	100%
Lamp flux	1094 lm	Total flux	1094 lm
Luminous efficacy	77 lm/W	Total power	14.3 W
CCT	2700 K		
CRI	90		



ALL UNITS IN METER



Select a utilization profile

Preset: USA

Template selection: Please select

Active utilization profile

Type of use

Space: 33 - LIGHTING FOR RESIDENCES

Application: Residential Interiors; Arts, Crafts and Collecting

Illuminance

Maintenance values

Visual task (Em): 300 lx

Uniformity (E_{avg}/E_{min}): 3.000

Use times

Day: 1398 Hours per year
~3.83 Hours per day

Night: 2 Hours per year
~0.01 Hours per day

Absence factor: 0.25

Partial operation factor of the building operation time for lighting: 0.90

Begin Time: 0 o'clock

End Time: 24 o'clock

Days per Week: 5

Maintenance and miscellaneous

Colour rendering index (Ra): 80

Height of working plane: 0.76 m

Reduction factor (task area): 0.97

Height for measuring the cylindrical illuminance: 1.20 m

Inspection interval: 1.0 Years

Ambient condition: Clean

Maintenance interval for luminaires: 3.0 Years

Replacement interval for lamps: 1.0 Years

Individual replacement of failed lamps: ☒

Design Studio (8), Technical
Project, 2024/2025

INSTRUCTORS:

Prof. M. Essam Shaawat Co.
Dr. Faris Almaziad
Dr. M. Farooqi
Dr. M. Al Mulhim
Dr. AL Tayeb
Dr. M. Al hater
Eng. Meqdad
Eng. Rehan Gamil
Eng. Noman Ashraf

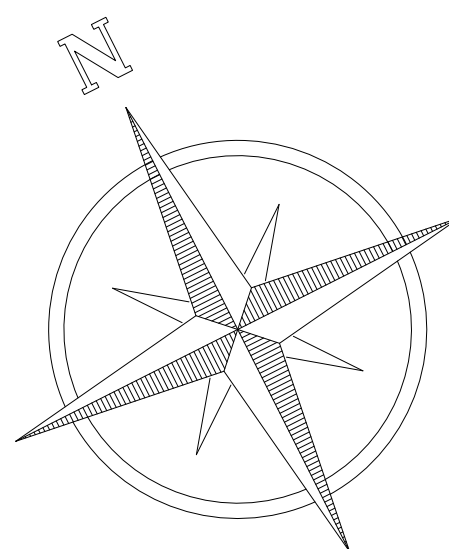
DONE BY:
ABDULELAH ALQABBANI

PROJECT:
Apartments Building

DRAWING NAME:
Site Plan

SCALE:
1:150

SHEET NUMBER:
A31



ALL UNITS IN METER

DB

LIGHT KEY

CIRCULAR LIGHTING
FIXTURE

Design Studio (8), Technical
Project, 2024/2025

INSTRUCTORS:

Prof. M. Essam Shaawat Co.
Dr. Faris Almaziad
Dr. M. Farooqi
Dr. M. Al Mulhim
Dr. AL Tayeb
Dr. M. Al hater
Eng. Meqdad
Eng. Rehan Gamil
Eng. Noman Ashraf

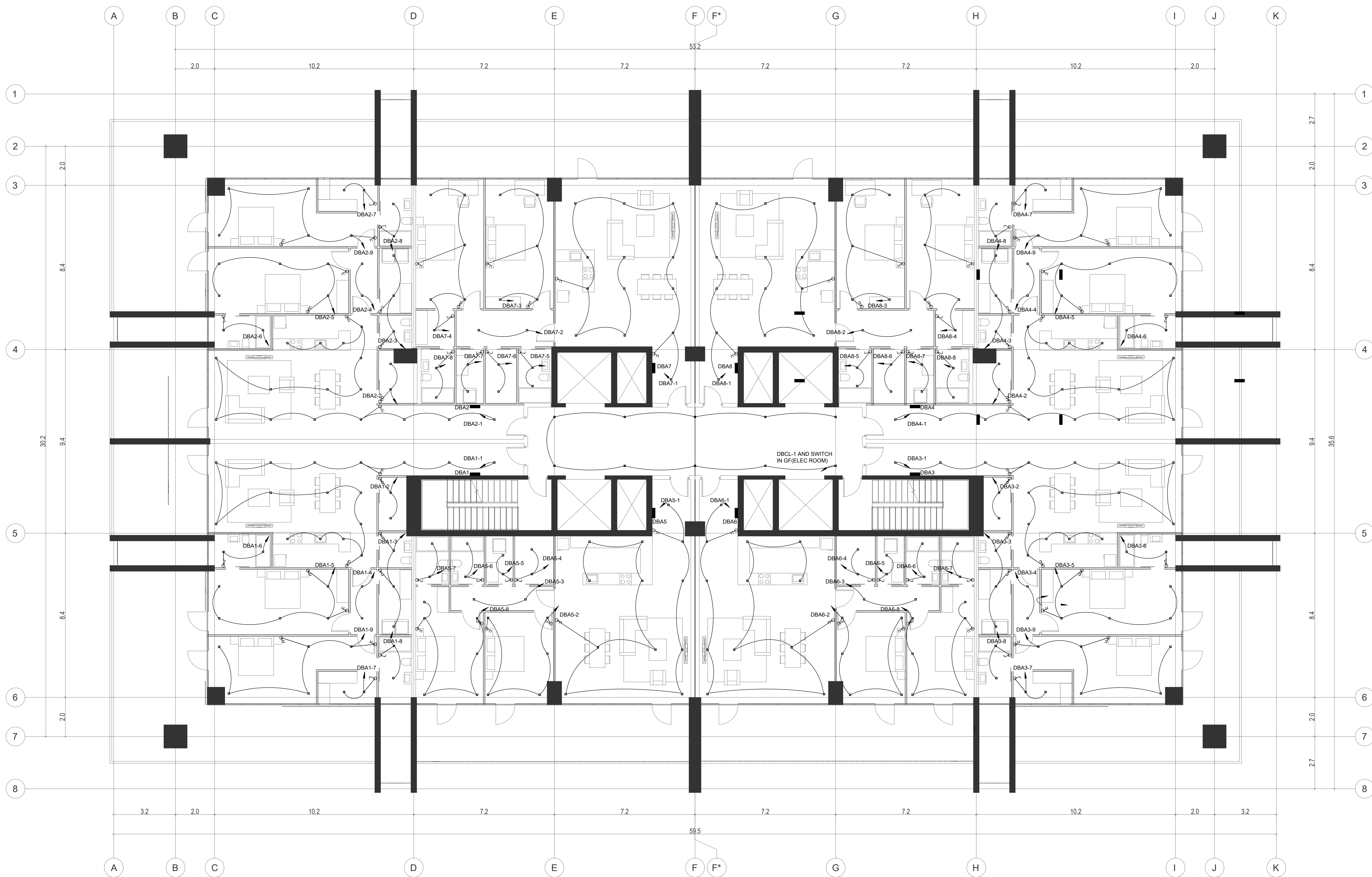
DONE BY:
ABDULELAH ALQABBANI

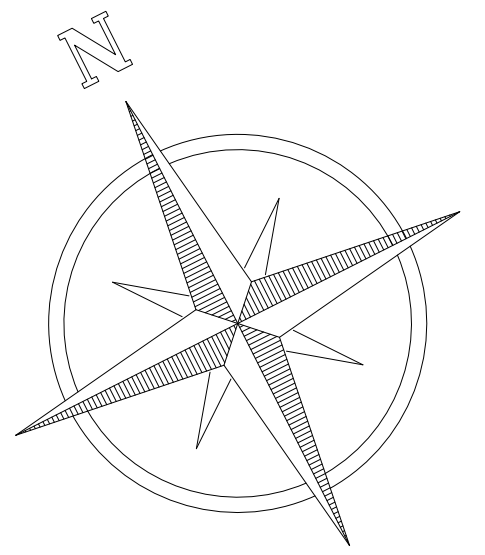
PROJECT:
Apartments Building

DRAWING NAME:
LIGHTING LAYOUT

SCALE:
1:75

SHEET NUMBER:
A32

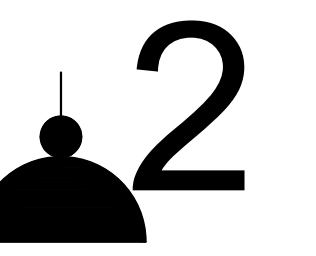




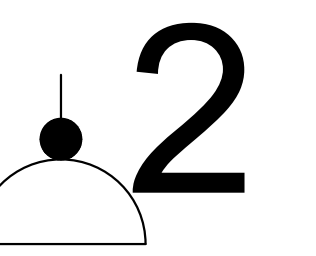
ALL UNITS IN METER

DB

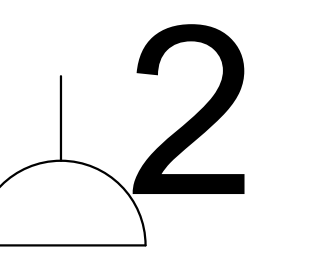
WP DOUBLE
SOCKET 400W
W/ NEON
INDICATOR



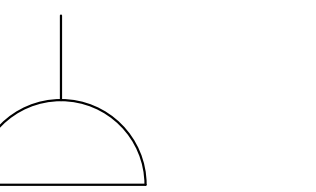
DOUBLE
SOCKET 400W
W/ NEON
INDICATOR



DOUBLE
SOCKET 400W



SINGLE
SOCKET 250W



Design Studio (8), Technical
Project, 2024/2025

INSTRUCTORS:

Prof. M. Essam Shaawat Co.
Dr. Faris Almaziad
Dr. M. Farooqi
Dr. M. Al Mulhim
Dr. AL Tayeb
Dr. M. Al hater
Eng. Megdad
Eng. Rehan Gamil
Eng. Noman Ashraf

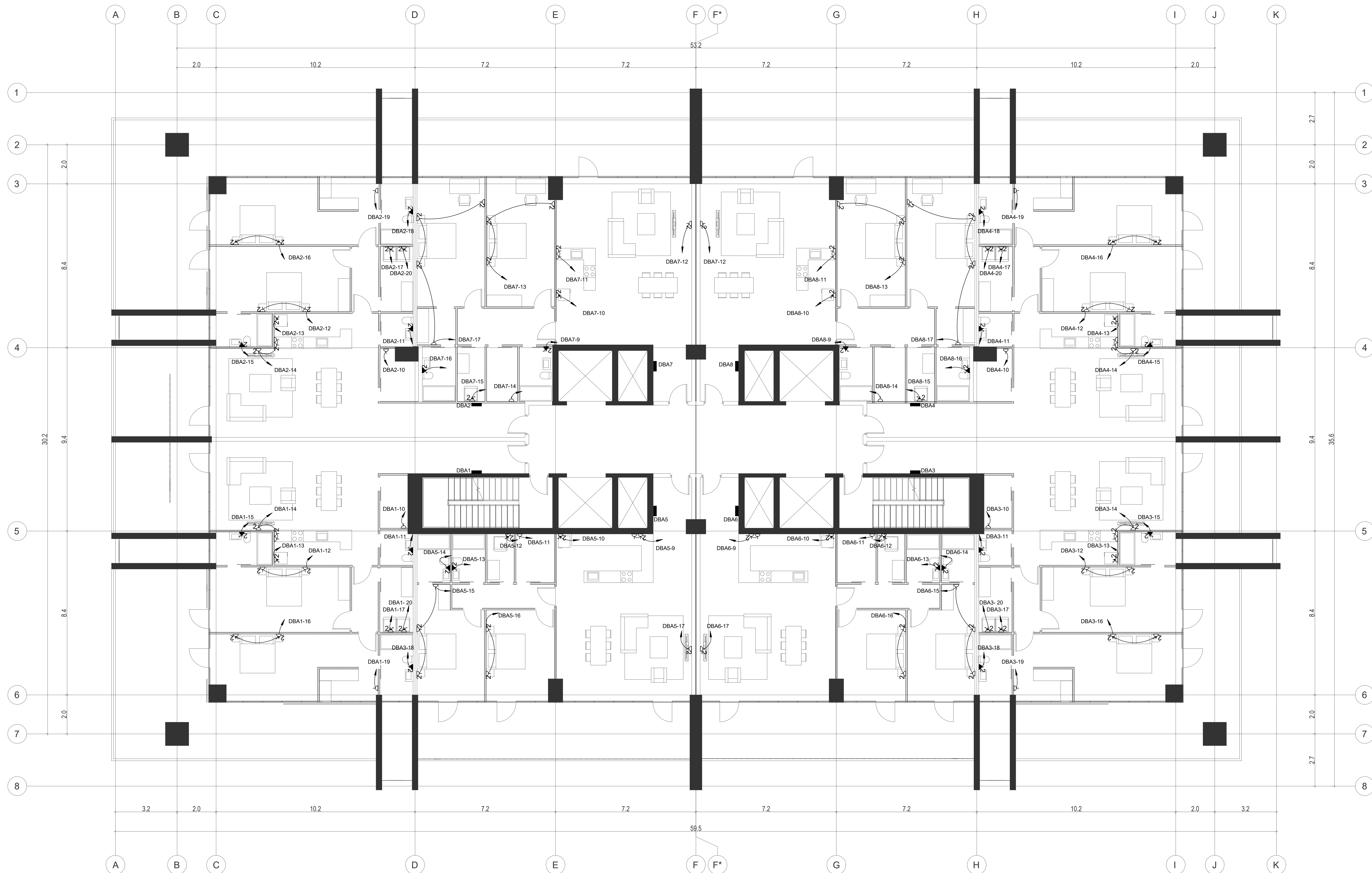
DONE BY:
ABDULELAH ALQABBANI

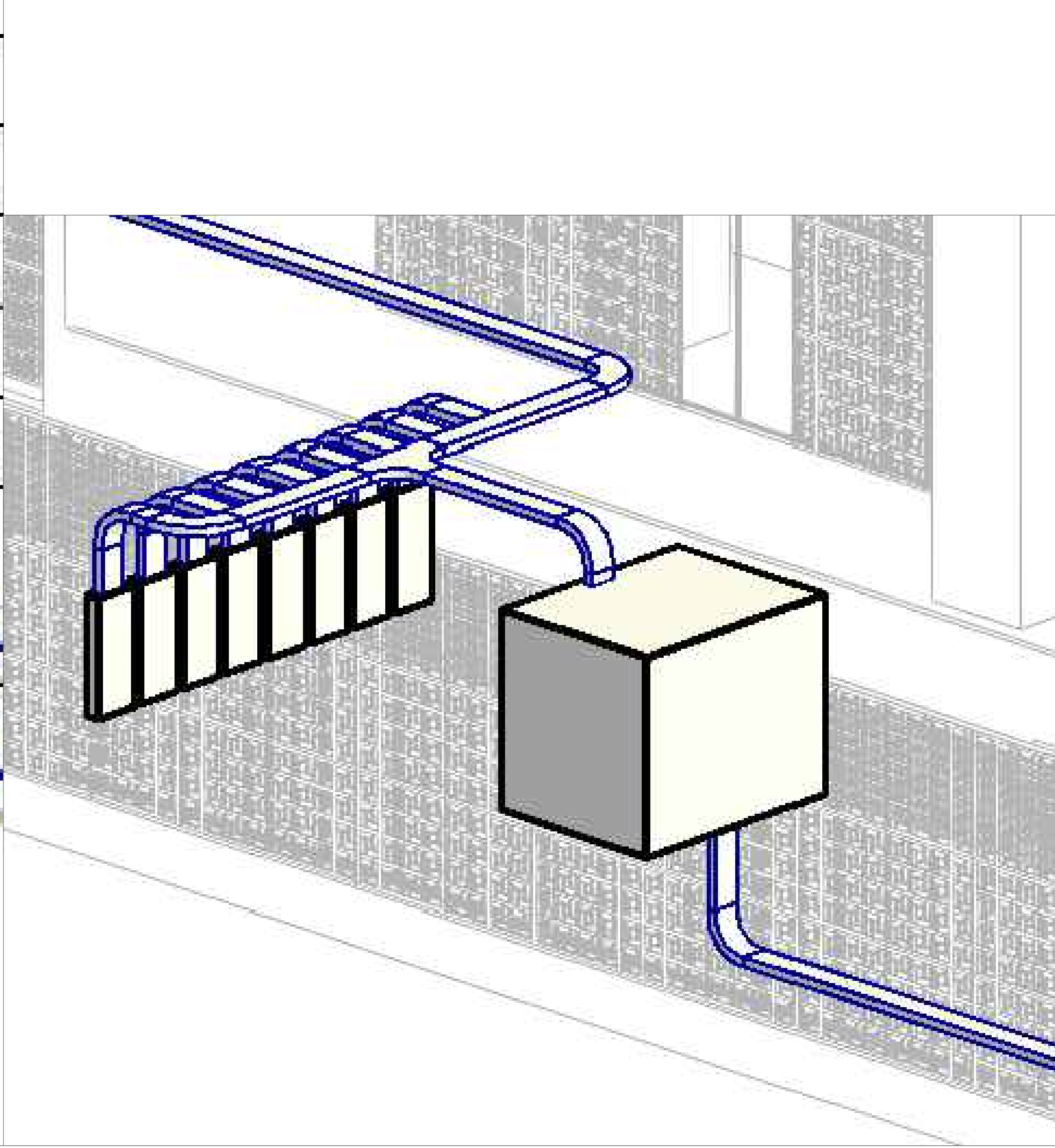
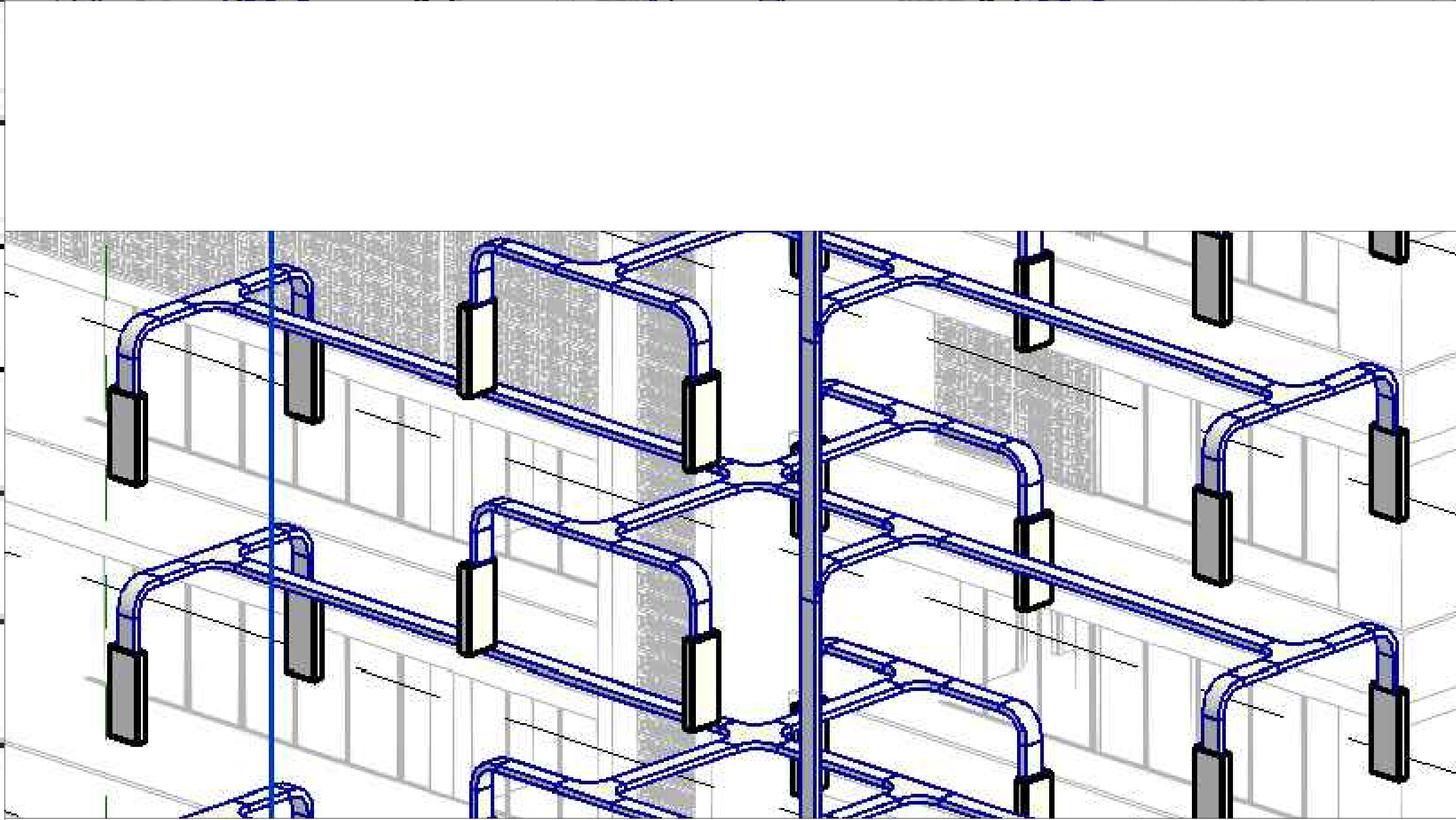
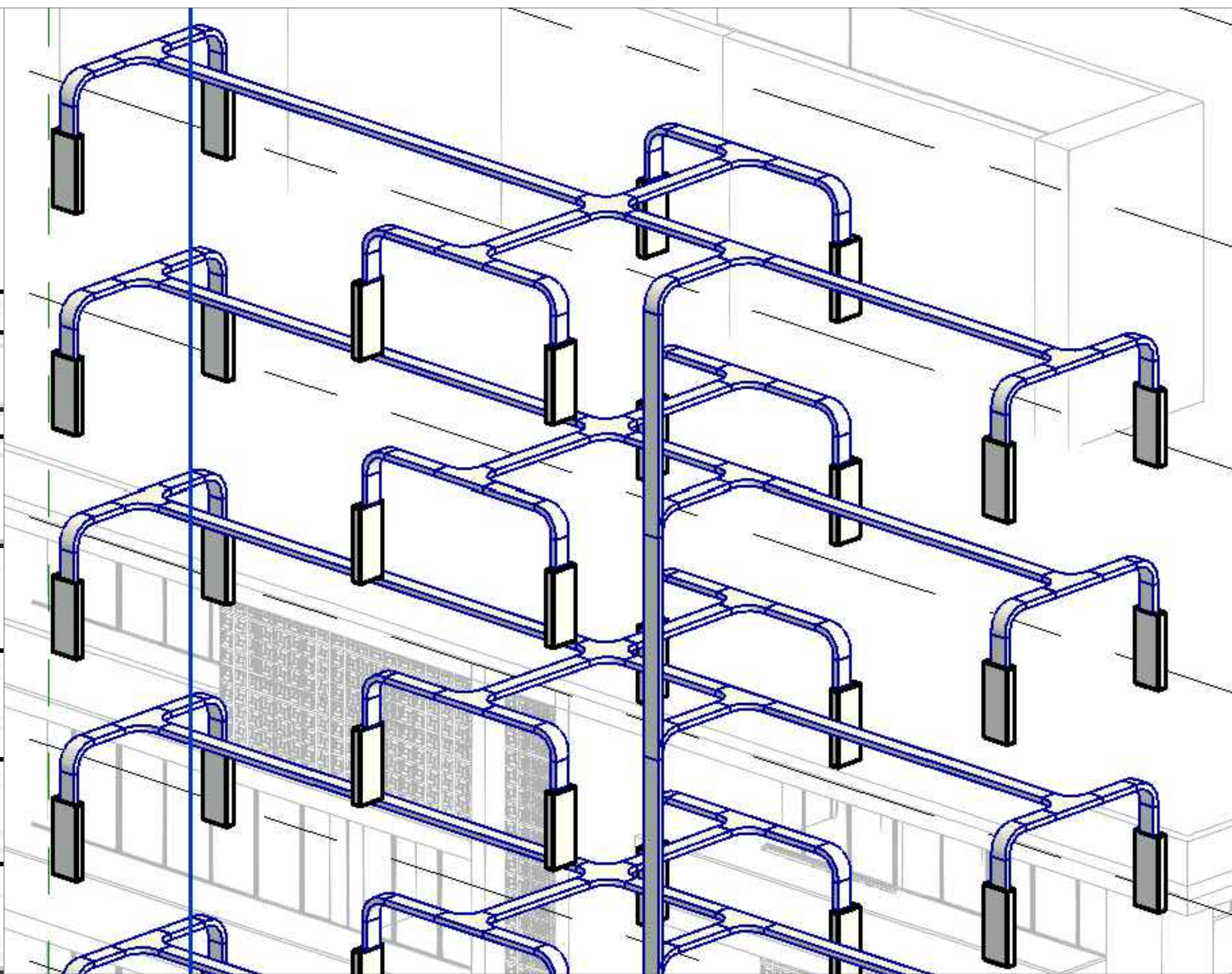
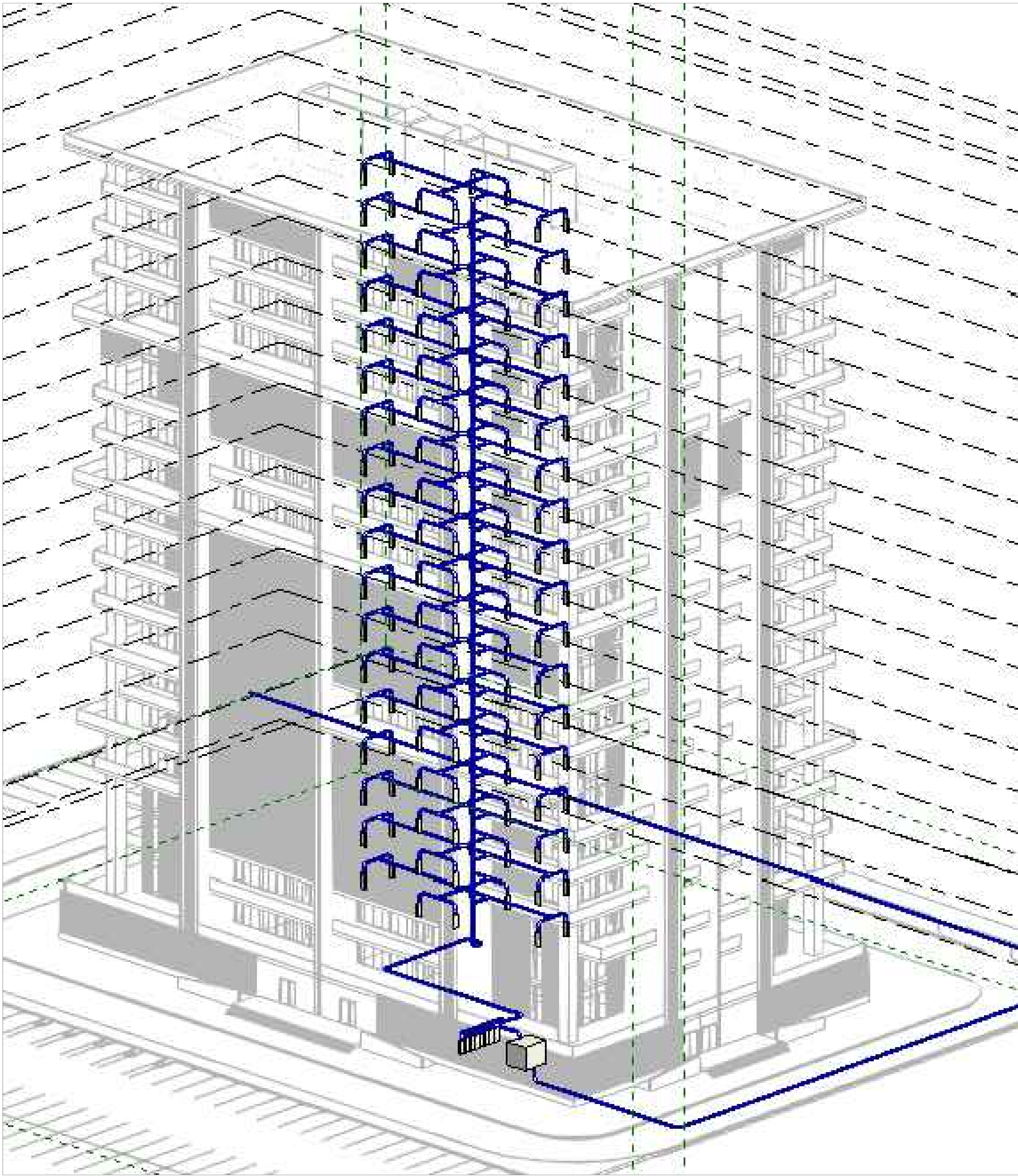
PROJECT:
Apartments Building

DRAWING NAME:
SOCKET LAYOUT

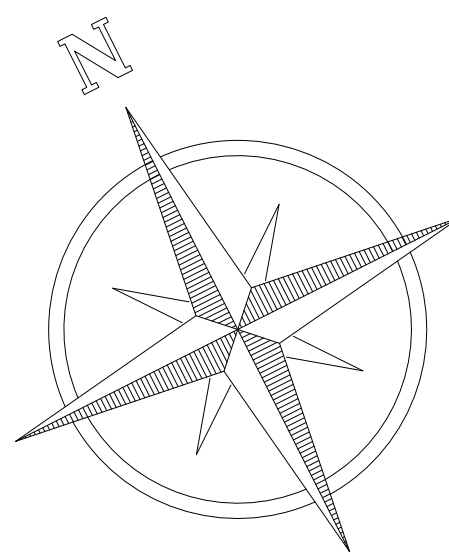
SCALE:
1:75

SHEET NUMBER:
A33





COLLEGE OF ARCH. & PLAN
DEPT. OF BUILDING ENGINEERING



ALL UNITS IN METER

Design Studio (8), Technical
Project, 2024/2025

INSTRUCTORS:

Prof. M. Essam Shaawat Co.
Dr. Faris Almaziad
Dr. M. Farooqi
Dr. M. Al Mulhim
Dr. AL Tayeb
Dr. M. Al hater
Eng. Meqdad
Eng. Rehan Gamil
Eng. Noman Ashraf

DONE BY:
ABDULELAH ALQABBANI

PROJECT:
Apartments Building

DRAWING NAME:
RISER

SCALE:
--

SHEET NUMBER:
A34